



Driftwood

Setup & Mod Guide

Minecraft 1.21.1 – NeoForge

Version 4.0.0

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Driftwood — Setup & Mod Guide

Wave -1 — Prerequisites

JDK 21

Tool	Role
Eclipse Temurin JDK 21 (LTS)	Java runtime and development kit for Minecraft 1.21.1

Install the **JDK** (not just JRE) — the JDK includes [javac](#) and [jlink](#) needed by NeoForge. Minimum version 21.0.0, any LTS build. On Windows, use the MSI installer and ensure `JAVA_HOME` is set to the install path (e.g., `C:\Program Files\Eclipse Adoptium\jdk-21.0.6.13-hotspot`). On Linux/macOS, use your package manager (`apt install temurin-21-jdk`, `brew install temurin21`).

Verify installation:

```
java -version
# Expected: openjdk version "21.0.x" ...
```

Performance Tuning

Recommended JVM arguments (for XMCL → instance settings → Java → JVM Arguments) — community consensus for Java 21 + NeoForge (AllTheMods guide, CleanroomMC, Distant Horizons' own FAQ):

```
-Xms12G -Xmx12G -XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:
+PerfDisableSharedMem -XX:+DisableExplicitGC -XX:
+UseDynamicNumberOfGCThreads
```

System RAM	Recommended allocation
16 GB	<code>-Xms8G -Xmx8G</code>
32 GB	<code>-Xms12G -Xmx12G</code>
64 GB	<code>-Xms16G -Xmx16G</code>

Why ZGC: Minecraft's default collector, G1GC, stops the world while collecting — that's the classic modpack stutter. ZGC's **generational** mode (JEP 439) collects concurrently with near-zero pauses and scales to large heaps; Distant Horizons' own FAQ recommends ZGC for Java 21+ and names G1GC as a stutter source. On Java 21 the generational mode must

be enabled explicitly with `-XX:+ZGenerational` — newer JDKs (23+) have it on by default. Temurin JDK 21 includes it.

-Xms must equal -Xmx: an equal initial/max heap stops the JVM from resizing the heap mid-game, which causes micro-stutters. `-XX:+AlwaysPreTouch` commits all heap pages at startup (slightly slower launch, no page-fault hitches later); `-XX:+PerfDisableSharedMem` skips `hsperfdata` writes; `-XX:+DisableExplicitGC` ignores `System.gc()` calls some mods make (with ZGC those force an expensive full collection); `-XX:+UseDynamicNumberOfGCThreads` lets GC threads scale with your CPU.

Memory sizing: don't over-allocate — a bigger heap means longer GC cycles even with ZGC. For a 200+ mod pack with Distant Horizons, 12–16 GB is the sweet spot on a 10+ core CPU (Distant Horizons' guidance: 6–8 GB for 4–8 cores, 12–16 GB for 10+ cores). ZGC also needs a little extra RAM headroom because it doesn't use compressed object pointers. `-XX:SoftMaxHeapSize` is optional: it caps ZGC's *typical* usage while allowing bursts up to `-Xmx` (e.g. `-Xms12G -Xmx16G -XX:SoftMaxHeapSize=12G`) — most players won't need it.

Distant Horizons is CPU-bound, not RAM-bound — if you see stutter while LODs generate, lower DH's CPU Load preset (Video Settings → DH menu) before raising the heap.

Fallback — tuned G1GC (if ZGC misbehaves on your hardware, e.g. an older CPU): keep the same `-Xms/-Xmx` from the table and swap the collector flags. G1GC is the safe default for 4–8 GB heaps but shouldn't be pushed past ~12 GB:

```
-XX:+UnlockExperimentalVMOptions -XX:+UseG1GC -XX:G1NewSizePercent=20 -XX:G1ReservePercent=20 -XX:MaxGCPauseMillis=50 -XX:G1HeapRegionSize=32M
```

Sources: [AllTheMods](#) — Java & Arguments, [CleanroomMC](#) — JVM Arguments, [Distant Horizons FAQ](#) — ZGC & stuttering

X Minecraft Launcher (XMCL)

Tool	Role
XMCL	Mod manager and launcher for the modpack

XMCL manages mod and modpack installation, updates, and launching for this modpack. It supports both **CurseForge** and **Modrinth** as mod sources in one launcher, handles NeoForge installation automatically, and manages Java versions per instance. Open source, free, with no premium tier.

Setup:

1. Download and install **XMCL** from xmcl.app for your OS.
 - Windows: App Installer, AppX, or Zip (x64)
 - macOS: DMG (Intel or Apple Silicon)
 - Linux: Deb, RPM, AppImage, Flathub
 - Or via winget: `winget install CI010.XMinecraftLauncher`
2. Open XMCL and follow the **NeoForge** instructions below to create your instance.

Once the instance is running, install mods through the **Mods** tab from either CurseForge or Modrinth, or by dropping `.jar` files into the instance's `mods/` folder. XMCL uses hard links — mods are stored once globally and linked per instance, avoiding duplicates. All mod links in this document link to their CurseForge pages — XMCL can install from CurseForge directly via the Mods tab.

For texture packs, drop `.zip` files into the instance's **Resource Packs** tab or folder. For shaderpacks, drop `.zip` files into the instance's **Shader Packs** tab or folder. Access the instance folder from the instance settings.

NeoForge

Tool	Role
NeoForge	Mod loader for Minecraft 1.21.1

NeoForge is the mod loader powering this modpack. **XMCL installs NeoForge and Minecraft 1.21.1 automatically** when you create a NeoForge instance (see [XMCL](#) above). Follow the XMCL steps — it handles the Minecraft version, mod loader, and Java runtime in one step.

If you prefer the official launcher:

1. Run **Minecraft 1.21.1** vanilla at least once from the official launcher to generate the game directory.
2. Download the **NeoForge installer** for 1.21.1 from neoforged.net.
3. Run it: `java -jar neoforge-1.21.1-<version>-installer.jar` → select “Install client”.
4. Launch the new **neoforge** profile once to generate the `mods/` directory.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.
 - Your Minecraft directory: Windows `%APPDATA%\minecraft`, Linux `~/.minecraft`, macOS `~/Library/Application Support/minecraft`

Important: Do not add any mod `.jar` files yet. Wave -1 is complete when the vanilla 1.21.1 client runs under NeoForge with no mods installed.

Skip this if you've played Minecraft before.

Controls (Default)

Action	Key
Move	W A S D
Jump	Space
Sneak	Shift
Sprint	Ctrl
Attack / Break	Left-click
Use / Place	Right-click
Inventory	E
Drop	Q
Hotbar slots	1–9

Your First 10 Minutes

1. **Punch a tree** — Hold left-click on a tree trunk until it breaks. Get 8+ logs.
2. **Make a crafting table** — Open inventory (E), place 1 log → 4 planks. Fill all 4 crafting slots with planks → crafting table.
3. **Craft tools** at the crafting table (3×3 grid):

Tool	Pattern
Wooden Pickaxe	3 planks top row, 2 sticks middle column
Wooden Axe	3 planks top-left, 2 sticks down
Wooden Sword	2 planks middle column, 1 stick below

1. **Gather stone** — Find grey stone, mine 8–12 blocks with your pickaxe.
2. **Build a shelter** — Dirt/wood walls + door (6 planks, 2 columns of 3) + torches (stick + coal/charcoal).
3. **Craft a furnace** — 8 cobblestone in a ring (center empty). Use it to smelt ore, cook meat, make charcoal.
4. **Survive the night** — Stay inside. Dig, expand, or sleep if you have a bed (3 wool + 3 planks).
5. **Day 2 priorities:** Find food, get iron (dig staircase-down until you see brown speckled blocks), find a village, make a bed.

Keybind to Set Immediately

Open **Options** → **Controls** → **Key Binds** and set these before anything else:

Action	Suggested Key	Why
Open Waystone Menu	Ctrl+Shift+W	Fast travel anywhere
Zoom	C	Sodium zoom
Toggle Dynamic Lights	Ctrl+L	Torch in off-hand = mobile light

Action		Suggested Key	Why	
Open Backpack		B		Sophisticated Backpacks
Open Fullscreen Map		J		JourneyMap

Wave 0 — Foundation

The essentials that make the game run well. Dependencies, performance, rendering, shaders, chunk loading, and texture packs. Install everything in this wave before launching.

Performance & Rendering

Mod	Role
Sodium 0.8.12	Rendering engine — pin 0.8.12+ (current 1.21.1 line; incompatible with the removed Sodium Options API chain — see STATUS). Fall back to 0.6.x only if DH/Entity Culling misbehave
Iris	Shader loader (v1.8.14, NeoForge native)
Lithium	Server-side game logic optimization
FerriteCore	Memory usage reduction
Distant Horizons	LOD rendering
ModernFix	All-in-one perf + bug fixes
ImmediatelyFast	Immediate-mode rendering
Entity Culling	Hide off-screen entities
Dynamic FPS	Background FPS reduction
FastWorkbench 1.21.1-9.1.3	Caches crafting-table recipes — eliminates late-game crafting lag spikes
MoreCulling v1.0.9	Block-face culling — skips rendering hidden faces (complements Entity Culling). Leaf culling: keep at default — Cull Leaves (Wave 0.5) owns leaves; enabling MoreCulling's leaf modes would duplicate it
BadOptimizations 2.4.1	Bundle of small perf fixes — removes stutter and redundant entity/block logic

ModernFix (configurable in-game via Mod List → ModernFix → Config, or `config/modernfix-mixins.properties`):

- `mixin.perf.dynamic_resources = true` — Load block/item models lazily instead of all at startup. **Biggest single launch-time speedup** in the pack. Reduces memory usage too. Disable if you see missing textures or broken models.

Shaderpack

Shaderpack	Description
Bliss Shaders	Main shader — fantasy-styled Chocapic13 V9 edit; scene variation, LabPBR + Distant Horizons support, iris-native

Bliss Shaders is the pack's main shader (X0nk's large edit of Chocapic13 V9) — a well-performing fantasy look built around scene variation and customization. Runs on Iris (NeoForge) with zero extra mods. Use **v2.1.0+** — required for full Distant Horizons support (DH LODs render through the shader pipeline). Quality presets run Low → Medium → High; colored lighting; LabPBR support (specular f0/reflectance, roughness/metalness, subsurface scattering, emissive, parallax occlusion) when the active resource pack ships PBR maps.

PBR note: Faithful 32x (the base layer) ships no PBR maps, so Bliss's LabPBR features idle on the base textures — the flat 32x look is the intended baseline. Pack-provided PBR maps (Fire Rekindled's Normal/Specular, RAY's 3D Rails' shader support) still apply where present.

Recommended baseline (NVIDIA RTX 4080 SUPER, 16GB VRAM, 4K — target stable 70-80 FPS): start with the **High** quality preset and the shader's defaults. If FPS dips below 70: drop the preset one step, then reduce Shadow Resolution to **1024**, then Shadow Distance to **8 chunks** (see Distant Horizons below).

Bliss Shader Settings — Recommended

Start from the **High** preset, then apply the table below. All options are in the shader's own settings screen: Options → Video Settings → Shaderpacks → Bliss → **Shader Settings**. Values are tuned for the pack's target (RTX 4080 SUPER, 4K, 70–80 FPS, DH LODs at 128–256).

Settings screen	Option	Recommended	Why
Ambient Light	Indirect Lighting	SSAO (default)	⚠ the one that matters — see note below
Ambient Light	AO Multiplier	1.0 (default)	baseline occlusion strength
Ambient Light	GI Multiplier	1.0 (default)	only affects SSRT modes
Ambient Light	Long Range Marching (SSRT) · High Quality Ambient Light (SSRT) · sample 1/4 resolution depth (SSRT)	leave at defaults	SSRT-only toggles — inert while Indirect Lighting = SSAO
Direct Light → Shadows	Shadow Resolution	2048 (default) → 1024 if FPS < 70	documented baseline lever

Settings screen	Option	Recommended	Why
Direct Light → Shadows	Shadow Distance	8 chunks (128)	DH handles far shading; drop to 8 if needed
Direct Light → Shadows	Shadows for Entities	ON	your own shadow always renders
Post Processing → Anti-Aliasing	TAA	ON (default)	SSAO sampling is temporal — TAA is what cleans the dither
Post Processing → Anti-Aliasing	Temporal Upscaling	leave at defaults	free 4K headroom: renders lower, TAA-upscales; keep Scale Factor ≥ 0.75
Mod Support → Distant Horizons	ambient occlusion on LOD chunks · sub-surface scattering · screen-space reflections	ON (defaults)	LODs render through the shader pipeline
Mod Support → Distant Horizons	DH shadowmap support	OFF — never enable	shader's own red warning: destroys performance, blocky/flickery shadows

Indirect Lighting — user-tested rule: keep it on **SSAO**. Any other mode — **GTAO**, **SSRT (AO only)**, **SSRT (AO + GI)** — introduces visible noise in the distance. Bliss's own UI marks both SSRT modes "LOTS OF VISUAL NOISE", and screen-space techniques can't sample DH LOD geometry, so distant areas get the noisy fallback; SSAO is the only mode with a dedicated Distant Horizons code path. If you want the GI look, only run it indoors/in caves — expect noise in open terrain. (If you ever see dithering noise specifically on LOD chunks even on SSAO, the DH settings screen has a noise texture — NOISE_INTENSITY / NOISE_DROPOFF.)

General MC settings: Graphics **Fancy**, Render Distance **10 chunks**, Simulation Distance **8 chunks**, Biome Blend **2 blocks**, Mipmap Levels **4x**, Entity Distance

100%, Fullscreen ON, VSync OFF, Max Framerate **120**. Always Defer Chunk Updates **Enabled**, Use No Error Context **Enabled**.

Distant Horizons handles far rendering — keep vanilla render distance low. DH LODs render through the shader pipeline; start LOD distance at **128** and push to **256** if the FPS budget allows. Monitor VRAM usage (shaders + DH + high-res texture packs can peak past 12GB at 4K). If you experience dips below 70 FPS, first reduce Shadow Resolution to **1024**, then lower Shadow Distance to **8 chunks**.

Texture Packs

Pack	Role
Improved Create 32x	32x32 texture extension for Create — items, blocks, GUI; designed to pair with Faithful 32x
Improved Applied Energistics 2	32x32 texture extension for AE2 — items, blocks, GUI; designed to pair with Faithful 32x
Armored Legacy	Custom armor models + working armor trims for vanilla sets (requires EMF + ETF mods)
Mod Descriptions	Field Guide companion — lore descriptions for 200+ mods' entries (lang-only, order irrelevant)
Fresh Animations	Living entity animation overhaul — mobs blink, look around, show emotion
Enhanced Boss Bars	Redesigned boss bars with themed designs for vanilla and modded bosses
Faithful 32x AppleSkin Addon	AppleSkin HUD icons (hunger, saturation) at 32x in Faithful style — syncs the food bar to the 32x base
RAY's 3D Rails	3D block & item textures for all rail types (incl. 3D ladders)
Better Lanterns	Enhanced 3D look for lanterns and chains
(Bee's) Fancy Crops	Reworked crop models with texture variations — compatible with Farmer's Delight

Pack	Role
Rekindled CTM	Connected textures for Fire/Soul Fire + moss carpet overhangs — official Fire Rekindled addon (needs Continuity)
Fire Rekindled	Animated fire, lava, water, and portal textures — includes Create mod support
Motsch's Better Leaves	Bushier, denser leaves with explicit mod support — pair with Cull Leaves for FPS
Bushy Pink Petals, Wildflowers & Leaf Litter	3D models for pink petals, wildflowers, and leaf litter on the ground
Modded Omelet — Modern Spawn Eggs	1.21.5-style spawn eggs backported + restyled for 204 mods — matches pack mob lineup
Mace Fusion	Cleaner 3D model + 32x texture for the vanilla mace
Fresh Waystones Texture	Improved textures for the Waystones mod block + item
Overlay's	Dynamic CTM overlays — dirt moisture, cracks, smooth block transitions (requires Continuity + Connector)
MissingSoundsFix	Suppresses vanilla empty-sound warnings (salmon.ambient, cod.ambient, etc.) — fixes log spam from MC-97521
Faithful 32x	Full 32x base pack — complete coverage, cohesive palette; load at bottom of pack order

- **Improved Create 32x** — 32x32 textures for Create (items, blocks, GUI); partly based on Faithful 32x assets — designed to pair with it. Coverage still incomplete; author requires top placement.
- **Improved Applied Energistics 2** — 32x32 textures for AE2 (items, blocks, GUI); same author + Faithful pairing as Improved Create 32x. Coverage still incomplete; author requires top placement.
- **Armored Legacy** — vanilla armor remodel with working armor trims (all styles/materials) and interchangeable helmet variants. Requires the [EMF](#) + [ETF](#) mods (Wave 0.5). Place above other packs.

- **Mod Descriptions** — Field Guide companion (resource pack, lang-only): adds lore descriptions for modded entries in the catalog — 200+ mods, incl. full coverage for Farmer's Delight, Brewin' And Chewin', Supplementaries, Spice of Life: Carrot Edition, Explorer's Compass, Waystones and partial for Cataclysm, Simply Swords, Upgrade Aquatic. Requires the Item Descriptions mod (already in pack as a required Field Guide companion). Load position irrelevant.
- **Fresh Animations** — idle animations, directional looking, sleep/blink, emotions for all vanilla mobs. Resource pack overlay (not a mod) — place high in pack order.
- **Enhanced Boss Bars** — themed boss bar designs for vanilla + Twilight Forest. For Cataclysm/Aether bars (custom rendering), also install the companion [Enhanced Boss Bars mod](#).
- **Faithful 32x AppleSkin Addon** — AppleSkin's hunger/saturation HUD icons redrawn at 32x to match Faithful's style. AppleSkin ships 16x icons from the mod, so this keeps the food bar crisp on the 32x base. Zero deps; order-independent (only touches AppleSkin's own icons) — grouped with the UI tier.
- **RAY's 3D Rails** — 3D rail models; shader support since v3.0. Compatible with Steam 'n' Rails and Create rails. Also covers 3D ladders — RAY's 3D Ladders is redundant.
- **Better Lanterns** — 3D lanterns + connected chain models. Compatible with Arcane Lanterns and Thin Air.
- **(Bee's) Fancy Crops** — vanilla crop rework with texture variations; built to work with Farmer's Delight (Wave 4).
- **Rekindled CTM** — official Fire Rekindled addon: connected textures for fire and soul fire (plus moss-carpet overhangs). Requires Continuity — already in the pack (Wave 0.5, for Overlay's), so zero new mods. **Load above Fire Rekindled**. Pairs well with Incendium's Nether (connected soul fire across the transformed biomes). If fire textures look desynced from their neighbors, disable Sodium's **"Animate Only Visible Textures"**. Slime/honey connected textures live in a separate pack (Viscous CTM) — not added.
- **Fire Rekindled** — animated fire, lava, water, portal textures with Normal/Specular maps for PBR shaders; supports Create (honey, steam particles). CTM addon: Rekindled CTM (previous bullet).
- **Motsch's Better Leaves** — extra layers on leaf blocks for denser, bushier foliage; explicit mod support (Aether, Deep Aether, Twilight Forest, Deeper and Darker, Upgrade Aquatic). Pair with [Cull Leaves](#) (Wave 0.5) for the FPS win. v9 ships static overlays — recompile the custom version if the 16x-on-32x leaf mismatch bothers you.
- **Bushy Pink Petals, Wildflowers & Leaf Litter** — 3D models for ground-cover flora; denser, more natural look.
- **Modded Omelet — Modern Spawn Eggs** — backports the 1.21.5-style spawn eggs to 1.21.1 and restyles modded eggs to match; 204 mods supported, incl. ~9 in this pack (Aether, Twilight Forest, Cataclysm, Hybrid Aquatic, Upgrade

Aquatic, Darker Depths, MineColonies, Northstar Redux, Guard Villagers). Zero dependencies.

- **Mace Fusion** — refined 3D model + 32x texture for the vanilla mace; subtle, vanilla-like, no conflicts.
- **Fresh Waystones Texture** — improved textures for the Waystones block + item (Waystones is Wave 2); no conflicts, no dependencies.
- **Overlay's** — dynamic OptiFine-format CTM overlays: dirt moisture, cracks, smooth block transitions. Requires [Continuity](#) + [Sinytra Connector](#) (Forgified Fabric API already in pack). Fills the CTM gap Patrix couldn't deliver under Iris.
- **MissingSoundsFix** — silences vanilla empty-sound log spam ([salmon.ambient](#), etc. — MC-97521) with dummy sound mappings; no audio changed.
- **Faithful 32x** — full 32x base pack with complete coverage (blocks, items, mobs, GUI, fonts). Replaces Patrix 32x: no PBR normal/specular maps, but lighter, complete, and cohesive. Load at the **bottom** of pack order as the base layer. BetterGrassify samples the active pack at runtime — connected grass works at 32x automatically.

Installation: Install via XMCL — drop [.zip](#) files into the instance's Resource Packs tab, or copy them into the instance's [resourcepacks/](#) folder (access via instance settings).

Resource Pack Load Order

Top = highest priority. Packs only win where they have content, so packs touching different assets (armor vs. blocks vs. items vs. language files) are order-independent; the constraints that matter are called out per item. The 32x base pack sits at the **bottom** so every overlay above it wins.

1. Improved Create 32x — author requires top placement; only overlaps the base pack
2. Improved Applied Energistics 2 — author requires top placement; only overlaps the base pack
3. Armored Legacy — author requires top placement; needs EMF + ETF mods
4. Mod Descriptions — Field Guide companion, lang-only; any position works
5. Fresh Animations
6. Enhanced Boss Bars
7. Faithful 32x AppleSkin Addon (AppleSkin HUD icons — 32x Faithful style)
8. RAY's 3D Rails (3D rails + ladders)
9. Better Lanterns
10. (Bee's) Fancy Crops
11. Rekindled CTM (Fire Rekindled addon — load above it)
12. Fire Rekindled
13. Motsch's Better Leaves (pair with Cull Leaves mod)
14. Bushy Pink Petals, Wildflowers & Leaf Litter
15. Modded Omelet (spawn egg items)

16. Mace Fusion (mace item model)
17. Fresh Waystones Texture (Waystones block + item)
18. Overlay's — needs Continuity + Sinytra Connector mods
19. MissingSoundsFix — no visual impact, order irrelevant
20. Faithful 32x — base layer (overlays above win)

Infrastructure

Mod	Role
Chunk Loaders	Keep chunks loaded across dimensions — essential for Phase 2+ automation
SuperMartijn642's Config Lib	Config library (Chunk Loaders dependency)
SuperMartijn642's Core Lib	Core library (Chunk Loaders dependency)

Install

Use **XMCL** — create a 1.21.1 NeoForge instance, install all Wave 0 mods via the Mods tab. See [Wave 0](#) for the full list.

Wave 0.5 (Player Experience) can be installed at the same time — they're independent mods that don't conflict.

First Launch Checklist

1. Video Settings

Setting	Recommended	Notes
Graphics	Fancy	Sodium makes this fast; Fancy (not Fabulous) avoids transparency conflicts with Iris shaders
Render Distance	10 chunks	Distant Horizons handles the rest
Simulation Distance	8 chunks	Fewer distant mobs ticking = better FPS
VSync	OFF	
Max Framerate	Uncapped	

2. Distant Horizons (LOD Mod)

Open **Video Settings** → **[colored tiles icon]** next to the FOV slider.

Setting	Value
Enable Rendering	ON
Enable Distant Generation	ON
LOD Render Distance Radius	128–256
Quality Preset	Medium

Setting	Value
CPU Load	Aggressive
Enable Cloud Rendering	OFF
	Show LOD Gen Progress

Tip: DH generates LOD data as you explore, in new chunks only. First visit to an area has temporary pop-in. Terralith + Tectonic terrain takes longer than vanilla. Monitor VRAM — shaders + DH at 4K can push past 12GB.

3. Shader Setup

Drop **Bliss Shaders** .zip into [shaderpacks/](#). In-game: Options → Video Settings → Shaderpacks → select Bliss. Works with Iris on NeoForge. Use **v2.1.0+** for full Distant Horizons support.

Bliss needs no add-on mods — it fully replaces the previous Complementary Unbound + Euphoria Patches setup.

Recommended baseline (RTX 4080 SUPER, 16GB VRAM): High preset, tune from there — see the [Bliss Shader Settings](#) table in [Shaderpack](#) above (incl. the **Indirect Lighting = SSAO** rule). General MC: Graphics Fancy, Render Distance 10, Simulation 8, VSync OFF.

Infrastructure

Chunk Loaders

Keeps chunks loaded when you're far away or in another dimension.

Type	Coverage	Use
Single Chunk Loader	1 chunk	One machine or farm
Spotloader	Handheld, temporary	Explore nearby
Multi Chunk Loader	3×3 chunks	Factory complex or colony

How: Craft (check EMI — iron + redstone + ender pearl), place, right-click to toggle. Colored border shows loaded area.

Placement priority (late game):

1. Main base (ME drive, Mechanism induction matrix, Create power core) — Multi Chunk
2. MineColonies colony — Single or Multi
3. Mining outposts (Digital Miner) — Single

4. Other dimensions — Single per outpost

No performance concern with a dozen loaders on a modern CPU.

Wave 0.5 — Player Experience

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great.

Visual & Client QoL

Mod	Role
LambDynamicLights	Dynamic lighting
BetterF3	Customizable debug HUD
Sound Physics Remastered	Realistic sound
AmbientSounds	Nature ambiance (birds, wind, water)
The Immersive Music Mod	147 new biome/structure-based songs, vanilla silence gaps preserved
Modern UI	Full UI framework — modernized screens, own text engine, fancy tooltips, Gaussian blur
Immersive Hotbar	Animated hotbar slot transitions on pickup/use/swap, XP bar + level-up effects, tooltip animations
Dark Mode Everywhere	Shader-based dark mode for all GUIs — 3 darkness levels, toggleable in-game
Status Effect Bars Reforged	Duration bars on status effect icons in HUD and inventory
Traveler's Titles	RPG-style titles entering biomes/dimensions
Better Modlist	Enhanced mod list screen with badges/categories
Better Third Person	Independent 360° camera rotation in third-person view
Countered's Smooth F5	Smooth third-person camera transitions on F5 — cinematic, configurable
3D Skin Layers	Renders player skin layers (jackets, hats) with 3D depth
Presence Footsteps	Block-aware footstep sounds for every surface type
Particular Reforged	Ambient particles: 3D water splashes, fireflies, falling leaves, cave dust, waterfall cascades

Mod	Role
Particle Rain	Replaces default weather with particle effects: angled rain, desert sandstorms, fog, wind
Swinging Lanterns	Subtle swinging movement on lanterns and attached chains — makes builds feel alive
Wakes Reforged	Realistic water wakes and splashes when traveling on water
Sounds	170+ new sound effects for UIs, items, blocks, actions
Better Days	Customizable day/night cycle length, enhanced sleep
Better Safe Bed	Prevents mobs that can't reach you from blocking sleep — chill-rule essential
Beautiful Enchanted Books [Mod Edition]	Unique textures per enchantment book type — standalone mod, no resource pack needed
Overflowing Bars	Expanded health, armor, and toughness bars beyond vanilla limits
Leave My Bars Alone	Keeps food/health/armor bars visible while riding mounts — requires Puzzles Lib
Cherished Worlds	Favorite/pin worlds to top of list, prevent accidental deletion
Eating Animation	First-person eating/drinking animation — food moves to mouth and shrinks
Camera Overhaul	Camera tilting, screen shakes, mouselook roll, idle sway — satisfying movement feedback
Real Camera	Realistic first-person camera — bind view to head/body part, FPV model rendering, configurable
Dynamic Crosshair	Context-aware crosshair — hides/shows/changes style based on what you're looking at or holding

Mod	Role
[DLS] Dark Loading Screen	Replaces the default bright-red loading screen with a dark version — easier on the eyes
Cosy Critters & Creepy Crawlies	Ambient atmospheric animals — birds, bugs, and critters that bring the world to life
Foxified Dense Flowers	Renders multiple flowers per block in flower fields for denser, more natural-looking flora
Effectual	Atmospheric effects and decorative particles — steam breath, footprints, cave dust, sparks
BetterGrassify	OptiFine Fancy + Fast better grass — connected grass/podzol/path/nylium sides, better snow. Requires Forgified Fabric API
Loot Beams: Refork neoforge-1.21.1-3.4.7	Colored rarity-tinted beams on item drops — makes Apotheosis rare gear visibly pop. Client-side
Auto HUD 8.11+1.21.1-neoforge	Auto-hides idle HUD elements (full health bar, status effects) for a cleaner screen; toggle the whole HUD with a keybind
[EMF] Entity Model Features	OptiFine-format custom entity models — powers Armored Legacy's armor models; works with Fresh Animations; requires ETF
[ETF] Entity Texture Features	Emissive/random/custom entity textures — required by EMF
Continuity	CTM connected textures (OptiFine format) — enables Overlay's; runs on NeoForge via Connector + Forgified Fabric API
Cull Leaves	Culls leaf interior faces — big FPS win in forests; recommended companion for Motsch's Better Leaves

Inventory & UI

Mod	Role
EMI	Recipe viewer with recipe trees
Jade	Block info HUD
Jade Addons (Neo/Forge)	Extra Jade info panels
JadeColonies	Jade tooltips for MineColonies — worker stats, hut info, DO blocks
JourneyMap	Real-time minimap + full-screen world map, waypoints, entity radar
JourneyMap Integration	Shows Waystones as icons on JourneyMap
AppleSkin	Food hunger/saturation overlay
Immersive Armor HUD	Armor type icons + durability above armor bar — shows what's about to break
Inventory Essentials	Drag-to-transfer, single-item moves, sorting
ClientSort	Sort (4 modes), Fill Stacks, Transfer Matching, customizable button positions — requires Cloth Config API
Advancement Plaques	Glowing plaque advancement popups with copper/silver/gold borders
Shulker Box Tooltip	Preview shulker box contents on hover
Item Borders	Colored borders on items based on rarity
Loot Journal	Animated item pickup notifications
Day Counter	Day counter HUD — displays at dawn via ActionBar, fades after. Milestone rewards at days 100/250/500/1000/2000. Calendar block available but disabled.
Hud Texts	HUD text framework — dependency for Day Counter (from V3.4)

Day Counter config (config/day_counter.toml — tracked override):

Setting	Value	Why
[UI."New Day Text"] Display	actionbar	New-day message appears and auto-fades after ~3s
[Sound] Enabled	false	No sound notification — just the text
[Rewards.1]–[Rewards.5]	set	Day-milestone rewards — see table below

Setting	Value	Why
[UI.Scoreboard] / [UI.Playerlist] / [UI.Name] Enabled	false	No persistent HUD or nametag element — only the flash
[Date] Enabled	false	No calendar date — just day count
[Block] / [Item] Enabled	false	No calendar block or item
[UI."Disabled Feature Text"] Enabled	false	No “feature is disabled” nag text

This gives you exactly: a brief “Day 42” (or similar) on the ActionBar at dawn that appears and fades, plus a small gift on milestone days. No HUD element, no sounds, no calendar block. Days count in the background.

Milestone rewards (tracked via the server day count; each fires once when the day is reached and posts a chat message):

Day	EXP	Item	Count
100	100	Diamond	8
250	250	Diamond block	2
500	500	Diamond block	5
1000	1000	Netherite ingot	4
2000	2000	Netherite block	2

Tweak payloads directly in `config/day_counter.toml` ([Rewards.1]–[Rewards.5]: Day / Message / Command / EXP / Item / "Item Count"). All items are vanilla — no pack gating bypassed.

| [Better Advancements](#) | Overhauled advancements UI with editor/pan/zoom | | [Polymorph+](#) | Choose crafting result when recipes conflict | | [Traveler Tool Belt](#) | Tool radial menu — hold R for quick-swap, 2→9 slots via upgrades | | [EMIeffect](#) | Status effects in EMI browser — descriptions, sources, colors for all modded effects | | [Searchables](#) 1.0.2 | Search bars in creative menu, JEI/EMI-style browsers, and other GUIs — find items by name instantly |

Storage & Travel

Mod	Role
Sophisticated Storage	Upgradable barrels/chests
Sophisticated Backpacks	Portable inventory
Waystones	Fast travel between waypoints
GraveStone Mod	Keep inventory on death

Mod	Role
Simple Recall Potion	Drink to teleport back to your spawn point — simple, stackable

Utility & QoL

Mod	Role
Easy Anvils	Overhauled anvils — persistent items, reduced prior work penalties, free renaming, repair with blocks

Easy Anvils requires **Puzzles Lib** — already in pack (also used by Leave My Bars Alone and Hang Glider). No additional mods needed.

Village & Trading

Mod	Role
Dynamic Villager Trades Neoforge 1.0.0	Dynamically changing villager trades — villagers specialize in what you buy from them, reducing the need for massive trading halls
Improved Village Placement	Forces villages to spawn on flat terrain — prevents cliff-side and underwater villages

Configuration — Village & Trading

Dynamic Villager Trades ([config/dynamicvillagertraders.properties](#)):

- `global_randomness = 0.7` (recommended, default 1.0) — Lower values make villagers settle on specific trades faster; higher values increase time to get desired trades. Multiplied by group randomness in the datapack.
- `refresh_delay = 1` (recommended, default 0) — Number of restocks between trade changes. 1 means trades change every other restock, giving you time to use existing trades before they rotate. Trades always change on level-up regardless.
- `no_book_duplicates = true` — Prevents librarians from selling the same enchant at different levels simultaneously.
- `enchant_repair_compat = false` — Adds Curse of Vanishing to every book sold by librarians (for Enchant & Repair compatibility). Keep disabled unless that mod is added.

Dependencies — Wave 0.5

Mod	Role	Required By
Architecture API	Cross-loader compatibility — lets Fabric mods run on NeoForge	Eating Animation, Effectual, multiple Wave 1-3+ mods
Iceberg	Library for config-driven UI mods	Advancement Plaques
Polytone	Custom block colors, lightmaps, biome colors — Optifine format support for resource packs	— (dependency orphaned — Rainbow's Foliage removed)
Lithostitched	Worldgen modifier API — handles village placement hooks	Improved Village Placement
TLib (Take's Lib)	Library for atmospheric effects	Effectual
Forgified Fabric API	Fabric API implemented on NeoForge (Sinytra) — lets Fabric-origin mods run on NeoForge	BetterGrassify
Nirvana Library 2.2.0	Library for config + networking abstraction	Loot Beams: Refork
Common Network 1.0.21-1.21.1	Networking library (Nirvana Library dependency)	Nirvana Library
Sinytra Connector	Lets Fabric mods run on NeoForge — translation/compat layer	Continuity
Resourceful Lib	Cross-loader utilities/API library (TeamResourceful)	The Bumblezone

Lithostitched is a worldgen library that Improved Village Placement uses to hook into the village generation system. Without it, IVP crashes at startup with a `NoClassDefFoundError` for `AddWorldgenModifiersEvent`. This mod was not declared as a dependency in IVP's metadata — install it manually.

Architecture API is a lightweight library with no user-visible features — install once and forget. It provides the abstraction layer that lets Fabric-origin client QoL

mods (Eating Animation) run seamlessly on NeoForge. Without it, these mods crash at startup with a missing-dependency error.

Install: In XMCL, search [Architecture API](#) in the Mods tab and add it. Alternatively, download from CurseForge and drop the `.jar` into `mods/`. No config needed. The game loads it silently — you’ll never see it in-game, but EMI’s mod list confirms it’s present.

Configuration — Wave 0.5

Waystones (`config/waystones-common.toml`):

- `inventoryButton = "any"` — Opens waystone menu from inventory without a keybind.
- `enableCosts = false` — Free teleportation.

Loot Journal (`config/obscuria/loot_journal-client.toml`):

- `trackItemPickups = true` — Logs all item pickups for per-session statistics.
- `trackXpPickups = true` — Logs all XP pickups for per-session statistics.

Better Days (`config/betterdays-common.toml`):

Better Days is the *engine* that lengthens the day/night cycle, but it’s installed at **default** and isn’t where you set the length. Its real modes are [RATIO](#), [MINUTES](#), and [REALTIME](#) — there is no [SEASON](#) mode. **Serene Seasons Plus** (Wave 2) drives Better Days at runtime through its `[seasonalDaylightCycle]` config (see *Serene Seasons Plus* in Wave 2) and overwrites `daySpeed/nightSpeed` automatically — leave Better Days’ own config untouched.

Better Days’ “time effects” all default to [NEVER](#) (except `weatherEffect = "SLEEPING"`), which is exactly the pack’s intent — no crop or block-entity desync with Serene Seasons.

Dynamic Crosshair (configurable in-game via Mod List → Dynamic Crosshair → Config, or `config/dynamiccrosshair.json`):

Dynamic Crosshair uses **YACL** (already in the pack) for its in-game config screen. Recommended settings:

Setting	Value	Effect
<code>hideWhenNotLookingAtAnything</code>	<code>true</code>	Hides crosshair when pointing at sky/air — clean UI during building/exploration
<code>showWhenLookingAtBlock</code>	<code>true</code>	Reappears when looking at a block — signals interactability

Setting	Value	Effect
<code>showWhenLookingAtEntity</code>	<code>true</code>	Reappears when looking at mobs/players — signals combat target
<code>showWhenHoldingTool</code>	<code>true</code>	Shows crosshair when mining with pickaxe/axe/shovel — shows exact breaking target
<code>showWhenHoldingWeapon</code>	<code>true</code>	Shows crosshair in combat with any weapon
<code>crosshairStyle</code>	<code>"CONTEXTUAL"</code>	Changes crosshair shape per context (default dot, block outline for mining, reticle for combat)
<code>crosshairColor</code>	<code>"ENTITY_BASED"</code>	White for neutral/block, red for hostile entities, green for interactable entities

No keybind changes needed — the mod works fully through context detection. YACL provides a searchable settings screen.

Dark Mode Everywhere (`config/darkmodeeverywhere-client.toml`):

- `darkModeLevel = 1` — Darkness level: 0 (light), 1 (medium, default), 2 (full dark). Change in-game via the button in the bottom-left corner of any GUI.
- `methodShaderDump = false` — Set to `true` temporarily if a GUI looks broken, to log which methods the shader is targeting.

Known issue — double text: Some mods render text shadows that get doubled by the dark shader. Fix: edit `darkmodeeverywhereshaders.json` (create in `config/` or via a resource pack) and change `darkColorRecplacement` from `16777215` to `4868682` (all 3 occurrences). Only needed if you see doubled text on specific screens.

Modern UI + ImmediatelyFast (`config/immediatelyfast.json` — tracked override):

Modern UI requires three ImmediatelyFast mixins disabled on 1.21.1 (Modern UI compat notes — otherwise random game crashes). The tracked `config/immediatelyfast.json` ships with these flipped; ImmediatelyFast's other options stay at defaults:

Setting	Value	Why
<code>hud_batching</code>	<code>false</code>	Modern UI requirement — HUD text batching conflicts with its tooltip/text rendering

Setting	Value	Why
font_atlas_resizing	false	Modern UI requirement — font atlas resize conflicts with its text engine
fast_text_lookup	false	Modern UI requirement — fast glyph lookup conflicts with its text engine

Known conflict to avoid: Legendary Tooltips (ver 1.16 ~ 1.21.4) fights Modern UI's tooltip renderer — if it's ever added, disable Modern UI's tooltip via the config screen (Action Center) instead. The pack's tooltip rendering is Modern UI's own; no tooltip-overhaul mod needed.

Modern UI itself runs on defaults. Tune it in-game via **Action Center** (**Ctrl+K**): fonts (Preferences), disable its zoom (Sodium owns **C**), background blur/colors. Its configs write to `config/ModernUI/`.

Visual & Client QoL — What to Expect

These mods change how the game looks and feels. All are client-side and work immediately.

- **LambDynamicLights** — held light sources emit light around you; off-hand works. Pro tip: torch in off-hand while mining. Toggle **Ctrl+L**.
- **BetterF3** — cleaner debug HUD; toggle FPS, coords, biome, light level. Customize in the Mods screen → BetterF3 → Config.
- **Sound Physics Remastered** — sound muffled by walls, echoes in caves, directional. Works automatically.
- **AmbientSounds** — biome-aware ambience (birds, wind, water). No config.
- **The Immersive Music Mod** — 147 biome/structure-aware songs with natural silence gaps.
- **Immersive Hotbar** — animated slot contents on pickup/use/swap, animated XP bar. Configurable via YACL.
- **Modern UI** — full UI framework: modernized screens, own text engine (fonts via Action Center **Ctrl+K** → Preferences), fancy rounded tooltips, Gaussian blur + fade-in backgrounds, borderless fullscreen, GUI-scale slider. Ships an OptiFine-style **C** zoom — disable it in the Action Center since Sodium owns **C**. Requires the ImmediatelyFast config override below.
- **Status Effect Bars Reforged** — duration bars on effect icons; auto-hides beacon effects. Config via Cloth Config.
- **Traveler's Titles** — RPG-style title cards entering biomes/dimensions. No config.
- **Better Modlist** — mod list with badges (library/core/addon), categories, search.
- **Better Third Person** — independent camera in third-person; bind a toggle key in Controls.

- **3D Skin Layers** — renders player skin layers (jackets, hats) with 3D depth.
- **Presence Footsteps** — block-aware footstep sounds; works with Sound Physics.
- **Particular Reforged** — ambient particles (splashes, fireflies, leaves, cave dust, cascades). Toggle per-effect in [config/particular.json](#).
- **Particle Rain** — particle-based weather (angled rain, sandstorms, fog). / [particlerain](#) or Mod List → Config.
- **Swinging Lanterns** — physics-based sway on lanterns and chains. No config.
- **Wakes Reforged** — water wakes and splashes; works with Small Ships and boats.
- **Sounds** — 170+ UI/item/block sound effects. Customize in Mod List → Sounds or [config/sounds/](#).
- **Better Safe Bed** — unreachable mobs no longer block sleep. Chill-rule essential.
- **Beautiful Enchanted Books [Mod Edition]** — unique texture per enchantment type. Standalone, no resource pack needed.
- **Overflowing Bars** — health/armor/toughness bars expand past vanilla limits — needed once Apotheosis bonuses stack.
- **Leave My Bars Alone** — keeps your bars visible while riding mounts. Requires Puzzles Lib (already in pack).
- **Cherished Worlds** — pin worlds to top, prevent accidental deletion. Right-click in world list.
- **Day Counter** — “Day N” ActionBar flash at dawn + milestone gifts (days 100–2000). Config: [config/day_counter.toml](#).
- **Eating Animation** — first-person eating/drinking animation; works with Farmer’s Delight foods.
- **Camera Overhaul** — camera tilt, screen shake, mouselook roll, idle sway. Config: [config/cameraoverhaul-client.toml](#).
- **Real Camera** — first-person body view bound to your head; **F6** toggles. Version pin **0.7.8-beta**. Compatible with Better Combat, Camera Overhaul; GeckoLib-rendered armor incompatible (none in pack). [VERIFY] Protection Pixel armor with model binding after test launch.
- **[DLS] Dark Loading Screen** — dark loading screen. No config.
- **Cosy Critters & Creepy Crawlies** — ambient birds/bugs/critters; client-side, toggleable via [/cosycritters](#). Zero perf impact.
- **Foxified Dense Flowers** — denser flower fields; client-side.
- **Effectual** — atmospheric particles (steam breath, bubbles, footprints, cave dust, sparks). Disable overlap with Particular Reforged in [config/effectual-client.toml](#). Requires Architectury API, Cloth Config, TLib (all in pack).
- **BetterGrassify** — OptiFine-style connected grass: grass blocks, snowy grass, podzol, mycelium, dirt paths, farmland, and both nylum blend their side textures into surrounding terrain (Fancy, default) or use the top texture (Fast). Adds Better Snow + Better Snowy Grass for snowy biomes. Ships no textures —

samples your active texture pack's sprites at runtime, so it works at any resolution; resource-pack compatibility mode is on by default. Set mode in config — YACL GUI (already in pack) or `config/bettergrass.json`. Requires Forged Fabric API.

Inventory & UI — Your Daily Tools

- **EMI** — your primary learning tool: hover any item, `R` for recipe, `U` for uses. Recipe trees break complex crafts (Create sequenced assembly, Mekanism ore chains) into every step.
- **Jade + Jade Addons** — block name, mod source, machine status, crop growth, chest contents; Addons adds Create (SU/RPM), Mekanism (energy/gas), AE2 (channel) panels. Keep enabled. Conflict fix: Jade → Config → Mod Settings → Display Bosses → OFF (avoids Enhanced Boss Bars conflict).
- **JourneyMap + Integration** — minimap with entity dots, fullscreen map (`J`), waypoints (`B` to set), entity radar. Waystones appear as icons via the Integration mod. `J` → right-click sets a temporary waypoint.
- **AppleSkin** — hover food for exact hunger/saturation; dashed outlines preview a meal. No config.
- **Immersive Armor HUD** — armor icons + durability above the bar; works with modded armor. No config.
- **Inventory Essentials** — drag-to-transfer and single-item moves: `Ctrl+click` moves one item, `Shift+Ctrl+click` moves all of a type. Sorting is superseded by ClientSort (below).
- **ClientSort** — sorting with 4 modes (creative order, quantity, alphabetical, item ID) plus Fill Stacks, Transfer Matching, and a Button Editor. No overlap with Inventory Essentials.
- **Shulker Box Tooltip** — hover a shulker for a contents preview. No config.
- **Item Borders** — rarity-colored borders (gray → gold) to spot drops instantly.
- **Loot Journal** — animated pickup notifications; session summary on disconnect. Config: `trackItemPickups`, `trackXpPickups`, `showSummaryOnDisconnect`.
- **Better Advancements** — overhauled advancements UI, open with `L`. Pairs with Advancement Plaques.
- **Advancement Plaques** — glowing plaques (copper/silver/gold per tier). Requires Iceberg (auto-installed).
- **Polymorph+** — when recipes conflict, click the button next to the crafting output to cycle. Persists per ingredient combination.
- **Traveler Tool Belt** — hold `R` for a radial quick-swap menu; 2 slots, expandable to 9. Integrates with the Curios belt slot.

- **EMIffect** — browse any status effect in EMI (description, source mod, colors) — a built-in wiki for Apotheosis/Relics/Runes effects. Complements Status Effect Bars Reforged.

Storage & Travel

Sophisticated Storage + Backpacks

- **Barrel** — craft: 1 chest + 8 planks. Upgrade with iron/gold/diamond/netherite upgrades for more slots, pickup, magnet, filter, composting
- **Backpack** — equip in chest armor slot, press **B** to open. Upgradable like barrels
- **Create Integration** (auto-installed) — Sophisticated barrels and backpacks work fully on Create contraptions: upgrades, automation, inventory settings all function while moving

Waystones

Craft a waystone → right-click to claim. Use a Warp Stone (ender pearl + amethyst) to teleport from anywhere. **No XP costs** (`enableCosts = false` in config). Set keybind to `Ctrl+Shift+W`.

GraveStone Mod

Die → grave spawns with all your items. Right-click the grave to retrieve. No item loss, no despawning, 24-hour protection.

Simple Recall Potion

Drink to teleport straight back to your **spawn point** — the classic Terraria-style escape potion, no save points to manage. Works within your current dimension by default; enable the mod's interdimensional config option if you want it to work across the Aether, Twilight Forest, etc. Zero dependencies.

Village & Trading

Dynamic Villager Trades

Villager trades change dynamically — villagers specialize in what you buy from them. No more massive trading halls.

Config (`config/dynamicvillagertraders.properties`):

- `global_randomness = 0.7` — faster specialization
- `refresh_delay = 1` — trades change every other restock
- `no_book_duplicates = true` — no duplicate enchantment books

Improved Village Placement

Forces villages to spawn on flat terrain. No more cliff-side or underwater villages.
No config.

Session Plan — Wave 0.5

Session	Goal
Session 1 — Setup	Launch, set keybinds, configure Distant Horizons (LOD 128-256, Medium quality). Install Bliss shader. Enable Fresh Animations texture pack. Walk around spawn to generate LOD data
Session 2 — Survival basics	Punch trees, build a starter shack, find food, get iron. Place your first Waystone. Craft a Sophisticated Storage barrel. Craft a backpack
Session 3 — Comfort	Explore nearby terrain. Find a village. Set up a bed. Light up your base perimeter. Start noticing Traveler's Titles when entering new biomes

Wave 1 — Tech (Create + Mekanism + AE2)

The classic tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting.

Core Tech

Mod	Role
Create	Mechanical automation — rotational power, water wheels, windmills, gear trains, conveyor belts, deployers, sequenced crafters, trains
Mekanism	Industrial processing — ore multiplication (2x→5x), digital miner, superdense energy storage, MekaSuit (flight, invulnerability)
Mekanism Generators	Power generation — heat, wind, solar, gas-burning, bio generators, fission reactor, fusion reactor, turbine
Mekanism: Ponders 1.0.3	Ponder scenes for Mekanism machines and multiblocks (fission/fusion reactors, turbine). Requires Mekanism
Mekanism Guide Book	Patchouli-based in-game guidebook for Mekanism. Requires Patchouli (already in pack via Apotheosis)
AE2 19.2.17	Digital storage — ME drives, autocrafting, P2P tunnels, spatial storage. The storage backbone
Applied Mekanistics	AE2-Mekanism integration — autocrafting for chemical/gas/fission/fusion recipes, ME chemical storage, P2P chemical tunnels

Note: Mekanism v10 for 1.21.1 NeoForge is split into separate modules.

Mekanism (core) provides machines, processing, cables, and armor. **Mekanism Generators** provides all power generation. Both must be installed.

Create Addons

Mod	Role
Create Crafts & Additions 1.6.0	Electric motors, alternators, rolling mill, wires/rods, silver/electrum — bridges kinetic SU and Forge Energy (FE)
Create: Power Grid 0.5.5.1	Physics-based electricity simulation — Ohm's law, AC/DC, generators, motors,

Mod	Role
	cables, FE conversion. Separates Create power from SU
Steam 'n' Rails NeoForge 0.2.1	Expanded train system — new tracks (spruce, monorail), semaphores, conductor mob, coupling/decoupling blocks
Create: The Factory Must Grow (TFMG)	Heavy engineering & oil — crude oil drilling, distillation, diesel/gasoline/LPG engines, steel, aluminum, electricity with voltage
Mekanism TFMG Compatibility	85+ recipe bridges — deduplicates lead/steel/sulfur, integrates aluminum/plastic into Mekanism chains
TFMG Energy Converter 26.03	Bidirectional TFMG electricity ↔ FE/AE2 bridge — lets TFMG diesel engines/generators power non-TFMG machines (Mekanism, AE2). Requires Kotlin for Forge
Create: Gunsmithing	Steampunk firearms integrated with Create — craftable using Create machines. Animated guns, attachments, ammo types. Optional Better Combat compatibility
Create Ornithopter Glider	Mechanical flapping glider with boost-assisted flight — integrates with Create rotational power
Sophisticated Storage Create Integration	Full-featured Sophisticated Storage on Create contraptions — upgrades, automation, inventory settings work while moving
Sophisticated Backpacks Create Integration	Full-featured Sophisticated Backpacks on Create contraptions — upgrades, automation, inventory settings work while moving
Create: Train Utilities 3.0.3	Sliding doors with realistic sounds, platform blocks, station building blocks

Mod	Role
Create Railways Navigator	Display boards, route navigation, new schedule entries — search routes and manage train networks
Create: Interiors 0.6.1	Colored chairs with armrest toggles, tray tables, kelp seats — functional furniture for train cars
Create Train Parts 0.4.1	Crossing gates, sliding windows, train steps, slides — decorative and functional blocks for trains and stations
Create: Threaded Trains	Performance — runs train pathfinding on a separate thread, prevents tick lag on complex rail networks
Create Better FPS	Create rendering optimization — up to 50% FPS gain with shaders. Handles addon models, speed controllers, factory gauges, and schematic previews
ColorWheel	Flywheel + Iris compatibility — renders Create contraptions under shaders. Required for Create to work with Iris (shader-agnostic)
Create Slice & Dice	Create + Farmer's Delight automation — Slicer (auto Cutting Board), automatic FD Cooking Pot via heated mixing, Sprinkler for fluid-based farming
Create: Dragons Plus 1.11.2b	Fan processing expansion — Bulk Coloring, Bulk Freezing, Bulk Ending, Bulk Sanding. Fluid Hatch for tanks. Adds Aether Bulk Enchanting. Library mod for Create addon devs
Create: Protection Pixel	Create-themed power armor — brass/alloy tiers, per-piece active abilities (AoE, debuff clear, speed→damage), combustion reactor (blaze rods + water). Bridges gear between Mekanism Tools and MekaSuit
Create: Central Kitchen 2.6.0	Mechanical Arm integration for Farmer's Delight blocks — automate

Mod	Role
	Cooking Pot, Cutting Board, Stove via Arms; extends to Brewin' And Chewin' kegs. Requires Create: Dragons Plus

Create Addon Details

- **Create Better FPS** — Create rendering optimization (addon models, speed controllers, gauges, schematic previews). Benchmarked 7→35 FPS with 512 waterwheels under shaders. Pairs with ColorWheel, which handles Flywheel-Iris compatibility.
- **ColorWheel** — required for Create contraptions to render under Iris shaders; install it before Create Better FPS.
- **Create Slice & Dice** — automates Farmer's Delight: Slicer (auto Cutting Board), FD Cooking Pot recipes as Create heated mixing, Sprinkler for fluid farming (water, lava, potions, Liquid Fertilizer). Meal production becomes fully pipeable in Phase 2.
- **Create: Dragons Plus** — fan-processing expansion (Bulk Coloring, Freezing, Ending, Sanding), Fluid Hatch for tank I/O, Aether Bulk Enchanting. Features toggleable in config; required by Create: Shimmer.
- **Create: Central Kitchen** — Mechanical Arms drive existing Farmer's Delight blocks directly (Cooking Pot, Cutting Board, Stove), plus Brewin' And Chewin' kegs and other Delight-ecosystem stations. Complements (not replaces) Create Slice & Dice: Slice & Dice converts recipes into new Create machines, Central Kitchen automates the FD blocks themselves. Requires Create: Dragons Plus (in pack).

Create: Protection Pixel

Steampunk power armor built with Create materials. Each armor piece has a unique active ability powered by a waist-slot Combustion Reactor (blaze rods + water as fuel).

Armor pieces (brass → alloy tier progression):

Piece	Ability	Best for
Plague Helmet	Release steam to dispel blindness, darkness, weakness, slowness	Exploration, debuff-heavy mobs
Lancer Helmet	Speed → damage scaling — the faster you move, the harder you hit	Mobility builds

Piece	Ability	Best for
Hunter Helmet	Sensors highlight nearby creatures while sneaking; +luck for finding treasure	Dungeon crawling
Breaker Chestplate	+damage +attack speed while maintaining protection	DPS builds
Magnetic Storm	AoE magnetic burst when taking heavy damage or surrounded; pulls ranged attackers	Crowd control
Float Shield	Energy shell around user when charged — absorbs incoming damage	Tank builds

How it works:

1. Craft armor pieces at Create machines (check EMI — requires brass-tier Create infrastructure)
2. Craft a **Combustion Reactor** (waist slot, check EMI) — this powers all equipped armor
3. Feed the reactor **blaze rods + water** (displayed in minutes in the GUI). More blaze rods = longer runtime
4. If reactor runs out: slowness + mining fatigue + weakness. Keep it fueled!

Armor plates: Stand on an **Armor Load Platform** (requires lava piped in — 100mB per assembly/disassembly). Right-click while sneaking to install/remove armor plates with different stat values. Plates also provide a repair method — no anvil needed.

External devices: Additional Create-themed accessories (back-mounted) that provide utility without consuming armor slots.

Compatibility: Works with Better Combat, Epic Fight, and backpacks (visual space reserved for backpack rendering). Apotheosis affixes and gems should apply — verify after test launch.

Dependencies: [Create](#) (already in pack) + [Kleiders Custom Renderer API](#). For versions below 1.9.0, requires [Curios API](#) instead (already in pack). Currently on v2.1.9+ using Kleiders.

Progression placement: Brass-tier armor is early Phase 2 (requires Create brass). Alloy-tier is mid-Phase 2 (requires Create alloy infrastructure). MekaSuit remains the Phase 3 endgame upgrade. This fills the Phase 2 power-armor gap between basic Mekanism Tools and the MekaSuit.

Mekanism Addons

Mod	Role
Mekanism Tools	Armor, shields, tools, and paxels for Mekanism materials — Osmium, Bronze, Refined Obsidian, Refined Glowstone, Steel. Fills gear gap between netherite and MekaSuit.
Advanced Finders	Ore detection devices — directional indicators for nearby ore veins, synchronized multiplayer support
Advanced Chimneys	Smoke ventilation systems — redirect factory emissions outdoors, customizable appearance and behavior

Dependencies: Both Advanced Finders and Advanced Chimneys require [ForgeEndertech](#) library.

Ores

Mod	Role
Create (built-in)	Zinc, brass, copper
Mekanism (built-in)	Osmium, tin, lead, uranium, fluorite

Configuration — Wave 1

Mekanism ([config/Mekanism/general.toml](#)):

- `allowChunkloading = false` — Disable Mekanism’s built-in chunk loading. Prevents conflict with the dedicated Chunk Loaders mod and keeps chunk management unified.

Ore generation: All default rates. Osmium at iron-like frequency, tin/lead moderate, uranium/fluorite rarer. Default distribution aligns with the pack’s progression curve.

Interdependency notes for Wave 1:

- Mekanism’s universal cables can power AE2 controllers via the Energy Acceptor. No adapter mod needed.
- **Create Crafts & Additions** electric motors and alternators create a direct kinetic↔FE bridge, replacing the Create Alternator workaround. Silver and electrum add new material tiers that integrate with Mekanism’s ore processing line.
- **Steam ‘n’ Rails** expands Create’s train system with new tracks, semaphores, conductor mobs, and coupling blocks. Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.
- Mekanism’s 5x ore processing (Chemical Dissolution Chamber → Washer → Crystallizer) outclasses Create’s Crushing Wheels (1.5x) and Millstone (1x) for ore multiplication. This is intentional — Create handles mechanical automation

and component crafting, Mekanism handles industrial processing. Neither obsoletes the other.

- AE2's Storage Bus on a Sophisticated Storage barrel makes early-mid game storage migration seamless. Place a Storage Bus on your barrel wall, configure it to show contents in the ME terminal, and transition gradually to full digital storage.
- **Sophisticated Storage Create Integration** and **Sophisticated Backpacks Create Integration** (both by P3pp3rF1y) enable full backpack and barrel functionality on Create contraptions — upgrades, automation features (pickup, magnet, feeding), and inventory settings all work while the contraption is in motion. Without these addons, Sophisticated inventories on contraptions are limited to basic item transport.
- **Applied Mekanistics** bridges AE2 autocrafting with Mekanism's chemical processing — ME Chemical Output Buses, ME Chemical Import Buses, and P2P tunnels for chemicals/gases enable autocrafting of Mekanism recipes (including fission/fusion products) directly through the AE2 crafting CPU. Without this mod, Mekanism chemical steps are manual bottlenecks.
- **Create: The Factory Must Grow** adds oil drilling, distillation towers, blast furnaces, and diesel engines — expanding Create's mechanical toolkit into heavy industry. Its steel replaces Mekanism's steel (TFMG's blast furnace is the intended steel source; Mekanism's steel recipe is disabled by the Mekanism TFMG Compat mod). TFMG aluminum and plastic feed into Mekanism's advanced processing chains.
- **Mekanism TFMG Compatibility** resolves 85+ recipe overlaps between TFMG and Mekanism — lead, sulfur, and steel are unified under TFMG's production lines, while Mekanism retains its chemical processing advantage. Both EMI item blacklist config and per-recipe toggles are available for pack tuning.
- **Create: Protection Pixel** adds steampunk power armor that bridges Create's brass-tier infrastructure with the power fantasy of Mekanism's MekaSuit. Brass-tier armor requires Create brass (early Phase 2); alloy-tier requires additional Create alloy processing (mid-Phase 2). The Combustion Reactor's blaze rod fuel can be farmed via Mekanism spawner systems, Apotheosis spawner upgrades, or vanilla blaze grinders. Blaze automation is a natural Phase 2 milestone that unlocks full power armor capability. MekaSuit remains the Phase 3 ultimate upgrade — Protection Pixel fills the gap, it doesn't compete.

Core Tech

Create — Mechanical Automation

Create is about **rotational power (SU)**. Machines consume SU to run. If consumption exceeds generation, machines stop.

Learning with Ponder: Hold **W** while hovering over any Create block/item in EMI. An animated tutorial plays showing exactly how it works. **Use Ponder constantly** — it's better than any written guide.

First Create setup:

1. Mine **andesite** — the most important early resource
2. Craft **andesite casing** (andesite + wood nugget in EMI)
3. Craft **cogwheel** and **large cogwheel** — transfer rotational power
4. Place a **hand crank** → attach cogwheel → attach **millstone** → right-click crank to grind wheat/ore manually
5. Build a **water wheel** in flowing water (2–3 block drop) → attach shaft → power 2–3 basic machines (256 SU)

Brass progression (mid-Create): Find zinc ore → mix zinc + copper in a **heated mixer** → brass ingots → brass casing → unlock:

- **Deployer** — places blocks/uses items from inventory. Automates crafting
- **Sequenced Crafter** — multi-step autocrafting sequence
- **Mechanical Arm** — moves items between inventories
- **Chute & Funnel** — vertical item transport with filtering
- **Train Station** — build tracks, assemble trains

Blueprint system: Craft **Blueprint and Quill**. Select an area, save as a blueprint. Place a **Deployer** with the blueprint to auto-build it. Or use the **Schematicannon** — load a blueprint, feed it materials from chests, and it builds the structure for you automatically. This is your “here’s a blueprint → build it” tool.

Keybinds:

- Ponder: **W** (hold over block)
- Wrench: right-click
- Goggles: **G** (hold over machine) — shows SU, RPM, stress info
- Toolbox: **T** (hold)

Goggles are essential — craft early, check EMI.

Mekanism — Industrial Processing

Mekanism v10 for 1.21.1 is split into two required modules:

- **Mekanism** (core) — machines, processing, cables, armor
- **Mekanism Generators** — all power generation (heat, wind, solar, gas-burning, bio, fission, fusion)

Finding Osmium: Generates at all underground levels, similar rarity to iron. Mine with stone pickaxe or better.

Tier system: Every machine has 4 tiers. Upgrade by shift-right-clicking with the tier upgrade item.

Tier	Color	Speed	Upgrade
Basic	Orange	1x	Starting
Advanced	Red	2x	Basic + osmium + redstone
Elite	Dark Red	4x	Advanced + diamond + obsidian
Ultimate	Purple	8x	Elite + refined obsidian + alloy

First Mekanism setup:

1. **Heat Generator** — place lava source below it for early power
2. **Metallurgic Infuser** — combine osmium + carbon (coal) → **steel ingots**
3. **Enrichment Chamber** — 1 ore → 2 dust → smelt → 2 ingots (2x processing)

Processing multipliers:

- **2x**: Enrichment Chamber only
- **3x**: Enrichment Chamber → Crusher → Enrichment Chamber again
- **5x**: Full chemical line — Chemical Dissolution Chamber (sulfuric acid) → Washer → Crystallizer → 5 dust

Gas infrastructure (for 5x processing):

- Hydrogen/Oxygen: Electrolytic Separator (split water)
- Sulfuric Acid: Chemical Infuser (sulfur + oxygen)
- Deuterium/Tritium: Fusion reactor fuel (late game)

Gases move through **pressurized tubes**. Use a **Rotary Condensentrator** to convert gas ↔ liquid.

Digital Miner (mid-game): Automated mining machine. Set radius (up to 64 blocks) and filters (whitelist ores). Needs significant power. Tip: set to mine only ores to avoid filling with stone.

Fusion Reactor (endgame): Millions of Joules per tick. Fuel chain: water → Electrolytic Separator → hydrogen → deuterium → tritium → feed both into reactor. Requires significant infrastructure — a Phase 3 project.

Config: `config/Mekanism/general.toml` → `allowChunkloading = false` (delegated to Chunk Loaders mod).

AE2 — Applied Energistics 2

Digital item storage and autocrafting. Everything lives in an electronic network accessed from a single terminal.

Finding Certus Quartz:

- **Meteorites** — circular surface depressions. Dig down to find certus quartz blocks + sky stone chest with free AE2 components
- **Ore veins** — certus quartz ore underground

Processing: certus quartz block → mine → item → craft into dust → smelt → crystal → charge in **AE2 Charger** → charged certus quartz

Channel system: AE2 limits devices per cable. This is the hardest concept for new players.

Cable type	Channels	Notes
ME Glass Cable	8	No indicator
ME Smart Cable	8	Colored lines show usage
ME Dense Smart Cable	32	Thicker, for backbone

- Each ME Controller face provides 32 channels
- Each device (drive, terminal, bus) consumes 1 channel
- Exceed channels → devices show “no power”

Use ME Smart Cable everywhere — colored lines instantly show usage.

First network:

1. **Power:** Build a **vibration chamber** or **energy acceptor**, connect any power source (Mekanism generator works)
2. **Controller:** Craft ME Controller, place it, connect power. One controller = 32 channels
3. **Storage:** Craft **ME Drive + 1k storage cells** (charged certus + redstone + glass), place next to controller
4. **Terminal:** Craft **ME Crafting Terminal** — essential, lets you craft from stored items

Connecting to the world:

- **ME Import Bus** — pulls items from adjacent inventory into network
- **ME Export Bus** — pushes items from network into adjacent inventory
- **Storage Bus** — treats adjacent inventory as part of network (no import/export needed)
- **ME Interface** — exports AND accepts items for autocrafting

Autocrafting:

1. Craft **Patterns** (blank pattern + items in grid)
2. Put patterns in **ME Pattern Provider**
3. Place provider next to **ME Molecular Assembler**
4. Request item from terminal — AE2 pulls ingredients, crafts, returns result

For Mekanism/Create machines: use **ME Interface** instead of pattern provider. Put pattern in interface, it exports ingredients to adjacent machine and accepts the result.

Subnetworks (advanced): When you exceed 32 channels, add a second controller connected via **ME P2P Tunnel** or **ME Quantum Link** for wireless.

Applied Mekanistics

Bridges AE2 autocrafting with Mekanism's chemical processing. Adds:

- ME Chemical Output Buses / Import Buses
- P2P tunnels for chemicals/gases
- Autocraft Mekanism recipes (including fission/fusion products) through AE2 crafting CPU

Without this mod, Mekanism chemical steps are manual bottlenecks.

Create Addons

Create Crafts & Additions

Electric motors and alternators create a direct **kinetic SU ↔ Forge Energy (FE)** bridge. Also adds:

- **Rolling mill** — processes metal into wires/rods
- **Silver and electrum** — new material tiers that integrate with Mekanism ore processing

Use: Connect an electric motor to your Create network to power it with FE from Mekanism. Or connect an alternator to generate FE from Create rotational power.

Create: Power Grid

Physics-based electricity simulation — Ohm's law, AC/DC, cable loss, transformers. Bridges to FE so Power Grid can power Mekanism/AE2 and vice versa.

Key concepts:

- Generators produce voltage, cables have resistance, machines draw current
- AC travels farther but needs rectification for DC machines
- Use transformers to step voltage up for long runs, then step down locally
- Built-in FE conversion via FE Converter block

When to build: Mid-Phase 2, after you have basic Create power and have outgrown SU. Diesel engines from TFMG feed into Power Grid naturally.

Steam 'n' Rails

Expanded train system:

- New track types: spruce, monorail, wide gauge
- Semaphores for signal control

- Conductor mob (rides trains, interacts with stations)
- Coupling/decoupling blocks

Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.

Create: The Factory Must Grow (TFMG)

Heavy engineering and oil processing. Drill for crude oil, refine it, burn fuels in diesel engines.

Oil processing chain:

1. Find oil — surface seepage in desert/plains biomes, or underground pools
2. **Pumpjack** — extracts oil automatically (crafted from brass and steel)
3. **Distillation Tower** (multi-block, 4+ blocks tall) — refines crude oil into diesel, gasoline, LPG, lubricant, asphalt
4. **Cracker** — breaks heavy fractions into lighter fuels
5. **Desulfurizer** — cleans fuel for more efficiency

Diesel engines:

Engine	Fuel	SU Output
Small Diesel	Diesel/gasoline	~2,000 SU
Large Diesel	Diesel/gasoline	~8,000 SU
Gas Turbine	LPG	~16,000 SU

Progression: water wheel → steam engine → diesel engine.

Steel Blast Furnace: Multi-block structure for industrial-scale steel. Faster than Mekanism's Metallurgic Infuser once you have oil power.

Aluminum: Bauxite ore (mid-to-high Y-levels) → multi-block smelter → aluminum ingots. Used in advanced TFMG machines and feeds into Mekanism alloy processing.

TFMG Energy Converter: Bidirectional bridge between TFMG electricity ↔ FE/AE2. Wire diesel generators into your ME system or Mekanism induction matrix. Requires Kotlin for Forge (auto-installed Wave 0 dep).

Mekanism TFMG Compatibility: Resolves 85+ recipe overlaps between TFMG and Mekanism. Steel, lead, sulfur unified under TFMG. Mekanism's steel recipe disabled — use TFMG Blast Furnace instead. Config in [config/mekanismtfgmcompat-common.toml](#).

Create: Gunsmithing

Steampunk firearms crafted using Create machines. Animated guns with attachments and ammo types.

Getting started:

1. Build a **Gunsmithing Table** (check EMI — uses Create brass + wood)
2. Craft **gun frames** — pistol, rifle, shotgun, each with different stats
3. Craft **barrels, grips, magazines** at the mechanical press
4. Assemble at the Gunsmithing Table
5. Craft **ammo** — each gun type uses different ammo (pistol rounds, rifle cartridges, shotgun shells)

Attachments: Scopes, extended magazines, suppressors, bayonets. Attach at the Gunsmithing Table.

Compatibility: Built-in Better Combat support — guns use appropriate animations.

Expansion path — NTGL gun packs: Create: Gunsmithing runs on NTGL, and the library supports *gun packs* (datapack-style content packs that add new guns, stats, and models via a [registry.json](#)). In practice, **no curated NTGL gun pack ecosystem exists** for 1.21.1 — the only published items are micro-fixes (renewable lead, breakable framed-glass trapdoors, an Alloyed compat pack, each a few hundred downloads at most). Authoring a custom gun pack is a DIY datapack task (NTGL wiki), not a drop-in download. Do not reach for TACZ/Scorched Guns to fill this gap — that violates the pack’s steampunk-only weapon rule.

Dependencies: NTGL (gun framework), GeckoLib (3D animation), Framework (utility library) — all auto-installed.

Create Ornithopter Glider

Mechanical flapping glider powered by Create rotational power.

How: Craft the glider (check EMI — brass + leather + feathers). Equip in chest slot. Hold jump to flap (consumes SU from a connected power source or internal buffer). Boost-assisted flight for short bursts.

Use: Early flight before MekaSuit. Pairs with Create windmills or water wheels for power.

Create Train Ecosystem

Six mods work together for a complete train network:

Mod	Role
Steam ‘n’ Rails	Track types, semaphores, conductor, coupling
Train Utilities	Station architecture — sliding doors, platform blocks
Railways Navigator	Display boards, route management, schedule entries
Interiors	Train car furniture — chairs, tray tables, kelp seats

Mod	Role
Train Parts	Crossing gates, sliding windows, train steps, slides
Threaded Trains	Performance — train pathfinding on separate thread

Together: tracks → stations → navigation → interiors → detailing → performance, with no overlap.

Mekanism Addons

Advanced Finders

Ore detection devices with directional indicators for nearby ore veins.

How: Craft a finder (check EMI — tiers for different ores). Hold it — arrows point toward nearest vein, distance shown on HUD. Synchronized in multiplayer.

Advanced Chimneys

Smoke ventilation systems for factories. Redirect emissions outdoors.

How: Craft chimney blocks, place above furnaces/generators. Smoke renders realistically and vents upward. Customizable appearance.

Dependencies: Both require ForgeEndertech library (auto-installed).

How Wave 1 Mods Work Together

Layer	Mod	What's Happening
Power generation	Create + Mekanism Generators	Water wheels → windmills → steam → heat/gas/solar → fission → fusion
Ore processing	Mekanism	2x→3x→5x ore multiplication
Factory automation	Create	Conveyors, deployers, sequenced crafters, mechanical arms
Storage	AE2	ME drives, storage busses, crafting CPUs
Transport	Create trains + AE2 P2P tunnels	Bulk item movement across bases

Layer	Mod	What's Happening
Endgame	Mekanism fusion + MekaSuit + AE2 autocrafting	God-tier power and automation
Oil/steel	TFMG	Distillation → diesel engines → blast furnace steel
Power bridge	Crafts & Additions + TFMG Energy Converter	Kinetic ↔ FE ↔ TFMG voltage
Rail expansion	Steam 'n' Rails + train ecosystem	Full train network with stations and interiors
Colony defense	Create Big Cannons (Wave 5)	Artillery for base perimeter

Session Plan — Wave 1

Session	Focus	What to Build
Session 1 — Create basics	Mine andesite. Hand crank + millstone. Water wheel. Grind wheat and ore	256 SU power, basic ore doubling
Session 2 — Mekanism start	Find osmium. Heat Generator + Enrichment Chamber + Metallurgic Infuser. 2x ore processing	Basic Mekanism line, first steel
Session 3 — AE2 discovery	Find certus quartz meteorite. Mine it. Craft dust → crystal → charge. Build AE2 Charger	First certus, charge infrastructure
Session 4 — AE2 network	ME Controller + ME Drive + 1k cells + ME Crafting Terminal. Storage Bus on Sophisticated barrel	First digital storage network
Session 5 — Mid Create	Find zinc. Make brass. Deployer + Sequenced Crafter + Mechanical Arm. Automate a simple recipe	Component automation, brass-tier

Session	Focus	What to Build
Session 6 — Mekanism expand	Add Crusher for 3x ore. Electrolytic Separator for gas. Upgrade to Advanced tier	3x multiplication, gas infrastructure
Session 7 — Integration	Import Bus on Enrichment Chamber → AE2. Export Bus to Create machines. Connect alternator to Mek power	Cross-mod automation bridge
Session 8 — TFMG start	Find oil (desert/plains pools). Pumpjack + Distillation Tower. Small Diesel Engine	Oil refining, massive SU upgrade
Session 9 — TFMG expand	Build Steel Blast Furnace. Mass-produce steel. Find bauxite for aluminum	Industrial steel, aluminum
Session 10+ — Scaling	Digital Miner. Expand AE2 drives. Add Create train between base and outpost. Work toward 5x processing	Automated mining, bulk transport

Pacing tip: You don't need to finish Wave 1 before starting Wave 1.1 (megabuilding), Wave 1.5 (colony), or Wave 2 (exploration). Dip into exploration between factory sessions. The colony takes real-world days to grow — start it early. TFMG is mid-game — start after you have brass Create and basic Mekanism.

Wave 1.1 — Megabuilding

Schematics and blueprint tools for building at scale in survival. Import community-made structures, place them layer-by-layer with Forgematica's holographic overlay, and construct anything from starter homes to mega-factories without creative mode or guesswork. This wave unlocks the Phase 3 megabuild fantasy early — you can plan and start placing mega-structures from day one, even if the resources take weeks to gather.

Megabuilding Mod

Mod	Role
Forgematica	Client-side schematic mod — import <code>.litematic</code> blueprints, display holographic overlay, place layer-by-layer in survival

Forgematica is the unofficial NeoForge port of Litematica. It lets you load schematics (`.litematic` files) and shows a translucent hologram of the build in-world, complete with material lists per layer. You place blocks manually while the overlay guides every position — no auto-build, fully survival-legal. Essential for mega-scale projects that would be impossible to freehand.

- **Version:** 0.4.3+mc1.21.11 (NeoForge, 1.8 MB)
- **Environment:** Client-side only
- **Dependencies:** None (standalone — no MaLiLib required)
- **License:** LGPLv3
- **Downloads:** 1.6M+ total

Shader compatibility: Forgematica rendering may conflict with shader packs. If the schematic overlay looks broken, toggle shaders off temporarily (`K` by default in Iris). On 1.21.1 (this pack's version), Iris compatibility is better than newer MC versions. Switch back when done placing.

Keybinds (set in Options → Controls → Key Binds after installing):

Action	Suggested Key	Purpose
Open Schematic Manager	<code>M</code>	Load, manage, and switch between schematics
Execute Schematic Placement	<code>Ctrl+Enter</code>	Locks in the current layer after verifying alignment
Toggle Schematic Rendering	<code>Ctrl+M</code>	Show/hide hologram (useful for screenshots)
Toggle Material List	<code>Ctrl+Shift+M</code>	Show/hide required materials HUD

How it works:

1. Download [.litematic](#) schematics and place them in your instance's [schematics/](#) folder.
2. Open Schematic Manager ([M](#)), select a schematic, choose placement position and rotation.
3. The holographic overlay appears at the target location. Use the material list to gather blocks per layer.
4. Place blocks manually following the overlay. Cycle through layers as you build up.
5. When a layer is complete, press [Ctrl+Enter](#) to lock it in and show the next layer.

No blocks are placed automatically — Forgematica is a guide, not a world editor. Every block must be placed in survival. This respects the pack's survival-only constraint while enabling builds at a scale that would normally require creative mode.

Schematic Recommendations

These schematics complement the Driftwood progression. All are free Litematica-format downloads from [Abfielder.com](#) — one of the largest schematic libraries (700+ schematics, all free). Each schematic is picked to serve a specific purpose in your journey from Cozy Roots through Industrial Ascent to Perpetual Ascent.

#	Schematic	ID	Phase	Purpose	Style
1	Starter Home	2818	1	Cozy first shelter — compact, complete, no wasted space	Oak & cobblestone cottage
2	Medieval Starter House	6457	1	Early game base with living space and storage	Timber-framed medieval
3	Cozy Windmill	5959	1	Decorative windmill — pairs with Create water wheels	Sails & stone tower
4	Simple Starter House	13174	1	Minimal starter — place in first 30 minutes	Spruce & stone
5	Windmill Farmhouse (FURNISHED)	10955	1–2	Farmer's Delight hub with enclosed mill, kitchen, and crop plots	Furnished medieval farm

#	Schematic	ID	Phase	Purpose	Style
6	Medieval Windmill	4216	1-2	Standing windmill — landmark and Create rotational power aesthetic	Tall stone & wood tower
7	Warehouse, Auto-sorting	11682	2	Multi-chest sorting hall — pre-AE2 bulk storage	Industrial timber frame
8	Medieval Stone Factory	13065	2	Factory shell for Create sequenced lines and Mekanism machines	Stone & deepslate industrial
9	Steampunk Factory	10175	2	Themed factory — fits Create brass-tier and TFMG oil refining	Brass, copper & brick
10	Industrial Factory <i>(Blackstone/ Nether Brick)</i>	2899	2-3	Late-game factory enclosure — Nether-themed Mekanism wing	Blackstone & nether brick
11	Large Underground Base <i>(Full Interior + Netherhub)</i>	12756	2-3	Massive underground complex with portal hub — ultimate tech bunker	Subterranean sprawl
12	Dark Fantasy Castle	7162	3	Endgame colony fortress — 62K+ DLs, fully detailed exterior	Dark stone & spires
13	Skyreach Castle	12188	3	Skyborne castle — Phase 3 “I’ve earned this” megabuild	Floating fantasy fortress
14	Wizard Watchtower	12509	1-3	RPG-themed tower — fits	Mage spire

#	Schematic	ID	Phase	Purpose	Style
				Apotheosis enchanting room	
15	Stone Bridge	10587	1–3	Decorated bridge — connect base sections or span colony rivers	Arched stone bridge
16	Wall + Tower	8559	2–3	Colony perimeter wall segment with integrated guard tower	Defensive medieval wall
17	Grand Dwarven Gate	11546	3	Epic mountain-face entrance — mega-scale statement piece	Dwarven stonework
18	Minecraft Wooden Mansion	12525	2	Large furnished mansion — Phase 2 upgrade from starter home	Spruce & dark oak estate

Download: Visit <https://abfielder.com/Products/ProductDetails.php?id=<ID>> for each schematic. Click “Download” for the `.litematic` file. Place all `.litematic` files in your instance’s `schematics/` folder.

Finding more: Browse <https://abfielder.com/Products/BrowseProducts.php?sort=trending&game=mincraft&type=Schematic> — filter by tag (House, Castle, Factory, Bridge, Mega Build, Farm, Medieval, Industrial) for additional options. Planet Minecraft (planetminecraft.com/projects/tag/litematica/) and MineMev (minemev.com/explore/) have thousands more schematics if you want alternatives. Search for tags: `medieval`, `survival`, `industrial`, `fantasy`, `steampunk`.

Style guide: When browsing additional schematics, stick to materials already present in the modpack — stone, deepslate, oak/spruce/dark oak, brick, terracotta, copper, iron, and Create decorative blocks. Avoid schematics that depend on mods not in this pack (looking for Quark, Chipped, or BYG blocks in the material list is a red flag). Medieval, steampunk, and dark fantasy styles all fit the Driftwood aesthetic. Modern, sci-fi, and glass-heavy builds don’t.

Interdependency Notes — Wave 1.1

- **Forgemtica × Create:** Factory schematics are designed as shells — you install Create machinery and Mekanism lines inside them. They provide the structure; you wire the automation.

- **Forgemata × MineColonies:** Castle and wall schematics make excellent colony defense perimeters. Wall + Tower (ID:8559) segments can be tiled to enclose an entire colony. Rebuild colony buildings inside them for a unified architectural style.
- **Forgemata × AE2:** The Warehouse Auto-sorting schematic (ID:11682) is designed for pre-AE2 storage. Once you transition to ME drives, repurpose it as bulk item overflow or decorative archive hall.
- **Forgemata × Apotheosis:** The Wizard Watchtower (ID:12509) includes space for an enchanting setup — drop your Apotheosis enchanting table and bookshelves in the top room.
- **Shader caveat:** Forgemata hologram rendering may flicker with Bliss shaders. Toggle shaders off (**K**) while placing schematics, then re-enable. The 1.21.1 compatibility is better than newer MC versions — distortion is usually minor.
- **Survival constraint:** Forgemata is purely a visual guide. No blocks are auto-placed. Every resource must be gathered and placed by the player in survival mode. This respects the pack's survival-only constraint.

Install: In XMCL, search [Forgemata](#) in the Mods tab, or download the [.jar](#) from CurseForge and drop into [mods/](#). Create a [schematics/](#) folder in your instance directory, then drop downloaded [.litematic](#) files there. Forgemata will pick them up automatically on next launch.

Session Plan — Wave 1.1

Session	Focus	What to Build
Session 1 — Pick a starter	Download 3 Phase 1 schematics. Place Starter Home (ID:2818).	First schematic build, learn Forgemata flow
Session 2 — Farm hub	Place Windmill Farmhouse (ID:10955). Plant Farmer's Delight crops around it.	Farm-to-table setup next to base
Session 3 — Bridge it	Place Stone Bridge (ID:10587) between base areas or over a river.	Connect your growing compound
Session 4+ — Factory prep	Download Phase 2 schematics. Scout a flat area for the factory district.	Site planning for industrial ascent

Session		Focus		What to Build	
Ongoing		As your tech tree advances, place larger schematics. Wall+Tower around colony. Underground base for AE2. Castle for endgame.		Megabuilding scales with progression	

Tip: Schematics eat materials fast. The Windmill Farmhouse alone needs several stacks of spruce and stone. Set up a Create tree farm and cobblestone generator early — you'll need them. Treat schematic building as a multi-session project, not a speedrun.

Wave 1.5 — Colony Management

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base.

Colony Automation

Mod	Role
MineColonies	Autonomous NPC colony — builders, miners, farmers, crafters, guards, research tree
Structurize	Structure building/placement library (MineColonies dependency)
MultiPiston	Multi-block piston library (MineColonies dependency)
BlockUI	UI library (MineColonies dependency)
Domum Ornamentum	Decorative block library (MineColonies dependency)

Colony Addons

Mod	Role
Byzantine Styles Pack	3 build styles — Byzantine, Shogun, Nile. Visual variety for colony buildings
Stylecolonies	Official MineColonies blueprint addon — 10+ styles (Steampunk, Frontier, High Magic, FairyTale, etc.)
Create: Colony Logistics	Bridges MineColonies warehouses to Create logistics — belts, chutes, trains for colony supply
ColonyLink	Connects MineColonies builders to AE2 — monitor resources, trigger autocrafting from ME terminal

Style packs coexist: Byzantine and Stylecolonies cover completely different themes (Byzantine/Shogun/Nile vs Steampunk/Frontier/FairyTale/etc.) — no overlap. Both can be installed simultaneously. FairyTale style requires Twilight Forest + Create (both in pack) and Quark (not in pack — that style simply won't appear).

Create: Colony Logistics and **ColonyLink** serve complementary roles — Colony Logistics handles Create-based colony restocking (belts, trains), while ColonyLink handles AE2-based resource monitoring and autocrafting. Both integrate MineColonies into the tech loop.

Interdependency notes for Wave 1.5

- MineColonies guards patrol your base perimeter; coverage scales with Guard Tower level and skill research — defense grows with the colony.

- Colony Research soft-gates Phase 2: “Industrial Revolution” (Colony level 3, 5000 points) signals you have the infrastructure to support Mekanism and AE2. Hard recipe gating is not implemented in this pack — playtesting determines if soft gating is sufficient.

Getting Started

1. Craft the **Town Hall** and place it on flat ground
2. Right-click → colony GUI opens
3. Hire a **Builder** from the Citizens tab
4. Place a **Builder’s Hut** near the Town Hall
5. Put wood, cobblestone, and tools in the builder’s chest
6. The builder starts constructing from blueprints

Core Worker Huts (build in this order)

Hut	Resource	What it does
Mine	Coal, iron, stone	Digs shaft mine. Passive ore supply
Lumberjack	Wood	Chops trees, replants. Infinite wood
Farmer	Wheat, carrots, potatoes	Plants, grows, harvests. Feeds colony
Guard Tower	Defense	Patrols base perimeter. Scales with colony size
Crafter	Recipes	Crafts items from storage materials

Research Tree

Research	Colony Level	Points	Unlocks
Basic Industry	2	2000	Better mines, furnace automation
Advanced Smithing	3	4000	Better worker tools
Industrial Revolution	3	5000	Atomic Alloy recipe (Mekanism gate)
Logistics	4	8000	Warehouse, courier routes

Tech gating: Atomic Alloy (Mekanism ultimate-tier) requires “Industrial Revolution” research. This prevents skipping from Phase 1 straight to Phase 3.

Guard Scaling

- 1 Guard Tower = 20-block radius coverage
- Each additional tower adds overlapping coverage
- Guard equipment upgrades with colony level (leather → iron → diamond)

Integration With Your Tech Base

- **AE2 Storage Bus** on colony warehouse → all colony resources in your ME terminal
- **ColonyLink** on AE2 network → monitor builder resource needs, trigger autocrafting when supplies run low
- **Create belts/trains via Colony Logistics** → automated material delivery to builder huts and worker chests
- **Create: Colony Logistics** on colony warehouse → colony resources flow through Create's logistic network (belts, depots, trains, stockpile switches)
- **Farmer's Delight meals** supply colony kitchen for worker happiness bonuses
- **Chunk Loaders** keep colony running when you're in other dimensions

Session Plan — Wave 1.5

Session	Focus	What to Do
Session 1 — Foundation	Town Hall, Builder's Hut, supply materials	Colony established, first build started
Session 2 — First workers	Mine hut + Lumberjack hut	Passive coal/iron/wood income
Session 3 — Farm + Guard	Farmer hut + Guard Tower	Colony self-sufficient, base defended
Session 4 — Research	Basic Industry research, Crafter hut	First tech unlocked
Ongoing	Check in every few sessions. Upgrade buildings, expand guards, unlock research	Atomic Alloy ready by Phase 2

Tip: The colony is a long-term background investment. Start it early, check periodically, let it grow.

Wave 2 — Exploration (World & Dimensions)

Overworld Worldgen

Mod	Role
Terralith	Overhauled Overworld biomes
Tectonic	Better terrain shape — deeper valleys, taller mountains, cliffs

Environment

Mod	Role
Serene Seasons	Seasonal foliage colors, temperature shifts, crop growth cycles
Serene Seasons Plus v5.1.1	Serene Seasons add-on — sub-season day/night speed + improved snow piling/melting
GlitchCore (NeoForge) v2.1.0.2 — REQUIRED	Library — required by Serene Seasons 10.1.0.3
Gabou's Libs v1.8.1	Library — required by Serene Seasons Plus

Underground & Ocean

Mod	Role
TerraBlender (NeoForge) — REQUIRED	Worldgen biome library — required by Darker Depths 2.1+ for biome placement
Darker Depths	Underground expansion — 3 new cave biomes, mobs, Forsaken Bronze, Living Crystals
Upgrade Aquatic	Ocean expansion — Thrasher mob, Ocean Ravines, Prismarine Coral, renewable sand
Hybrid Aquatic	Marine creatures — sharks, crustaceans, otters, nautilus, diving armor, corals

Interdependency note — Hybrid Aquatic + Upgrade Aquatic: Both expand oceans but cover different niches. Upgrade Aquatic adds mob mechanics (Thrasher) and ocean-floor features. Hybrid Aquatic adds creature variety

(sharks, otters, crustaceans) and diving gear. They layer without conflict — more ocean life, more reasons to dive.

Interdependency note — Darker Depths + TerraBlender: Darker Depths 2.1+ requires a worldgen API to register its cave biomes. Either **TerraBlender** (NeoForge) or **Biolith** must be installed. TerraBlender NeoForge is the recommended choice (available for 1.21.1 on both CurseForge and Modrinth). Without it, Darker Depths crashes on startup with [Missing Worldgen API!](#) [Install TerraBlender](#) or [Biolith](#).

Interdependency note — Darker Depths + YUNG's Better Caves: Darker Depths adds its own cave biomes (Molten Cavern, Sandy Catacombs, Crystal Cavern) that generate alongside YUNG's Better Caves' enlarged cave systems. Darker Depths biomes replace vanilla cave sections with themed variants, while YUNG's Better Caves changes the shape and connectivity of caves — they layer without direct conflict.

Interdependency note — Upgrade Aquatic + YUNG's Better Ocean

Monuments: Upgrade Aquatic adds ocean mobs, coral variants, and ocean-floor features. It does not modify Ocean Monuments — YUNG's Better Ocean Monuments replaces monument structure generation without conflict. Upgrade Aquatic items (thrasher teeth, prismarine coral) integrate naturally with Ocean's Delight (Wave 4) for seafood recipes.

Structures

Mod	Role
YUNG's Better Dungeons	Overhauled dungeons
YUNG's Better Mineshafts	Improved mineshafts
YUNG's Better Strongholds	Larger, more complex strongholds
YUNG's Better Desert Temples	Multi-room desert temples
YUNG's Better Jungle Temples	Jungle temples with traps
YUNG's Better Ocean Monuments	Redesigned ocean monuments
YUNG's Better Witch Huts	Expanded witch huts
YUNG's Better Nether Fortresses	Improved nether fortresses
YUNG's Better Caves	Larger, more varied caves
Structury	New hand-crafted structures
YUNG's Bridges	15+ natural bridges across terrain
YUNG's Extras	Desert wells, obelisks, ruins, pillars

Nether Worldgen

Mod	Role
Incendium v5.4.4	Nether biome and structure overhaul — 8 biomes, 9+ structures, 25+ custom weapons, Hovering Inferno boss

Incendium is a server-side data pack (packaged as a NeoForge mod) by Starmute that completely revamps the Nether using only vanilla blocks. The Nether generation height is extended to 192. Eight new biomes (Quartz Flats, Withered Forest, Infernal Dunes, Ash Barrens, Toxic Heap, and more) replace vanilla Nether biomes with 3D terrain — jagged mountains, twisting caves, and themed landscapes. Nine new structures range from the massive Forbidden Castle (one of Minecraft’s largest random structures) to the Sanctum (illager fortress), Nether Reactor, and Quartz Kitchen. Over 25 custom weapons and items (Trailblazer flame bow, Greatsword of Sacrifice, Hefty Pickaxe, Hazmat Suit) are rewarded from mobs and structures. New mobs include Toxic Slimes, Restless Spirits, a tamable Ghastling, and the summonable Hovering Inferno boss. The optional [Sparkles](#) resource pack gives Incendium items custom textures — without it, they use vanilla sprites with custom behavior and remain fully functional.

 **New world or Nether reset required:** Incendium overhauls Nether terrain generation. Do not add to an existing world with an explored Nether — reset the Nether dimension or create a new world.

Note: Incendium is listed as an alpha/unsupported port for 1.21.x while the Stardust Labs team works on a full rewrite. The mod is mature (34M+ downloads across all versions) and the featured v5.4.4 build is well-tested. The upcoming rewrite may require a Nether reset to upgrade — plan accordingly.

Dimensions

Mod	Role
The Aether	Sky dimension — safe islands, gear progression, dungeons with bosses
Deep Aether	Aether addon — 5 new biomes, Skyjade/Stratus gear, new mobs, poison swamps. Gives the Aether the “Nether Update” treatment
Umbral Skies	Aether × Twilight Forest fusion — Aether-themed structures in TF, Aether boss trophies, TF-material gloves

Mod	Role
Deeper and Darker	The “Otherside” dimension — deep dark themed, warden gear
Twilight Forest	Classic dimension — 6+ bosses with progression gates, unique loot, enchanted forest biomes
The Bumblezone	Bee dimension — honey/wax biomes, giant bees, cozy opt-in exploration; enter via bee nest/hive

End Access & Structures

Mod	Role
End Remastered v6.3.0	Portal progression overhaul — 16 custom Ender Eyes hidden across vanilla structures and bosses. Find 12+ to open the End portal
Moog's End Structures v2.0.3	New structures to fill the End dimension
YUNG's Better End Island	Overhauled dragon island — redesigned pillars, gateways, bell-tower summoning, new portal structure

End Remastered replaces vanilla Ender Eyes with 16 custom eyes scattered across vanilla structures and bosses. The Cryptic Eye comes from enchanting, the Evil Eye from master clerics, the Wither Eye from the Wither — each eye has a unique source. Once you have 12+, follow them to a stronghold and insert all 12 to open the portal. This turns reaching the End from a grind (kill Endermen for pearls) into an exploration-driven scavenger hunt.

Nullscape: A data pack that transforms the End’s terrain into alien shapes (floating islands, spires, arches, voids) with unique sky colors per biome. Nullscape handles terrain shape and sky only — it does not add biomes. Install as a data pack in the world’s [datapacks/](#) folder.

YUNG's Better End Island: Overhauls the main End island where the dragon fight takes place. Redesigned obsidian pillars, End gateways, spawn platform, and central portal structure. The dragon no longer spawns automatically — approach the bell tower in the center to begin the fight. Resummoning uses crystals on four bedrock blocks in the tower wings (vanilla positions also work). Dragon AI and difficulty are unchanged — this is a visual and atmospheric upgrade only. Requires YUNG’s API.

Navigation

Mod	Role
Explorer's Compass	Craftable structure/biome finder

Mobility

Mod	Role
Immersive Aircraft 1.1.5	Flyable aircraft — biplane, airship, cargo airship, warship, gyrocopter. Upgradable, colorable
Small Ships 2.0.0-b2.1	Sailable ships — brigantine, cog, drakkar, galley. Build, crew, and navigate; moveable bases on water. Beta (NeoForge)
Hang Glider v21.1.0	Craftable hang glider — glide from any height. No fuel, no power. Phase 1 accessible exploration aid
Man of Many Planes 0.2.1	Immersive Aircraft addon — two Create-themed aircraft. Requires Immersive Aircraft
Immersive Machinery 0.2.0	Rustic utility machines — Copperfin submarine, Tunnel Digger, Redstone Sheep, Bamboo Bee item drone

Field Guide

Mod	Role
Field Guide v1.14.0	Spyglass exploration catalog — scan plants, animals, and monsters to fill an in-game encyclopedia

Opens a new way to experience the world. Craft a spyglass, then look at any plant, animal, or monster to catalog it in your Field Guide. Each entry you discover adds to your collection with lore about the creature or plant. Works with modded mobs and biomes — automatically discovers content from every mod in the pack.

Companion mods — Immersive Overlays and Item Descriptions are **required dependencies** of the NeoForge 1.21.1 build: install them alongside Field Guide. **Mod Descriptions is an optional resource pack** — not a mod: it's language data only (no textures); it's listed in the [Resource Pack Load Order](#) (position irrelevant):

Mod	Feature Added
Immersive Overlays	Biome displays — shows spawn details for entities
Item Descriptions	Entry descriptions — auto-fills entry lore
Mod Descriptions (resource pack)	Optional — lore descriptions for 200+ mods' entries

Interdependency note — Field Guide + pack content: Field Guide auto-discovers entities, plants, and biomes from every mod in the pack — Aether and Twilight Forest mobs, Hybrid Aquatic creatures, and Incendium's Nether life are

all catalogable with zero configuration. The optional Cloth Config API dependency is already in the pack (Wave 0.5) — nothing extra to install.

Configuration — Wave 2

Aether (`aether-common.toml`):

- `"Use default Curios' menu" = true` — Uses Curios API menu instead of Aether's separate accessories screen.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Grants Book of Lore and Golden Parachutes on first entry.
- `"Show Patreon message" = false` — Removes Patreon nag screen.
- `"Add Temporary Freezing automatically" = false` — Disables Aether's temporary freezing datapack.
- `"Add Ruined Portals automatically" = false` — Disables ruined portal datapack.

Interdependency note — Deep Aether + Aether: Deep Aether is an addon that expands the Aether dimension with 5 new biomes, Skyjade/Stratus gear, and new mobs. It requires The Aether and Aeroblender (worldgen biome API) — Aeroblender comes **pre-bundled** with Deep Aether; no separate install needed. Aeroblender itself depends on TerraBlender (already in Wave 0.5 dependencies). Stratus armor grants slow-fall and dash abilities, extending the Aether's power fantasy progression. All content is within the Aether dimension — no new dimension added.

Interdependency note — Umbral Skies + Aether + Twilight Forest: Umbral Skies connects Aether and Twilight Forest. Aether-styled Druid Huts and Wells generate in Twilight Forest. Aether boss trophies appear in TF's trophy display style. Gloves can be crafted from TF materials (ironwood, fiery, knightmetal, phantom). No config needed — install and both dimensions gain crossover content.

Interdependency note — Immersive Aircraft + Aether: Aircraft can fly in the Aether dimension. An airship is a natural way to navigate Aether islands — it pairs well with the dimension's sky-island geography.

Interdependency note — Man of Many Planes + Immersive Aircraft: Man of Many Planes is a direct addon for Immersive Aircraft that adds two Create-themed aircraft. It requires Immersive Aircraft to function. The addon planes share the same fuel, parking, and dimension traversal behavior as base Immersive Aircraft vehicles.

Interdependency note — Hang Glider + Create Ornithopter Glider: The Hang Glider provides simpler, earlier flight than the Create Ornithopter Glider (Wave 1).

The Hang Glider requires only wood and leather (Phase 1 accessible, no power), while the Ornithopter Glider needs Create rotational power (Phase 1.5+). Both coexist as a natural progression: Hang Glider for early-Phase-1 exploration, Ornithopter Glider once Create infrastructure is online, Immersive Aircraft for powered mid-game flight.

Interdependency note — Immersive Machinery + Immersive Aircraft: Four Phase 1 utility machines that fill gaps before Create/Mekanism/AE2 arrive: Copperfin submarine (conduit-powered, independent of IA fuel), Bamboo Bee item drone, Redstone Sheep early farming automation, Tunnel Digger piloted mining vehicle.

Interdependency note — Twilight Forest + Apotheosis: Apotheosis boss affixes (Wave 3) apply to Twilight Forest bosses. A Naga with the “Tough” affix is significantly harder. This is intended — it keeps dimension bosses relevant even with late-game gear. If a boss feels too strong, the player can return after better equipment or use Apotheosis gems/runes for the edge.

Interdependency note — Incendium + YUNG’s Better Nether Fortresses: Incendium replaces Nether biome terrain and adds its own structures; YUNG’s Better Nether Fortresses overhauls the vanilla fortress structure generation. They don’t conflict — Incendium handles biomes and new structures, YUNG’s handles fortress interiors. Incendium fortresses will use YUNG’s layout if the fortress is in an Incendium biome.

Interdependency note — Incendium + Apotheosis: Incendium’s custom weapons (Trailblazer, Greatsword of Sacrifice, Hefty Pickaxe, etc.) are vanilla item variants — Apotheosis affixes, gems, and reforging apply to them. Incendium mobs and the Hovering Inferno boss also receive Apotheosis boss affixes (Wave 3), scaling their difficulty with player progression.

Interdependency note — Incendium + My Nether’s Delight: Incendium’s new Nether biomes introduce unique block palettes and mob drops — some may serve as My Nether’s Delight ingredients. Check EMI for new recipes once both mods are installed.

Interdependency note — Incendium + Cataclysm: L_Ender’s Cataclysm’s Netherite Monstrosity and Ignite bosses spawn in the Nether. Incendium’s biome overhaul changes where these bosses can appear — they’ll spawn in Incendium biomes instead of vanilla Nether wastes/basalt deltas. Use the Explorer’s Compass to locate them in the transformed Nether.

Interdependency note — Incendium + Fire Rekindled: Fire Rekindled’s animated Nether textures (soul fire, lava, magma, nether portal) apply to Incendium’s Nether terrain — the biome overhaul benefits from the enhanced visuals automatically.

Interdependency note — Bumblezone + Neo Bee Fix + Farmer's Delight:

Bumblezone's bees are vanilla bees at heart, so Neo Bee Fix's AI repairs (leave hives, no fall-from-sky, no wander bias) apply inside the dimension too. Honey, royal jelly, and bee bread from the Bumblezone feed Farmer's Delight honey recipes (e.g., Honey Glazed Ham) — a cozy resource chain with no gating. Field Guide auto-catalogs its mobs and plants.

Tectonic ([config/tectonic.json](#)):

Setting	Default	Recommended	Why
<code>oceans.ocean_depth</code>	-0.22	-0.35	Standard ocean floor ~45 blocks deep (vs vanilla ~28)
<code>oceans.deep_ocean_depth</code>	-0.45	-0.7	Deep ocean trenches ~90 blocks deep — into deepslate at Y=-28
<code>oceans.monument_offset</code>	-30	-55	Ocean monuments sink to match deeper ocean floor
<code>caves.ore_fix</code>	false	true	Redistributes ores when terrain extends below vanilla Y=-64
<code>global_terrain.min_y</code>	-64	-96	Lower bedrock to give 68 blocks of crust below the trench floor

How depths work: The noise value maps linearly — $|value| / 0.5 \times 64 = \text{blocks}$ below sea level (Y=62). With `deep_ocean_depth`: -0.7: $0.7 / 0.5 \times 64 = 90$ blocks deep, trench floor at Y=-28 (solid deepslate). With `ocean_depth`: -0.35: $0.35 / 0.5 \times 64 = 45$ blocks deep, floor at Y=17. `monument_offset`: -55 places ocean monument bases at Y=-16, hovering 12 blocks above the trench floor.

Ore generation with `min_y`: -96 + `ore_fix`: true: Lowering bedrock expands the vertical range ore must cover. `ore_fix` remaps ore distributions to the new depth envelope — diamond still clusters near the bottom, iron and coal fill the expanded crust. Without `ore_fix`, the extra stone between Y=-64 and Y=-96 would be barren.

 **New world required:** These are worldgen changes. Existing chunks keep their old terrain — explore fresh areas to see the new depths.

Quick verification: Stand at ocean surface (Y=62), look down with F3 open. Deep oceans should show the floor at Y=-28 (± 5). Standard oceans at Y=17 (± 5). Monuments should have their base at Y=-16 (± 5) — floating ~12 blocks above the trench floor. If values are off, adjust in-game via the Mods screen → Tectonic → Configure.

YUNG's Better Dungeons — Rarer Dungeons ([datapacks/rarer-better-dungeons/](#)):

Rebalances YUNG's Better Dungeons density for a desolate, vanilla+ Overworld. The 1.21.1 NeoForge build no longer exposes separation in its config — dungeon spacing is set by the mod's structure-set data files, so the override ships as a small datapack (pack_format 48).

Dungeon	Average separation (chunks)	Default	Recommended	Applied
Small Dungeons	spacing in small_dungeons.json	10	20–24	22
Spider Caves	spacing in spider_dungeons.json	44	55–60	58
Undead Fortresses	spacing in skeleton_dungeons.json	46	55–65	60
Catacombs	spacing in zombie_dungeons.json	48	55–65	60

- **Spacing** = average distance between structures in chunks (the old `averageSeparation` value). **Separation** (minimum distance) is raised in step to ~50% of spacing, matching the mod's own default ratio.
- **Why:** at default spacing (10) the classic small dungeons feel common. Rarer finds make each dungeon feel earned and keep the Overworld desolate between landmarks.
- **Install:** copy [datapacks/rarer-better-dungeons/](#) (or its zip) into the world's `datapacks/` folder — or `%APPDATA%\minecraft\datapacks\` for all worlds. Applies to newly generated regions only.
- **Verify:** `/locate structure betterdungeons:small_dungeon` should show ~22-chunk average spacing with the datapack active.

Overworld

Terrain — Terralith + Tectonic

Aspect	What changed	What to look for
Mountains	Taller, steeper. Painted mountains (orange/white layers), volcanic peaks	Build in mountain valleys
Oceans	90-block abyssal trenches, 45-block ocean floor, monuments floating in the deep	Use Small Ships (below) to cross, dive with Hybrid Aquatic gear
Caves	YUNG's Better Caves — wider tunnels, deeper chasms	Easier to navigate, more impressive
Rivers	Deeper valleys from Tectonic, shattered savannahs	Follow rivers for scenic travel

Pro tip: These worldgen mods work best in newly generated chunks. Every direction has something different.

Environment — Serene Seasons

Four seasons with visual foliage changes and crop growth modifiers:

- **Spring/Summer/Autumn/Winter** — foliage colors shift realistically
- **Crop impact ON** — planting in wrong season reduces yield. Plan greenhouses for off-season crops
- **Temperature damage OFF** — no freezing or heat damage (chill rule preserved)

Integration: Farmer's Delight crops are affected by seasons. Build glass greenhouses or use Season Sensor blocks to track optimal planting times.

Serene Seasons Plus (add-on): refines snow piling/melting and adds sub-season day/night speed — drives Better Days via its `[seasonalDaylightCycle]` config (below). Requires Serene Seasons + Better Days + Gabou's Libs (all in pack).

Serene Seasons Plus config (<config/sereneseasonsplus-common.toml>):

Day/night length lives here, not in Better Days. For a fixed **40-minute cycle** (2× vanilla → ~20 min day + ~20 min night):

Setting	Value	Why
<code>enableSeasonalDaylightCycle</code>	<code>false</code>	Fixed length, not seasonal variation
<code>enableBetterDaysDynamicTimeCompat</code>	<code>true</code>	Let SSP write <code>daySpeed/nightSpeed</code> into Better Days (reflection)

Setting		Value	Why
<code>customCycleLength</code>		<code>true</code>	Use the custom day/night values below
<code>customDayLength</code>		<code>0.5</code>	Better Days day-speed ratio (1.0 = vanilla 10 min day) → 0.5 = 20 min day
<code>customNightLength</code>		<code>0.5</code>	Better Days night-speed ratio (1.0 = vanilla 10 min night) → 0.5 = 20 min night

`customDayLength`/`customNightLength` are **speed ratios, not “ticks”** (the in-file comment is misleading): 1.0 = vanilla, lower = longer. Leave Better Days and Serene Seasons at their defaults — Serene Seasons Plus owns day/night length.

YUNG's Overhauled Structures

Every vanilla structure is expanded:

Structure	What's new	Loot
Better Dungeons	Multi-room with spawners, traps, treasure	Early iron, gold, enchanted books
Better Mineshafts	Multi-level with support beams, loot caches	Rails, minecarts, ore samples
Better Strongholds	Deeper portal room, more libraries/prisons	Enchanted books, mid-game exploration
Better Desert Temples	Multi-room underground, traps, vault	High-value vault loot
Better Jungle Temples	Vertical temple, redstone puzzles	Puzzle-solving loot
Better Ocean Monuments	New rooms, more elder guardians	Better sponge room loot
Better Witch Huts	Expanded hut with potion lab	Brewing supplies, potions
Better Nether Fortresses	Districts (blaze, wither skeleton, strider)	Nether wart, blaze rods, skulls
Better End Island	Redesigned pillars, bell tower, new portal	Dragon egg, gateway access

Structory & YUNG's Additions

- **Structory** — dozens of hand-crafted structures: camps, shrines, castles, towers, walled towns

- **YUNG's Bridges** — 15+ types of naturally generated bridges with biome-specific variants, spanning rivers and ravines
- **YUNG's Extras** — small vanilla+ additions: improved Desert Wells (rare Wishing Wells with treasure), Desert Obelisks, Flame Outposts, scattered ruins and pillars

All require **YUNG's API** (already in Wave 0.5 Dependencies). Structory is independent of the YUNG's mods.

What to do: Explore! Mark interesting structures on your JourneyMap (fullscreen map → right-click to set waypoint). Return with an Explorer's Compass.

Explorer's Compass — Navigation Tool

1. Craft: 1 compass + 4 iron ingots in a + pattern
2. Right-click → GUI opens
3. Search tab: type biome or structure name (partial matching — “cherry” finds all cherry biomes)
4. Select an entry → compass points to nearest instance, distance shown
5. Follow the arrow in real-time
6. When found, add a JourneyMap waypoint

Configuration ([config/explorerscompass-common.toml](#) → [General] section):

Setting	Default	Recommended	Why
<code>maxRadius</code>	10000	5000	Halve search radius — you must explore ~5K blocks before the compass finds a structure. Prevents instant discovery from spawn
<code>maxSamples</code>	100000	50000	Fewer search samples = lower search accuracy at range. Combined with reduced radius, structures won't be found until you're meaningfully close
<code>maxNextSearches</code>	100	10	Limit “find next” to 10 — once you've cleared the 10 nearest instances, explore further and search again
<code>displayCoordinates</code>	true	true	Show X/Z coords and distance on HUD — essential QoL
<code>defaultXpLevels</code>	0	0	No XP cost for searching — chill pack, navigation shouldn't gate

Effect: With `maxRadius = 5000`, the compass scans a $\sim 10K \times 10K$ area centered on you instead of $20K \times 20K$. You'll need to explore new regions to find new structures — no more “find the nearest End City from spawn.”

Underground & Ocean

Darker Depths — Expanded Underground

3 unique cave biomes deep underground:

Biome	Depth	What to expect	Key resources
Molten Cavern	Deep (Y < 0)	Lava pools, geysers, amber clusters	Living Crystals (feed diamond to grow), Crystal Melons
Sandy Catacombs	Mid (Y 0-20)	Petrified roots, aridrock, Void Soul mobs	Void Souls (bottle them), Forsaken Bronze
Crystal Cavern	Any depth	Growing crystals, glowing ambience	Crystal materials for decorative/utility

Living Crystals: Found dead in Molten Cavern. Feed a diamond → reawakens, grows, spreads to nearby diamond ore. Crystal Melons (off-hand) = unbreakable tools with +20% speed/damage for 5 minutes.

Void Souls: Kill a Void Soul mob → becomes invulnerable orb (60 seconds). Capture in glass bottle. Use amber to prevent expiration. Forsaken Bronze from void-touched materials has death-related abilities.

Compatibility: Darker Depths biomes generate inside YUNG's Better Caves systems — they layer without conflict.

Required dependency: TerraBlender NeoForge (auto-installed). Darker Depths 2.1+ crashes without it.

Upgrade Aquatic — Richer Oceans

Feature	What it does
Thrasher	Fantasy shark-like mob. Uses sonar, thrashes prey. Drops thrasher teeth for trident crafting
Ocean Ravines	Underwater ravines with Prismarine Coral — harvest for decorative blocks
Nautilus	Passive mob. Drops shells for conduit crafting
Renewable sand	Bubble columns below sandstone convert stone to sand

Integration: Thrasher teeth and prismarine coral integrate with Ocean's Delight (Wave 4) for seafood recipes. Compatible with YUNG's Better Ocean Monuments.

Hybrid Aquatic — Marine Creatures

Expands underwater life with realistic marine creatures and diving equipment.

Creatures added:

Category	Examples
Fish	Reef fish, clownfish, groupers, puffers
Sharks	Hammerhead, great white — neutral until provoked
Crustaceans	Crabs, lobsters, shrimp — drop food ingredients
Critters	Otters (tameable!), nautilus, seahorses
Mini-boss	Deep sea guardian — found in ocean caves

Blocks & items:

Item	Use
Diving Armor	Full set — extended underwater breathing, faster swim speed
Sea Sponges	Absorb water, decorative blocks
Fishing Hooks	Specialty hooks for better catches
Crates	Storage blocks with ocean aesthetic
Anemones	Decorative coral-like blocks
New Corals	Additional coral variants for ocean decoration

Getting started:

1. Explore oceans — creatures spawn naturally in rivers, swamps, and oceans
2. Kill crustaceans → food ingredients for Farmer's Delight recipes
3. Find deep sea guardian in ocean caves → challenging mini-boss fight
4. Collect coral and anemones → decorate your underwater base
5. Craft diving armor → extended ocean exploration becomes viable

Integration with Starcatcher (Wave 4): Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

Integration with Ocean's Delight (Wave 4): Hybrid Aquatic crustacean drops (crab meat, lobster tail) integrate as seafood ingredients.

Mobility

Immersive Aircraft

Vanilla-style flyable aircraft. Mid-game (requires iron + basic Create/Mek materials).

Vehicle	Fuel	Seats	Cargo	Best for
Biplane	Coal/charcoal	1	None	Fast personal flight, combat
Airship	Coal/charcoal	3	9 slots	Exploration with friends
Cargo Airship	Coal/charcoal	1	27 slots	Material transport
Warship	Coal/charcoal	3	9 slots	Armed combat, defense
Gyrocopter	Coal/charcoal	1	None	Agile short-range flight

How: Craft at table (check EMI — iron + wool/leather), place on ground, right-click to enter. [W](#) to ascend, [S](#) to descend, [A/D](#) to rotate. Any furnace fuel works.

Integration: Aircraft work in the **Aether** — fly between sky islands without bridges. Park indoors or on high platforms (mobs damage parked aircraft).

Small Ships

Buildable, sailable ships. **Phase 1 accessible** — requires wood, wool, iron.

Ship	Crew	Cargo	Best for
Brigantine	1	18 slots	All-purpose sailing
Cog	2	27 slots	Cargo transport
Drakkar	2	9 slots	Fast travel, combat
Galley	1	9 slots	Early personal vessel

How: Craft hull (wood) + mast (wood + wool) + helm. Right-click water to place. Right-click helm to board. [W/S](#) to move, [A/D](#) to steer. Speed depends on wind direction. Dismount to park — ships float where left. Right-click with lead to tow.

Note: Beta for NeoForge (2.0.0-b2.1). Core features work; minor bugs expected.

Hang Glider

Simple fabric-and-wood glider. **Phase 1 accessible** — requires only wood and leather, no fuel or power.

- Craft the glider (check EMI — wood + leather + string)
- Equip in your hand and jump off any height to glide
- Hold forward to maintain speed, steer with mouse
- Dismount by pressing sneak — you'll drop safely to the ground
- Requires **Puzzles Lib** (already in pack — Easy Anvils, Leave My Bars Alone)

Progression: The Hang Glider is the earliest flight option. It provides gentle gliding for early exploration before the player has Create power for the Ornithopter Glider (Wave 1) or fuel infrastructure for Immersive Aircraft. It cannot gain altitude — it's pure glide, not powered flight. Ideal for scouting Terralith biomes from mountain peaks.

Config: No changes needed — default behavior is ideal. Gliding is inherently chill (no fuel, no maintenance, no risk of crashing unless you fly into terrain).

Man of Many Planes — Immersive Aircraft Addon

Adds two Create-themed aircraft to Immersive Aircraft. **Phase 2 accessible** — requires Immersive Aircraft (iron + wool/leather) plus Create materials (andesite alloy, brass).

- **Two new aircraft** — brass-and-mechanical themed designs
- Uses the same fuel, parking, and controls as base Immersive Aircraft
- Works in the Aether dimension like all Immersive Aircraft
- Craft recipes available in EMI

Integration: These aircraft offer a visual and thematic bridge between Create (Wave 1) and Immersive Aircraft (Wave 2). A player deep in Create will appreciate aircraft that match their factory's aesthetic.

Dimensions

The Aether — Sky Dimension

Safe, beautiful sky islands. Gentle adventure curve.

Portal: 4×5 glowstone frame (like Nether portal) → fill with water bucket → step through.

Progression:

- **Phase 1:** Spawn on sky island. Collect **ambrosium** (Aether coal) and **zanite** (Aether iron — mining level increases as durability drops)
- **Phase 2:** Bronze dungeon → defeat **Slider** (hit when blue, avoid when red) → Golden Amber
- **Phase 3:** Silver dungeon → defeat **Valkyrie Queen** → Valkyrie tools (flight-enabled weapons), **Gravitite** (floats upward when right-clicked)
- **Phase 4:** Highest crystal islands → **Sun Spirit** — bring fire resistance

Tip: Not required for any tech progression. Visit when you want a change of pace. MekaSuit trivializes all Aether combat.

Deep Aether expansion: Installed alongside — it adds 5 new biomes (Aerglow Forest, Aerviolet Fields, Yagroot Swamp, Golden Heights) that generate

organically in the Aether. **Skyjade** tools/armor (stronger at full durability) appear in holystone. **Stratus** armor (slow-fall + dash — press R with full set) is crafted from Sterling Aerclouds found above Y=200. The Yagroot Swamp biome introduces poison liquid, Virulent Quicksand, and the Spooky Ring (night damage bonus). All content is purely additive — the base Aether progression is unchanged.

Config (`aether-common.toml`):

- "Use default Curios' menu" = true — uses Curios API menu
- "Gives player Aether Portal Frame item" = false — build portal manually
- "Gives starting loot on entry" = true — Book of Lore + Golden Parachutes on first entry
- "Show Patreon message" = false
- "Add Temporary Freezing automatically" = false
- "Add Ruined Portals automatically" = false

Twilight Forest — Progression Dimension

Dark, enchanted dimension with strict boss progression. Each boss kill unlocks the next area.

Portal: Find a 2×2 water pool → surround with 12 flowers (any type) → throw a **diamond** into the pool → lightning strikes → step into the water.

Note: One-way down. To return, craft a **Twilight Portal** block or use a Waystone (place one immediately on arrival).

Boss progression:

Order	Boss	Location	Strategy	Unlock
1	Naga	Naga Courtyard	Strafe charges, hit tail segment	Lich Tower
2	Lich	Lich Tower	Dodge projectiles, hit during cooldown. Shield reflects projectiles	Twilight Swamp
3	Minoshroom	Labyrinth	Find minotaur room, kill it, grab Meef Stroganoff	Firefly Forest
4	Hydra	Hydra Lair	Dodge fireballs, melee heads. Drops Fiery Tears	Ur-Ghast fight

Order	Boss	Location	Strategy	Unlock
5	Ur-Ghast	Dark Tower	Shoot tentacles, then main body. Dark Forest has blindness — bring milk	Snowy Forest + Highlands
6	Snow Queen	Aurora Palace	Push into pillars to break shield. Avoid ice arrows	Final Castle
7	Final Castle	Highlands	Multi-boss fight. Needs weapons from all prior bosses	Giant's Pick (5×3 mining area)

Key items:

- **Ironwood** gear — smelt ironwood ingots from iron + magic beans. Good starter armor
- **Fiery** gear — iron + blaze powder. Mid-tier weapons
- **Knightmetal** — from Lich Tower
- **Phantom armor** — creative flight for short bursts. From Ur-Ghast tentacles
- **Giant's Pick** — 5×3×5 mining area. From Final Castle

Tips:

- Always twilight — no day/night cycle. Mobs spawn any time
- Place **Waystones** frequently
- **Twilight Forest Maze Map** (paper + torch) reveals maze/castle layouts
- Bring **milk buckets** for Dark Forest blindness
- Your AE2 wireless terminal, Mekanism gear, and Create tools all work here

Integration: Apotheosis boss affixes apply to Twilight Forest bosses. A Naga with “Tough” affix is significantly harder — intended to keep dimension bosses relevant even with late-game gear.

Deeper and Darker — The Otherside

Deep-dark-themed dimension accessed through the Deep Dark.

How to enter:

1. Find a **Deep Dark** biome (Y=-40 or below)
2. Locate an **Ancient City** (Explorer's Compass → search “ancient city”)
3. Kill the **Warden** — drops a **Warden Heart**
4. Craft: Warden Heart + 8 sculk → **Otherside Portal Catalyst**
5. Build a 4×5 frame of **reinforced deepslate** → right-click catalyst on frame → step through

Warden strategy: Detects by sound. Use wool to silence footsteps. Sneak near sculk shriekers. Throw snowballs to distract. 500 HP, slow attacks — hit and run.

What to expect:

- 4 new biomes — sculk forests, frozen deep, expanded ancient cities
- 8 new mobs — sculk variants, stalker (ranged), shriek worm (ambush), **Sculk Leech** (heals on damage)
- **Miniboss:** Shattered Warden — drops **Sculk Transmitter** components
- **Key loot: Sculk Transmitter** — Elytra upgrade for mid-flight teleportation.
Combine at smithing table

Gear: Warden-themed armor/tools ≈ diamond tier with unique abilities (sound-based attacks, silence fields).

Difficulty: Harder than Aether, easier than Twilight Forest's Dark Forest. Come with iron-tier armor + decent weapon.

The Bumblezone — Bee Dimension

A cozy bee-themed dimension entered directly from any **bee nest or beehive** — no portal structure to hunt for or build.

How to enter: Right-click a bee nest/hive (vanilla or modded) to teleport in. Leaving returns you to the Overworld.

What to expect:

- Honeycomb caves, sugar-water seas, wax roads, and pollen fields — a warm, glowing, golden dimension
- Mobs: giant **Bumble Bees**, Honey Slime, Beehemoth, and bee-adjacent creatures
- Structures: Mite Fortress, Hanging Gardens, Candle Parkour, shrines and sanctums
- Key loot: **Royal Jelly**, bee bread, honey crystals, wax, the **Comb Cutter** enchantment (shear honeycombs from hives), and the **Crystalline Flower** (combine/transfer enchantments from books)
- A **Honey Compass** locates Bumblezone structures

The bees' one rule: they're peaceful until you take their honey — harvest and they turn hostile for a while, then calm down. Be a polite beekeeper.

Integration:

- **Neo Bee Fix** (Wave 4) — its vanilla bee AI repairs apply inside the dimension too (Bumblezone's bees are vanilla bees at heart)
- **Farmer's Delight** — honey, royal jelly, and bee bread feed FD honey recipes (e.g., Honey Glazed Ham)
- **Field Guide** — auto-catalogs Bumblezone mobs and plants

- **Bundled resource packs** (from the mod, enable in Resource Packs):
Bumblezone - Shader Emissive for glowing blocks under Bliss (works with ETF/Continuity now in the pack); **Bumblezone - Anti Tripophobia** replaces the trippy honey patterns with plain textures — enable if they bother you

Chill fit: fully opt-in, no gates, no structure hunting — enter any hive whenever you like. Honey/wax builds a cozy supply chain; deeper structures (Mite Fortress) reward mid-game gear.

Difficulty: gentle by default — the only real threat is provoking the bees. Visit any time from Phase 1 onward.

End Access — End Remastered

The End portal now requires a scavenger hunt instead of a grind.

Old method: Kill Endermen → craft Ender Eyes → find stronghold → insert 12 eyes.

New method: Find 16 unique custom Ender Eyes scattered across the world:

Eye	Source	Eye	Source
Old Eye	Desert Pyramids	Cursed Eye	Bastion treasure
Nether Eye	Nether Fortresses	Exotic Eye	Craft (aquatic items)
Cold Eye	Igloos	Evil Eye	Master Cleric trade
Rogue Eye	Jungle Pyramids	Undead Eye	Craft (Undead Soul from skeleton horse)
Black Eye	Buried Chests	Cryptic Eye	Random chance when enchanting
Magical Eye	Evoker drop	Wither Eye	Wither drop
Lost Eye	Mineshafts	Guardian Eye	Elder Guardian drop
Corrupted Eye	Pillager Outposts	Witch Eye	Craft (Witch Pupil)

How it works: Find 12+ unique eyes. Hold one to locate the stronghold (like vanilla Ender Eyes). Insert all 12 into the portal frame to open it. If a vanilla eye is already in a frame, right-click with a custom eye to replace it.

Tip: This naturally paces your journey to the End — you'll need to explore deserts, swamps, nether fortresses, ocean monuments, woodland mansions, and bastions before you can fight the dragon. By the time you open the portal, you'll have gear, experience, and world knowledge that makes the End a natural next step rather than a rushed target.

End Dimension

The End is now a layered destination — each mod handles a distinct piece:

Mod	What it does
End Remastered	How you get here — 16 custom Ender Eyes scattered across the world
YUNG's Better End Island	The main island — redesigned dragon arena with bell-tower summoning
Nullscape (data pack)	The outer islands — surreal floating terrain, spires, arches, unique sky colors
Moog's End Structures	Structures on outer islands — towers, ruins, dungeons with better-than-vanilla loot

Ender's Delight (Wave 4) adds food cooked from End ingredients — useful status effects for End exploration.

Progression:

- Post-Dragon: Explore outer islands for new structures and loot
- Mid-End: Tackle Moog's structures for unique weapons
- Endgame: Build an End base — Nullscape terrain makes for dramatic architecture

Tip: Place a Waystone immediately. Build a small AE2 subnet with wireless terminal to access main base storage.

Session Plan — Wave 2

Session	Focus	What to Do
Session 1 — Scout	Craft Explorer's Compass. Search for cherry grove or redwood forest. Mark waypoints on JourneyMap. Loot a YUNG's dungeon. Craft a Hang Glider for early aerial scouting	First taste of transformed world
Session 2 — Nether	Build Nether portal. Scout Incendium biomes. Find a fortress. Rescue a Ghastling from an Abandoned Tower. Mark	Nether foothold, ally, blaze rods

Session	Focus	What to Do
	a Waystone at your portal	
Session 3 — Aether	Gather glowstone. Build Aether portal. Explore first sky island, collect ambrosium and zanite	Safe new dimension
Session 4 — Twilight Forest	Find 2×2 pool, surround with flowers, throw diamond. Enter, place Waystone. Defeat the Naga	First Twilight boss
Session 5+ — Deep exploration	Work through Twilight Forest bosses. Visit Deeper and Darker (find Ancient City). Explore End post-Dragon	Boss loot, dimension resources
Session 6 — Aircraft	Craft a biplane or airship (iron + wool). Explore Terralith biomes from the air. Craft Man of Many Planes aircraft for Create-themed flight	Fast travel, scenic views

Pacing tip: Explore between factory builds. Grab your compass and walk in a random direction when waiting for Mekanism to process or Create to assemble.

Wave 2.5 — Building Your Home

Building is a core pillar of this modpack — at any point you can stop optimizing and just build something beautiful. This guide works at any stage: no creative mode or artistic talent required, just a few principles that separate a “dirt hut” from a home you want to come back to.

Finding Your Build Site

Before placing a single block, find a location that inspires you. You have tools for this:

Tool	How	Why
Explorer’s Compass	Craft: 1 compass + 4 iron (+pattern). Right-click → search “cherry”, “redwood”, “meadow”	Find specific biomes by name
JourneyMap	Press J → explore the fullscreen map	See terrain, biomes, and your marked waypoints
Small Ship	Sail along coastlines	Discover waterfronts you’d miss on foot
Biplane	Fly low over terrain	Spot valley builds, mountain perches, river mouths

What to look for:

- **Waterfront** — river banks, ocean cliffs, lake shores. Water reflects builds beautifully with shaders
- **Mountain valleys** — enclosed, dramatic, defensible. Terralith makes these spectacular
- **Forest clearings** — surrounded by trees feels cozy and sheltered
- **Cliff faces** — cut into the hillside for a build that looks permanent, not placed
- **Cherry groves** — pink petals + mountain backdrop = instant atmosphere

Pro tip: Place a **Waystone** at your chosen site before you start building. You’ll want to teleport back with materials.

The Block Palette Rule

The single biggest upgrade from “meh” to “wow” is using **3–5 complementary blocks** instead of one. Every great build you’ve ever seen uses a palette. Here’s the formula:

Role	Percentage	Purpose
Primary	~50%	Dominant wall material — defines the build
Secondary	~30%	Complementary texture — similar tone, different surface
Accent	~20%	Detail material — corners, trim, highlights, pops of color

Finding Palettes: BlockPalettes.com

BlockPalettes.com — the single best resource for block combination inspiration. Browse hundreds of community-created palettes, each showing exactly which blocks work together. Filter by style, color, or mood.

How to use it:

1. Open blockpalettes.com/palettes
2. Browse featured palettes or filter by color/theme
3. Click a palette to see all blocks in the combination
4. Test it: place a 5×5 section in a flat area, step back, see if it works
5. If any block fights the others, swap it out

Starter Palettes for This Modpack

These palettes use blocks you can gather in Waves 0–2. Each fits a different biome and vibe:

Medieval Stone — stone brick + cracked stone brick + mossy stone brick + cobblestone (corners/base)

Cozy Cottage — oak planks + spruce planks + cobblestone (chimney/base) + oak leaves (roofline trim)

Desert Sandstone — sandstone + smooth sandstone + cut sandstone + terracotta accents

Nordic Cabin — spruce planks + stripped spruce log + stone + dark oak stairs (roof)

Cherry Zen — cherry planks + cherry log + stone brick + pink petals (garden)

Mountain Fortress — stone brick + deepslate tiles + tuff bricks + copper (accents/patina)

Tropical — jungle planks + bamboo mosaic + mangrove propagules + sand

Endstone Palace — purpur block + purpur pillar + end stone bricks + chorus flower (End builds)

Tip: Open BlockPalettes.com in your browser while you build. When you're stuck on what goes together, browse palettes until something clicks.

Depth and Layering

Flat walls are the second most common beginner mistake (after single-block builds). A flat wall looks like a placeholder. A layered wall looks **built**.

Five Techniques That Transform Flat Walls

Technique	How to do it	Why it works
Inset windows	Push glass back one block into the wall instead of flush with the surface	Creates shadow lines that frame windows
Outcropped corners	Extend corner blocks one block forward from the main wall plane	Breaks up 90-degree edges, adds structural feel
Alternating depth	Every 3–4 blocks along a wall, push one column back or pull one forward	Wall reads as textured rather than uniform
Stair/slab detailing	Upside-down stairs under windowsills, stairs as roofline trim, slabs as half-height ledges	Catches light and shadow in ways full blocks can't
Overhanging roofline	Always extend the roof 1–2 blocks past the wall on all sides	Creates shadow line that defines where building ends and sky begins

Fastest single improvement: Add vertical **log pillars** every 3–5 blocks along exterior walls. Logs imply structural support and instantly break up flat surfaces. Works with every style.

Roof Design

A bad roof ruins a good build. A good roof makes even a simple box look intentional.

Three Roof Styles (Pick One)

Style	Shape	Best for	How
A-frame (pitched)	Triangle cross-section	Cottages, cabins, medieval	Stairs ascending from each wall meeting at a ridge. Overhang 1 block on all sides
Flat with parapet	Flat top, raised edges	Modern, desert, fortress	Slabs or full blocks, 1-block

Style	Shape	Best for	How
			wall around perimeter
Stepped pyramid	Layered squares getting smaller	Asian, temples, fantasy	Each layer 2 blocks smaller than the one below

Material rule: Roof material should **contrast** with wall material. Dark roof on light walls, or vice versa. Spruce stairs on oak walls. Deepslate on stone brick. Blackstone on sandstone.

Common mistake: Roof too small. The roof should be wider and taller than feels “necessary.” An oversized roof looks dramatic; an undersized one looks like a hat that doesn’t fit.

Interior Design

An empty interior is as bad as a bad exterior. Minecraft has no furniture mod yet (Wave 6 adds Handcrafted), but vanilla blocks create convincing rooms.

Lighting (Critical)

Never place torches on interior walls. They look improvised. Better options:

Light source	Look	How
Lanterns	Warm, architectural	Hang from ceiling via fence post
Glow berries	Organic, trailing	Place on ceiling, let vines trail down
Sea lanterns	Clean, neutral	Modern or sci-fi builds
Jack o’lanterns	Hidden under carpets	Floor lighting without visible sources
Froglights	Subtle, colorful	Pearlescent/verdant/ochre variants from Aether or Overworld

Functional Rooms

Room	Key blocks	Details
Kitchen	Smoker, campfire, barrels, chests	Barrel = cabinets. Chest with food item frame = pantry. Cooking pot over campfire
Bedroom	Bed, crafting table, bookshelf, lantern	Banner on wall above bed as headboard. Carpet at foot of bed
Library	Bookshelves, lectern, quill	Cover walls with bookshelves for instant warmth

Room	Key blocks	Details
Storage	Sophisticated Storage barrels, chests	Organized rows. Sign labels on each barrel
Workshop	Crafting table, furnace, anvil	Industrial feel — stone walls, iron accents

Pro tip: Place **item frames** on walls with maps, banners, or shields inside. They function as paintings with more variety.

Blending Into the Landscape

A build that looks dropped from the sky is unfinished, no matter how detailed the structure itself is.

Key principles:

- **Use biome-appropriate materials** — spruce in cold biomes, sandstone in desert, dark oak near jungle edges. The build should look like it was built from local materials, not imported
- **Never leave flat grass at the base** — add a slight slope toward the entrance, gravel/cobblestone paths leading away, small dirt mounds or rock outcroppings at corners
- **Plant trees near the build** — a cluster of 2–3 trees on one side breaks the silhouette and gives context. Use the biome’s native trees
- **Cut into hillsides** — a build emerging from a hillside looks permanent; one sitting on top looks temporary
- **Paths and surroundings** — gravel paths, stone brick walkways, and lantern posts leading away from the entrance sell the idea that people live here
- **Never run more than 7 blocks in a straight line** when shaping terrain — curves look natural, straight lines look artificial

Modpack-Specific Building Tools

You have tools that vanilla players don’t. Use them:

Tool	What it does for building
Waystones	Place one at your build site. Teleport home for materials, teleport back. No more carrying 3 inventories of blocks across 500 blocks
Sophisticated Storage	Barrels near your build site hold materials organized by type. Label with signs
Backpack	Carry your palette blocks in your backpack while building. Press B to swap materials on the go

Tool	What it does for building
Create Blueprint	Craft Blueprint and Quill. Select your finished build → save as blueprint → use Schematicannon to auto-rebuild it elsewhere with materials from chests
Building Wands (Wave 6)	Place walls, fill areas, build circles/cylinders in bulk. Wait for Wave 6 for this one
JourneyMap waypoints	Mark your build site, mark material locations, mark inspiration you find while exploring
Dynamic Lights (Ctrl+L)	Hold a lantern while building at night. No more stopping to place torches

Quick-Start: Your First Real House

If you've never built anything beyond a dirt hut, follow these steps for your first proper home:

1. **Pick a palette** — Open [BlockPalettes.com](https://blockpalettes.com), find 4 blocks you like. Test in a 5×5 section
2. **Lay the foundation** — Cobblestone or stone brick ring, 7×9 interior minimum (small but not cramped)
3. **Raise the walls** — Use your primary block. Leave gaps for windows (2-block wide minimum)
4. **Add depth** — Push windows back 1 block. Add log pillars at corners and every 4 blocks
5. **Build the roof** — A-frame with stairs. Overhang 1 block. Use a contrasting material
6. **Add the door** — Center of one wall. Stairs as a small porch step in front
7. **Light it up** — Lanterns from ceiling, not torches on walls
8. **Furnish basics** — Bed, crafting table, furnace, 2–4 barrels for storage
9. **Blend the exterior** — Paths leading away, trees nearby, flowers at the entrance, leaves around the roofline
10. **Step back and admire** — You just built something worth coming home to

Community Resources

Resource	URL	What it's for
BlockPalettes	blockpalettes.com	Block color combinations and palette inspiration
Building Guide App	buildingguide.app	Step-by-step tutorials with materials lists (128+ builds)
r/MinecraftBuilds	reddit.com/r/MinecraftBuilds	Community showcase, inspiration, feedback

Resource	URL	What it's for
r/MinecraftHouses	reddit.com/r/MinecraftHouses	House-specific builds and ideas
r/MinecraftTutorials	reddit.com/r/MinecraftTutorials	How-to guides for specific techniques
Minecraft Wiki — Building	minecraft.fandom.com/wiki/Building	Official guides on terraforming, furniture, layout

Session Plan — Wave 2.5

Session	Focus	What to Build
Session 1 — Site selection	Explore with Explorer's Compass. Find a biome you love. Place Waystone. Mark on JourneyMap	Your future home location
Session 2 — Palette + foundation	Browse BlockPalettes.com. Pick 4 blocks. Lay foundation ring. Raise walls to 4 blocks high	Shell of your first real house
Session 3 — Roof + depth	Build A-frame roof. Add log pillars. Inset windows. Add door	Recognizable house shape
Session 4 — Interior	Lanterns from ceiling. Bed, crafting table, furnace. Barrels for storage. Item frames on walls	Livable interior
Session 5 — Landscaping	Paths, trees, flowers, leaf trim around roofline. Step back and see the difference	Home in a world

Tip: Don't aim for perfection on your first build. Aim for "better than a dirt hut." You can always expand, renovate, or rebuild later. The skills transfer — your second build will be twice as good.

Wave 3 — Equipment Magic & RPG

RPG Progression

Mod	Role
Skills Mastery Reimagined	71-node skill tree with unique tradeoff abilities
Pufferfish's Skills	Underlying skill system framework
Pufferfish's Attributes	Custom attributes powering skill tree node effects

Equipment

Mod	Role
Simply Swords	60+ unique melee weapons with passive/activated abilities
Simply More	10 new weapon types + 33 unique weapons — Simply Swords addon
Too Many Bows	30+ unique bows with special abilities and custom attributes
Relics (RPG Series)	Legendary accessories with unique abilities — levels up with use
Enchanting Runes	Socketable runes for weapons/tools — early-game enchanting alternative
Immersive Armors	40+ unique armor sets with custom effects — berserk, double jump, hostile radar, spikes, divine block

Loot & Affixes

Mod	Role
Apotheosis	Rarity tiers, random affixes, gem socketing, enchanting overhaul. Replaces Savaru's Affixology with more depth
Apothic Compats 0.2.4.2	Datapack-driven Apotheosis bridges — affixed loot, gear sets, affixes, and invaders for 11 in-pack mods (AE2, Aether, Cataclysm, Create, Curios, Deep Aether, Deeper and Darker, Farmer's Delight, Mekanism Tools, Supplementaries, Twilight Forest). Requires Apotheosis

Configuration — Wave 3

Apotheosis — the most config-heavy major mod. Apotheosis is split into modules, each with its own config:

Adventure module — affixes, gems, boss modifiers, rogue spawners:

- Keep all affix types enabled: Ruthless (crit), Arcane (spell), Sturdy (armor), Nimble (speed), Plentiful (mining), Titanic (HP). Each supports a different playstyle.
- Gem tiers enabled: Rough, Cut, Flawless. The gem progression (Rough → Cut → Flawless) provides a satisfying vertical upgrade path. Each tier unlockable through Apotheosis's reforging mechanic.
- Gem types cover all primary stats: melee damage, projectile damage, magic damage, armor, max HP, mining speed, movement speed, luck. This ensures every build archetype benefits from gem-socketed gear.
- Socket count: Gear drops with 1-2 sockets by default. Reforging at the Smithing Table can add a 3rd socket at high rarity (Mythic+). This prevents socket overload while keeping the system meaningful.
- Boss modifiers: all active. Apothic Invaders and Elites spawn naturally with random affixes. They always drop affix items — this is the primary source of mythic gear.
- Rogue Spawners: enabled. Underground spawner+chest combos. Risk vs reward.
- Affix item probability: default rates (~5% base for rare+, scaling with luck and difficulty). Common-enough to feel rewarding, rare-enough to keep each upgrade exciting.

Enchanting module — [config/apotheosis/enchanting.cfg](#):

- Keep all enchantment overhauls enabled: higher level cap (up to level 150+), tiered bookshelves (Hell, Seashelf, Endshelf), Library of Alexandria, new enchantments (Chromatic, Mounting, Stable Footing, etc.).
- The Library of Alexandria stores all enchanted books in one block — essential for enchanting organization at scale.
- Enchantment max levels increased for many vanilla enchantments: Protection can reach level 8, Sharpness level 6, Fortune level 5, Looting level 5, etc. These higher caps match the power fantasy curve — gear enchanted at max level should feel mythic.

Potion module:

- Potion Charms enabled: craft a charm from blaze powder + potions, equip in offhand for passive effect. Reduces inventory clutter from potion bottles.
- Potion augmentation allowed: glowing, brewing, changing base type into splash/lingering at standard ratios. Augmented potions can have higher level caps (e.g., Potion of Strength III).

Spawner module:

- Spawner capture and modification enabled. Silk-touch a spawner with a pickaxe, then modify it (entity type, spawn delay, player range, min/max entities). Key for XP farms and mob-based automation.

- Spawner stats are fully configurable after placement through the spawner's GUI (right-click with an empty hand). Players can fine-tune delay, count, range, and conditions per spawner — no external tool needed.

Village module:

- Wandering trader improvements enabled. Traders can sell Apotheosis items (gems, affix gear, enchanted books) at elevated prices. Adds a secondary acquisition path for players who prefer trading over combat.

Deadly module: Enabled. Apotheosis's Deadly module adds randomly generated death-traps to the world (sudden spikes, poison darts, instakill rooms, lava traps). This adds risk-reward tension to structure exploration — the deadliest rooms hold the best loot. The Deadly module's traps are telegraphed and avoidable with caution, and GraveStone Mod keeps your items safe if you do trigger one, so the chill rule is preserved.

Reforging costs — Apotheosis's reforging mechanic (reroll affixes with gem dust) uses default material costs. As the player progresses, reforging costs scale:

- Rare→Epic: moderate (dust + some ingots)
- Epic→Mythic: expensive (dust + rare materials like netherite scrap/echo shards)

This cost curve encourages the player to settle for “good enough” gear until they're ready to invest in perfect rolls.

Interdependency notes for Wave 3

- **Spell Engine** is required by Relics (RPG Series) — it was previously considered for removal but must be retained as a mandatory dependency of this core RPG mod.
- **Ranged Weapon API** is shared between Relics (RPG Series) and Pufferfish's Skills ecosystem — install once, used by both.
- **Curios API** provides the accessory slot system used by Relics (RPG Series) and Immersive Armors. Adds an equipment GUI (keybind [Y](#)) with ring, amulet, belt, gloves, back, and charm slots. Auto-installed dependency — no separate config needed.
- **Apothic Compats** extends the affix/gem/gear-set/invasor system to other in-pack mods' gear and loot: AE2 (affixed loot + gear sets), The Aether (affixes, gear sets, invaders, affixable dart shooters), Cataclysm (affixed loot + gear sets + invaders), Create (affixable potato cannons), Curios (per-slot loot categories + special affixed curio chest loot), Deep Aether + Deeper and Darker + Twilight Forest (affixes, gear sets, invaders), Farmer's Delight (affixed loot), Mekanism Tools (affixed loot + gear sets), and Supplementaries (enchancing stats on candle holders). Each bridge is a datapack that activates only when its target mod is present — so only the 11 in-pack targets take effect; the rest stay inert.

This widens the Apotheosis loot chase across tech, food, and exploration without adding any standalone content that conflicts with the pack’s own progression.

Immersive Armors adds 40+ unique armor sets with custom effects: Wooden (arrow/explosion protection), Skeleton (speed, Wither variant, skeleton neutrality), Divine (dyeable, blocks 1 attack/min), Heavy (high armor, anti-knockback, slow), Slime (bounce attackers), Warrior (berserk — more damage at low HP), Robe (dyeable wool, fire thorns, fireproof), Steampunk (hostile radar, enhanced UI, double jump, no fall damage), and Prismarine (spikes, depth strider). All sets are enchantable, repairable, and dyeable where applicable. Recipes in vanilla recipe book and EMI. **Apotheosis affixes and gems apply to all Immersive Armors pieces.** Mekanism Tools fills the material-tier progression (Osmium→Bronze→Steel→MekaSuit); Immersive Armors fills the unique-effect side grade slot — different armor sets for different situations rather than linear upgrades.

Skills Mastery Reimagined — Character Growth

71 nodes. 36 unlock slots. Every point is a commitment — you can’t max everything, so you build a character, not a spreadsheet.

Design philosophy: Unlike simple “numbers go up” trees, Skills Mastery Reimagined uses tradeoff design. Some unique nodes carry **negative effects** alongside their power (e.g., more damage but less speed, higher HP but slower regen). Each unlock plays a unique sound effect, giving satisfying feedback. The tree pattern is wide and branching — no linear “must-take” paths.

Getting started:

1. Controls → Pufferfish’s Skills → set a key (K recommended)
2. Press key → skill tree GUI opens
3. Browse the tree. Early nodes are cheap. Plan your path — you can only unlock 36 of 71 nodes.

XP Sources — XP comes from diverse activities, not just combat:

Activity	XP Value	Chill-friendly
Killing mobs	Standard	
Mining ore blocks	2 per block	✓
Catching fish	5 per catch	✓
Enchanting books	2 × enchanting level cost	✓
Killing Evoker	100	
Killing Ravager	150	
Killing Elder Guardian	200	

Activity		XP Value		Chill-friendly
Killing Wither		400		
Killing Warden		500		
Killing Ender Dragon		600		

Unique attributes: Skills Mastery Reimagined unlocks attributes you won't find in vanilla or Apotheosis — Gravity (lower = floaty jumps), Scale (character size), Burning Time, Sneak Speed, Water Efficiency, Oxygen Bonus, Crossbow Projectile Speed, and weapon-specific damage (Mace, Trident, Sword, Axe). These change *how you interact with the world*, not just how hard you hit.

Weapon synergy: Weapon-specific nodes (Sword Damage, Axe Damage, Mace Damage, Trident Damage) create a direct feedback loop with Simply Swords and Simply More drops. Find a cool unique sword? There's a node that amplifies it. The skill tree and your loot chase feed each other.

Pufferfish's Skills: The framework powering the tree — handles XP tracking and attribute modification behind the scenes. Auto-installed dependency.

Pufferfish's Attributes: Custom attribute registry required by Skills Mastery Reimagined. Adds 35+ dynamic attributes (Stamina, Fortune, Melee/Ranged/Magic Damage, Life Steal, Stealth, Jump, Mining Speed, Sprinting Speed, etc.). Auto-installed dependency — no separate config needed.

Tip: Set the keybind and start earning XP from minute 1. Fishing and enchanting reward chill sessions — you progress even when you're just relaxing at home.

Simply Swords — Unique Weapons

60+ unique weapons with passive and activated abilities. Drop from mobs (~2% chance), chests, and bosses.

Category	Examples		Playstyle	
Heavy	Greathammer, Greatsword, Glaive		Slow, high damage, knockback. Two-handed	
Medium	Longsword, Katana, Claymore		Balanced speed/damage. Versatile	
Light	Rapier, Dagger, Sai, Cutlass		Fast, low damage per hit. Armor penetration on rapiers	
Polearms	Spear, Halberd, Pike		Extended reach (2-3 blocks). Throwable spears	
Special	Scythe (AoE), Soul Harvester (lifesteal)		Unique mechanics	

Weapon abilities (hold [Ctrl](#) while hovering to see):

Ability	Effect	Found on
Whirlwind	Spin attack, hits all nearby	Greathammer, Greatsword
Impale	Charge + release for piercing thrust	Spear, Pike
Backstab	+300% damage from behind	Dagger, Sai
Lifesteal	Heal 10% of damage dealt	Soul Harvester, Scythe
Execute	+50% damage to targets below 30% HP	Axe variants

Finding weapons:

- Mob drops: ~2% chance. Zombies = common, skeletons = ranged-adjacent
- Chests: Dungeon/mineshaft/YUNG's structure chests have higher rates
- Trading: Some from weaponsmith villagers
- Bosses: Higher guaranteed drops from Twilight Forest and Cataclysm bosses

Too Many Bows — Unique Ranged Weapons

30+ unique bows with special abilities and custom attributes.

How they work: Each bow has a unique passive or activated ability. Check EMI for recipes and stats. Some shoot multiple arrows, some inflict status effects, some have special draw mechanics.

Finding them: Dungeon chests, mob drops, and boss loot. Combine with Apotheosis gem socketing for even more damage.

Integration: Simply Swords + Simply More covers melee (60+ unique weapons across 20+ types). Too Many Bows covers ranged. Better Combat (Wave 5) provides appropriate animations for all three.

Simply More — Exotic Weapon Types

A Simply Swords addon that adds 10 new weapon types — not just more swords, but entirely new combat styles. Weapons are balanced around Better Combat (Wave 5) for full animation support.

Weapon Type	Playstyle
Great Katana	Two-handed, wide sweeping arcs
Grandsword	Massive two-hander — disables enemy shields on block
Backhand Blade	Fast, dual-wield friendly, close-range
Lance	Mounted combat specialist — massive boost while riding
Khopesh	Curved blade — bonus against unarmored targets
Dagger	Fastest attack speed, backstab bonus
Pernach	Armor-piercing mace — shreds tough enemies
Quarterstaff	Extended reach blunt weapon, knockback-focused

Weapon Type	Playstyle
Great Spear	Longest reach in the game — hit from 4+ blocks
Deer Horns	Dual-wield fist weapons — rapid flurry strikes

Plus 33 unique weapons with special abilities. All weapons use Simply Swords' unique drop system — loot from mobs, chests, and bosses.

Dependencies: Simply Swords (already in pack). Cloth Config API and Architecture API are already installed from Wave 0.5 — no new mods needed.

Finding them: Same as Simply Swords — mob drops (~2% chance), structure chests, boss loot. EMI shows all recipes and drop sources.

Relics (RPG Series) — Legendary Accessories

Unique accessories for **Curios slots**. You have 6 slots: Ring (2), Amulet (1), Belt (1), Gloves (1), Back (1), Charm (1). Press Curios keybind (**Y** recommended) to open inventory.

Relic	Slot	Ability	Where to find
Monkey Talisman	Charm	Double jump	Vanilla dungeon chests
Holy Water	Charm	Right-click cleanses negative effects	Desert temples, churches
Lucky Coin	Charm	+luck for better loot	Mineshafts, outposts
Captain's Hook	Gloves	Pull entities toward you	Shipwrecks, ocean ruins
Sacred Wardstone	Amulet	Absorbs X damage then breaks	Stronghold chests
Magic Mirror	Back	Teleport to spawn/home	End cities, libraries
Heart of the Void	Ring	+max HP, slow regen	Nether fortresses, bastions
Blazing Feather	Back	+15% movement speed	Jungle temples, pagodas
Drowned Belt	Belt	Underwater speed boost	Ocean monuments, shipwrecks

Stacking: Two rings = two ring relics. Wear multiple different types.

Finding strategy: Use Explorer's Compass → find YUNG's structures → loot them. Relics have higher drop rates in YUNG's structures than vanilla ones.

Enchanting Runes — Socketable Weapon Modifiers

Swappable modifications for weapons/tools. Change them between fights.

How to get:

- Loot chests in any structure
- Craft at **Rune Altar** (4 cobblestone 2×2 + lapis block on top, right-click with rune fragment)
- Low drop chance from hostile mobs

Rune	Effect	Best on
Fire	On-hit fire damage (3 sec)	Fast weapons (daggers, rapiers)
Ice	On-hit slow (2 sec)	Kiting weapons (spears, scythes)
Lightning	Chance to strike lightning	Heavy weapons (single big hits)
Poison	On-hit poison (5 sec)	DoT builds, boss kiting
Lifesteal	Heal 5% of damage dealt	Tank builds, slow weapons
Critical	+10% crit chance	Medium weapons

Using runes:

1. Craft **Rune Pouch** (leather + string + lapis) — stores all runes
2. Place weapon + rune at **Rune Altar** (4 stone + 1 lapis block)
3. Rune applied. To swap: place weapon back with different rune

Tip: Keep 2-3 loadouts. Fire for general combat, Lifesteal for bosses, Ice for kiting.

Apotheosis — Loot & Enchanting Overhaul

The single largest change to how gear feels.

Rarity System

Every item drop rolls with rarity:

Rarity	Affix slots	Color	Source
Common	0	White	Basic mobs, mining
Uncommon	1	Yellow	Hostile mobs, basic chests
Rare	2	Cyan	Dungeon chests, mini-bosses
Epic	3	Purple	Bosses, late-game chests
Mythic	4	Gold	Endgame bosses, Cataclysm

Higher rarities = more gem sockets + better affixes.

Affixes — Random Stats

Affix	Effect	Gear type
Ruthless	+crit damage, +crit chance	Melee weapons
Arcane	+spell damage, +mana	All weapons
Sturdy	+armor, +knockback resist	Armor

Affix	Effect	Gear type
Nimble	+movement speed, +attack speed	Boots, leggings
Plentiful	+mining speed, +fortune	Tools
Titanic	+max HP, +natural regen	Chestplate

Tip: Check every drop. A Common helmet with a good affix can beat an Epic with useless stats. Salvage unwanted gear at a salvaging table for gem dust.

Gem System

Socketable permanent stat boosts:

Tier	How	Effect
Rough	Mob/chest drops	Basic stat boost
Cut	Rough + Gem Cutting Station + dust	+50% stat value
Flawless	2 Cut gems of same type + station	+100% stat value

Gem Cutting Station: Craft (check EMI), place rough gem + dust → cut gem. Dust from breaking unwanted cut gems.

Gem Socketing Table: Craft (check EMI), place item + gem in socket slot. Limited sockets per gear piece — expand with socketing tools.

Enchanting Overhaul

- **Higher level cap:** Sharpness VIII, Protection VIII, Fortune V, Looting V possible
- **Tiered bookshelves:** Hell, Seashelf, Endshelf enable higher enchantment levels
- **Library of Alexandria:** Stores all enchanted books in one block
- **30+ new enchantments:** Check EMI for full list

Boss Affixes

Boss mobs spawn with modifiers under their health bar:

Modifier	Effect	Strategy
Tough	+50% HP, damage reduction	High-damage weapons + Lifesteal
Blinding	Blinds on hit	Blindness-resistant gear or milk
Clumsy	Disarms on hit	Speed — don't get hit
Plagued	Poison/wither on hit	Antidotes, regeneration
Huge	2× size, more knockback	Stay mobile
Chaotic	Random effects per hit	Unpredictable — bring all potions

Tip: Boss affixes are the main source of Mythic gear. Hunt affixed bosses for endgame upgrades.

Spawner Module

Silk-touch a spawner with a pickaxe → modify it (entity type, spawn delay, player range, min/max entities). Right-click placed spawner with empty hand to fine-tune. Essential for XP farms and mob automation.

Deadly Module

Enabled. Adds telegraphed death-traps to the world (spikes, poison darts, instakill rooms, lava traps). Deadliest rooms hold the best loot. Traps are avoidable with caution. GraveStone Mod preserves your items if you trigger one.

Session Plan — Wave 3

Session	Focus	What to Do
Session 1 — Skill tree	Set keybind (P). Spend first points on Strength or Vitality	Permanent progression started
Session 2 — First affix loot	Check every gear drop. Look for Rare+ with useful affixes	Learn the loot system
Session 3 — Weapon hunt	Explorer's Compass → YUNG's dungeon → loot for Simply Swords weapon. Try Rapier (fast) or Greathammer (AoE)	First unique weapon
Session 4 — Gem cutting	Collect rough gems. Build Gem Cutting Station. Cut Rough → Cut. Socket into gear	Permanent gear upgrade
Session 5 — Relic hunting	Monkey Talisman (dungeons) + Lucky Coin (mineshafts)	Accessory slots filled
Session 6 — Boss affixes	Search for affixed bosses (colored name above health bar). Best source of Mythic gear	Endgame gear progression
Session 7 — Rune loadout	Craft Rune Pouch + Rune Altar. Apply Fire rune to fast weapon, Lifesteal to slow weapon	Swappable combat modifiers

Tip: Wave 3 gear changes how every other wave feels. A Rare+ weapon with the right affix makes Twilight Forest and Cataclysm bosses dramatically easier. Invest in gear before tackling tough content.

Wave 4 — Food, Farming & Comfort

Mod	Role
Farmer's Delight 1.3.2	Cooking hub — cutting board, cooking pot, stove, new crops
Extra Mod Integrations	Adds compat recipes for 50+ mods — Farmer's Delight, Create, Mekanism, AE2
Brewin' And Chewin' 4.5.0	Drinks and fermentation — beer, wine, mead, whiskey
Storage Delight 1.3.0	Kitchen furniture — drawers, cabinets, shelves for ingredient storage
Crate Delight v26.07.01	Bulk ingredient storage — crates (apples, berries, eggs, fish) and bags (cocoa, sugar, gunpowder)
More Delight 26.04.18a+1.21-neoforge	Extra meals, sandwiches, toasts, and knife variants
My Nether's Delight 1.10.2	Nether-themed food ingredients and recipes
Ocean's Delight 1.0.4	Ocean-themed seafood dishes
Ender's Delight v1.3.0	End-themed food for Farmer's Delight (moved from Wave 2)
Twilight's Flavors & Delight 3.2.2	Twilight Forest × Farmer's Delight crossover — 35+ FD-style TF foods, 4 TF knives, maze stove, 5 potion effects, trophy dismantling
Neo Bee Fix	Fixes bee AI bugs — bees leave hives, don't fall from sky, no NW wander bias
Comfortable Campfires	Campfires grant configurable status effects — sit by the fire, get a buff
Smarter Farmers	Farmer villagers replant correctly — incl. modded seeds (FD crops via tags)
Create: Café	Coffee, tea & boba drinks via Create machines — roasting, grinding, brewing
Starcatcher	Fishing overhaul — 100+ fish, skill minigame, trophies, tournaments, catalogue
Starcatcher's Delight v3.0.1	Starcatcher × Farmer's Delight crossover — fish fillets, seafood dishes, sushi

Mod	Role
Better Fishtanks	Functional aquariums — fish bags, decorations, feeding system, Create/AE2 compat
Spice of Life: Carrot Edition solcarrot-1.21.1-1.16.6	Food-variety reward — eating new foods permanently raises max health up to a cap. No punishment, only reward

Interdependency note — Better Fishtanks + Hybrid Aquatic (Wave 2): Better Fishtanks can house any fish mod’s creatures. Use Fish Bags to catch Hybrid Aquatic mobs and place them in tanks as living displays.

Interdependency note — Twilight’s Flavors & Delight + Twilight Forest (Wave 2) + Farmer’s Delight: Bridges TF and Farmer’s Delight — TF mob drops and foraged items cook into 35+ FD-style meals, 4 TF-material knives with unique abilities, Maze Stove accelerates cooking in TF biomes, boss trophies dismantle into precious loot.

Spice of Life: Carrot Edition — the *gentle* variant of food variety: each distinct food eaten once permanently raises your max health (up to a configurable cap of hearts), and it never *reduces* nutrition for repeat meals (unlike the original Spice of Life). This pairs directly with the pack’s Farmer’s Delight ecosystem — 100+ distinct meals = steady, voluntary max-HP growth feeding the power-fantasy pillar, with zero chill penalty for eating the same thing. **Config lives in [saves/<world>/serverconfig/](#) (per-world), not the tracked global [config/](#) folder** — tune `solcarrot-server.toml` (milestone cap, heart counts) after first world launch; set the cap so the bonus stops short of feeling mandatory.

Farmer’s Delight — Cooking Hub

New Crops

Crop	Grows on	Use
Tomato	Farmland. Bush grows, pick when red	Soups, sandwiches, salads
Cabbage	Farmland. Full head when ready	Coleslaw, cabbage wraps
Onion	Farmland. Pick when shoots show	Savory dishes, soups
Rice	Shallow water (1 block deep on dirt/mud)	Hearty meals, sushi

Wild crops: Break wild variants in the world for seeds:

- Wild tomatoes — plains/forests
- Wild cabbage — beaches

- Wild onions — taiga
- Wild rice — swamps

Cutting Board

Craft: 1 knife + 2 planks (horizontal row). Place on ground, right-click to open GUI.

Place ingredient in left slot → processed output in right slot.

Input	Output	Use
Cooked meat	Meat strips	Sandwiches, salads
Cabbage	Cabbage leaves	Wraps, coleslaw
Bread	Bread slices	Toast, sandwiches
Carrot	Carrot slices	Salads, soups
Apple	Apple slices	Desserts, snacks

Tip: Knife in off-hand while chopping lasts longer (uses knife durability). Iron knife = dozens of chops.

Cooking Pot

Craft: 2 iron + 2 planks + 1 bowl. Place over a heat source:

- **Stove** (fastest) — 4 bricks + 4 iron nuggets
- **Campfire** (medium) — 3 logs + 3 sticks + 1 coal
- **Lava** (slowest) — pot must sit directly above lava source

Cooking process:

1. Right-click pot → open GUI
2. Fill water slot (bucket or bottle)
3. Add up to 6 ingredients in any order
4. Wait for progress bar
5. Collect meal — most give **6 servings** with bowl stacking

Meal categories:

- **Soups** — Tomato soup, beef stew. Good saturation, easy ingredients
- **Sandwiches** — Chicken sandwich, BLT. On-the-go eating
- **Salads** — Mixed salad, fruit salad. Good for hot biomes
- **Pasta** — Pasta with meat sauce. Filling, complex ingredients
- **Rice meals** — Fried rice. Requires rice farming

Kitchen Storage

Craft **oak/pantry cabinets** (check EMI — varies by wood). Function as chests with kitchen aesthetics.

Tip: Set up kitchen near your AE2 network. Cook manually or stock ingredients in ME drives for autocrafting.

Brewin' And Chewin' — Drinks & Fermentation

Alcohol brewing with aging mechanics. Grow hops, barley, grapes on farmland.

Brewing Station

Craft (check EMI — barrel-like recipe). Right-click to open GUI.

1. Add **water** + main ingredient:
 - Barley → beer base (wort)
 - Grapes → wine base (must)
 - Honey → mead base
 - Potatoes → spirits base (mash)
2. Wait for fermentation
3. Collect **fermented mixture**

Aging Barrel

Craft (check EMI — wooden barrel). Place fermented mixture inside, right-click to seal.

Ages one stage per in-game day: Young → Matured → Aged → Vintage (4 stages). Each stage increases status effect potency and duration.

Optimal: Let everything reach **Aged** (stage 3). Push to Vintage (stage 4) only for special occasions.

Drink	Matured effect	Vintage effect
Beer	Resistance I (2 min)	Resistance II (5 min)
Wine	Haste I (2 min)	Haste II (4 min)
Mead	Speed I (2 min)	Speed II (4 min)
Whiskey	Strength I (30 sec)	Strength II (1 min)
Vodka	Night Vision (3 min)	Night Vision + Fire Resist (5 min)

Practical use: Brew a batch of each, age to Matured/Aged, store in kitchen. Pop a drink before boss fights for free buffs. Mead stacks with Swiftess potions.

Bottle return: Empty bottles are returned after drinking. Reuse at brewing station.

Storage Delight — Kitchen Furniture

Decorative kitchen storage that works as functional inventory:

Block	Slots	Use
Cabinet	27	Counter-height storage
Drawer	18	Fits under counters
Shelf	9	Wall-mounted spice display

Tip: Use for early-game ingredient storage before AE2. Later, place an ME Storage Bus on a cabinet to bridge into your digital network.

Crate Delight — Ingredient Storage

Bulk storage for Farmer’s Delight ingredients by axperty (same author as Storage Delight). Crates and bags keep your kitchen organized and look great.

Type	Examples	Capacity
Crates	Apples, Sweet Berries, Eggs, Raw Salmon, Raw Cod, Glow Berries, Golden Apples	Bulk single-item storage
Bags	Cocoa Beans, Sugar, Gunpowder	Bulk single-item storage

Why it’s useful: Instead of a barrel full of mixed ingredients, you have a Crate of Apples next to your cooking station and a Bag of Sugar near your brewing setup. Thematic, organized, and satisfying to look at.

Integration: Same author as Storage Delight — identical art style, zero compatibility issues. All crates and bags are accessible via Create pipes and AE2 Storage Bus. Complements Storage Delight’s furniture (cabinets/drawers/shelves) — Crate Delight handles bulk ingredient storage, Storage Delight handles kitchen furniture. Both are intentionally outclassed by AE2 in Phase 2 but remain useful for localized micro-storage near cooking and brewing stations.

More Delight — Extra Meals & Knives

- **Extra meals:** Sandwiches, toasts, stuffed mushrooms
- **Knife variants:** Diamond and netherite knives (much longer lasting than iron)
- **New cooking pot recipes:** Stews, soups, pasta dishes

All recipes visible in EMI. Uses existing cutting board and cooking pot.

My Nether’s Delight — Nether Food

Nether-themed ingredients and recipes:

- **Nether crops:** Grow in soul sand or warped/crimson nylium
- **Ingredients:** Glowstone berries, soul berries, nether-root vegetables
- **Recipes:** Spicy stews (Fire Resistance), soul-force snacks (Night Vision)

Tip: Set up a Nether farm for combat-relevant status effects. Glow berry pie gives Haste II — useful before mining expeditions.

Ocean's Delight — Seafood

Seafood dishes using ocean mob drops:

- **Ingredients:** Squid tentacles, fish fillets, seaweed, turtle eggs
- **Recipes:** Seafood stew, grilled fish, kelp salads, sushi rolls
- **Effects:** Water Breathing + Dolphin's Grace — perfect for ocean monument raids

Tip: Build a fish farm or farm kelp for steady supply. Combined with My Nether's Delight, prepare meals for any environment.

Ender's Delight — End Food

End-themed food for post-Dragon exploration:

- **Ingredients:** Chorus fruit variants, end-themed crops
- **Recipes:** End stews and snacks
- **Effects:** Status boosts useful for End dimension exploration

Integration: End ingredients cook into Ender's Delight meals. Plan an End farming outpost with Farmer's Delight crops.

Neo Bee Fix

Fixes bee AI — bees properly leave hives, don't fall from sky, no northwest wander bias. No config. Install and forget.

Integration: Bees produce honey for Brewin' And Chewin' mead. Farm bees near your kitchen for renewable brewing ingredients.

Smarter Farmers — Villagers Replant

Fixes and upgrades villager farming: farmers replant the **correct** seed (no more wheat seeds on carrot rows) and can use **modded seeds** — Farmer's Delight crops included, via tag-based detection. A `harvest_blacklist` tag lets you exclude crops from auto-replanting.

Integration: Keeps village farms and FD crop plots self-sustaining — less manual replanting, more chill living. Works alongside MineColonies (separate villager system — no overlap). Requires Moonlight Lib (already in pack, Wave 6).

Create: Café — The Cozy Drinks Track

Coffee, tea, and boba milk tea made through **Create machines** — beans and leaves go through milling, crushing, and mixing processing (1.4.3, Jan 2026). Coffee is the one drink category Brewin' And Chewin' doesn't cover (B&C = fermentation/alcohol), so this is a second, non-overlapping drinks track that runs on Create power rather than the FD kitchen.

Integration: Feeds Create automation chains — roast/grind/brew as mechanical processing steps, pipeable and deployer-friendly. Phase 1 accessible (water wheel era). Drinks stack alongside FD meals and B&C drinks for buff-food variety. Only dependency is **Create** (already in pack) — zero new mods. [VERIFY] at test launch: drink effect strength (overpowered check), Create version compat, no runtime conflict with B&C.

Starcatcher — Fishing Overhaul

A complete fishing overhaul with 100+ fish, skill-based minigame, progression system, and tournaments.

Core mechanics:

Feature	What it does
Fishing Minigame	Skill-based reel-in — timing matters, not just patience
Catalogue	Records every catch — biome, weather, time conditions documented
Trophies	Mount your best catches as wall decorations
Tackle Box	Store hooks, bobbers, baits, and fish — portable fishing station
Aquariums	Multi-block fish tanks — place caught fish inside, scaled by size/weight
Tournaments	Compete against friends for biggest catch
Selling Bin	Sell fish for emeralds — price based on size, weight, rarity
Cosmetics	Fisherman hats, rod skins — found in shipwrecks and treasure

Getting started:

1. Craft a **fishing rod** (3 sticks + 2 string) — Starcatcher replaces vanilla fishing
2. Cast near water → minigame triggers on bite
3. Complete the minigame → fish caught, added to catalogue
4. Craft a **Tackle Box** (check EMI) → store fish and accessories
5. Find **baits** in loot chests or craft them → attract specific fish types

Fish conditions: Each fish has specific requirements:

Condition	Examples
Biome	Ocean fish, river fish, swamp fish
Weather	Rain-only, clear-only, storm-only
Time of day	Day fish, night fish, dawn/dusk
Elevation	Surface vs. deep water

Golden Fish: Rare variants of every fish — higher value, trophy-worthy. Hunt these for the ultimate collection.

Configuration: In `config/starcatcher-server.toml`, set `giveGuideOnFirstJoin = false` to prevent the fishing guide book from being handed to the player on first login. The guide is still craftable (check EMI for recipe) — just not auto-given.

Integration with Hybrid Aquatic (Wave 2): Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

Better Fishtanks — Aquariums

Functional aquariums for displaying your catches. Build the tank of your dreams.

Core blocks:

Block	Function
Fishtank	Glass tank — house fish, configurable water levels
Aquarist Table	Crafting station for tanks, decorations, and supplies
Fish Bag	Catch and transport fish — right-click fish in world, right-click tank to place
Food Container	Store fish food in bulk — flakes, kelp, shrimp, pellets
Decorations	Coral, shipwrecks, stones, branches, towers, kelp — 30+ variants

Getting started:

1. Craft an **Aquarist Table** (check EMI — iron + glass + planks)
2. Craft a **Fishtank** (glass + iron ingots) — place it down
3. Craft a **Fish Bag** (leather + string) → right-click any fish in the world to catch it
4. Right-click the Fishtank with a filled Fish Bag → fish placed inside
5. Add **decorations** from the Aquarist Table → customize the tank

Fish Eggs: Tanks passively produce fish eggs in 6 rarity tiers (Common → Legendary). Crack them in the Aquarist Table for loot.

Aquarist Villager: A new villager profession. Trades fish supplies and rare eggs. Assign by placing an Aquarist Table near an unemployed villager.

Tech integration (Create, AE2):

- Fishtank and Aquarist Table inventories are **fully accessible** via Create pipes
- Use fluid pipes to fill tanks with water automatically
- Automate fish feeding with Create mechanical systems
- Connect to AE2 network for remote inventory access

Config options:

Setting	Default	What it does
Require Food	ON	Fish need feeding to survive — disable for pure aesthetics
Fish Bag Captures	Normal fish only	Enable to capture Guardians, Drowned, Whales
Rendering Mode	Vanilla Waterlogging	Toggle experimental entity rendering for water

Integration with Starcatcher: Better Fishtanks displays what Starcatcher catches. Starcatcher provides the fishing gameplay; Better Fishtanks provides the aquarium showcase. Different systems, complementary purposes.

Starcatcher's Delight — Seafood Cooking

Starcatcher × Farmer's Delight crossover by wdiscute (same author as Starcatcher). Every fish you catch becomes a cooking ingredient.

Food Type	Ingredients	Notes
Fish Fillets	Any Starcatcher fish → Cutting Board	Fish quality (Common → Legendary) determines fillet quality
Sushi	Fish fillet + rice + seaweed	Rice from Farmer's Delight, seaweed from oceans
Fish Stew	Fish fillet + FD vegetables + Cooking Pot	Hearty meal with saturation buff
Fried Fish	Fish fillet + FD oil + stove	Quick crispy meal
Legendary Dishes	Legendary fillet only	Always grants absorption hearts — save these for boss fights

Quality system: The rarity of the fish you use determines the quality of the food. A legendary fish is cut into legendary fillets, which cook into legendary dishes. Each food has unique hand-written descriptions that change based on quality tier.

Getting started:

1. Catch any fish via Starcatcher's minigame
2. Use a Farmer's Delight Cutting Board with a knife to produce fillets
3. Combine fillets with FD ingredients for meals
4. Hunt for legendary fish → legendary dishes with absorption hearts → boss fight food

Integration: All Starcatcher's Delight recipes appear in EMI. Fillets and dishes can be stored in AE2. Quality-based dishes stack separately (Common Sushi ≠ Legendary Sushi). Zero config needed — install and fish.

Wave 4.5 — Quests

Authored quest book with chapters covering every major mod in the pack. Guides progression without forcing it — quests are optional breadcrumbs, not mandatory checklists.

Questing Foundation

Mod	Role
FTB Library	Core library for all FTB mods — configuration, GUI framework, team management
FTB Quests	Quest book framework — chapters, task types, rewards, in-game visual editor
FTB Teams	Quest progress tracker — in single-player, you're automatically a team of 1
KubeJS	JavaScript scripting engine — recipe modification, worldgen tweaks, event hooks
FTB XMod Compat	Bridges FTB Quests with KubeJS, JEI/REI compat layer (works with EMI)

Tip — hide the FTB Teams button (single-player): the Teams/Chunks/Quests buttons on the inventory screen are rendered by **FTB Library**, not FTB Teams — and they're pure GUI shortcuts (quest progress tracks fine without them). To hide just the Teams buttons, edit `config/ftblibrary-client.toml`:

```
[sidebar.buttons]
"ftbteams:my_team" = { enabled = false, x = 0, y = 0 }
"ftbteams:team_lives" = { enabled = false, x = 0, y = 0 }
```

To remove the whole sidebar instead, set `[sidebar] enabled = false`. Either way the quest book still opens with the - key.

Quest Addons

Mod	Role
MineColonies Questline	Pre-written 40+ quest chain covering MineColonies from Town Hall to endgame colony

MineColonies Questline covers colony progression from first Town Hall through Industrial Revolution research. The pack's own quest book is pre-authored too — 10 chapters, 211 quests covering every major mod (see *Installing the Quest Book* below). No in-game editing required.

How It Works

1. Press the quest key (configurable, default -) to open the quest book
2. Chapters organized by mod: **Mekanism, Create, Steam 'n' Rails, Electric Power, Combat, AE2, Apotheosis, Farmer's Delight, Northstar, MineColonies**
3. Quests auto-detect progress — craft an item, enter a dimension, kill a boss → quest completes
4. Rewards include items, XP, and knowledge — some powerful items are quest-gated to soft-enforce progression
5. The quest book is **always available** but **never mandatory** — ignore it when you want to chill

Installing the Quest Book

The pre-authored quest book lives in the repo at [minecraft/config/ftbquests/quests/](#). To load it in a Minecraft instance:

1. Copy the `quests` folder into your instance's `config/ftbquests/` folder (merge/overwrite the existing `quests` folder)
2. Start the game — or, if it's already running, run `/ftbquests reload`
3. Press the quest key (default -) to open the book

Updating after a change: re-copy the `quests` folder and run `/ftbquests reload`. The book is plain data files (SNBT) — there's nothing to edit in-game.

Pillar Alignment

- **Power fantasy:** Quests celebrate milestones. “You built a fusion reactor!” with a meaningful reward feels earned.
- **Easy curve:** New players follow quest breadcrumbs. “Craft enrichment chamber → craft crusher → 2x ore processing” teaches Mekanism step by step.
- **200h+ content:** Quests surface hidden content. “Have you visited the Aether?” “Try sequenced crafting with Create.” Players discover systems they might miss.
- **Chill living:** The quest book key is configurable. There's no on-screen HUD element. No time-limited quests. No penalties for ignoring it. The book waits for you — not the other way around.

Chapter Coverage

Chapter	Quests
Mekanism	37
Create	35
Steam 'n' Rails	9
Electric Power	13
Combat	14

Chapter	Quests
AE2	22
Apotheosis	12
Farmer's Delight	52
Northstar	18
MineColonies (pre-written)	~40

Total: 212 authored quests + ~40 pre-written colony quests. The pack is fully playable without quests; they're enhancement, not foundation.

Session Plan — Wave 4 & 4.5

Session	Focus	What to Build
Session 1 — Kitchen tools	Cutting board + cooking pot + stove. Find wild crop seeds	Functional kitchen
Session 2 — Farm	Plant tomato, cabbage, onion, rice. Cook first meal (tomato soup is easiest)	Sustainable ingredients
Session 3 — Brewing	Find barley/grapes/honey. Build brewing station. Start first batch. Place aging barrel	First drink fermenting
Session 4 — Kitchen expansion	Storage Delight cabinets/drawers. Try More Delight recipes. Collect Nether ingredients	Full kitchen
Session 5 — Questing	Open quest book (- key). Browse MineColonies questline. Complete "Town Hall" quest	Questing unlocked
Ongoing	Age drinks for boss buffs. Explore oceans for seafood. Stock AE2 with ingredients for autocrafting	Buff food always ready

Wave 5 — Combat & Mobs

Mod	Role
Better Combat	Melee combat animations — weapon reach, sweeping, attack speed
L_Ender's Cataclysm	Epic boss fights — Leviathan, Netherite Monstrosity, Ender Guardian (CurseForge, NeoForge 1.21.1)
Enchantment Descriptions	Shows enchantment effects in tooltips
Create Big Cannons	Artillery engineering — siege cannons for colony defense and spectacle
Create Big Cannons: Advanced Technologies	CBC addon — twin/heavy autocannons, rocket pods & rails, muzzle brakes, silencers, rifled barrels
Torchmaster	Territory pacification — Mega Torch suppresses hostile spawns in a radius; each torch gated behind a Cataclysm boss conquest (see below)

Apotheosis also adds boss modifiers and elite mob affixes — partially covers what Champions would have done.

Better Combat — Weapon Feel

Replaces Minecraft's single swing with weapon-specific movesets:

Weapon	Speed	Reach	Animation	Best for
Daggers	Very fast	2 blocks	Quick stab	Debuff stacking, fast crit
Swords	Fast	3 blocks	Horizontal arc	General combat
Rapiers	Fast	3.5 blocks	Thrust (ignore shields)	Shield-breaking
Greatswords	Slow	4 blocks	Wide sweep	Group clearing
Axes	Medium	3 blocks	Overhead chop	Armor penetration
Spears	Medium	4.5 blocks	Forward thrust	Kiting
Hammers	Slow	3.5 blocks	Overhead slam	Knockback, stun
Greathammers	Very slow	4 blocks	Ground pound	AoE knockback

Config: Mods screen → Better Combat → Config. Tweak reach, sweep radius, attack speed, visual effects per weapon type.

Integration: Simply Swords weapons have built-in Better Combat data files — each category uses correct animation, reach, and speed automatically. Too Many Bows and Simply More also integrate.

Enchantment Descriptions

Hover over any enchantment in tooltip → detailed description of what it does.
Works in enchanting table GUI, on items, and in books. No config.

L_Ender's Cataclysm — Epic Boss Fights

Hand-animated bosses with unique mechanics. **Opt-in** — you must build a summoning structure or find a specific location.

Boss	Location	How to Summon	HP	Recommended Gear
Leviathan	Deep ocean (Y<40)	Prismarine + sea lantern altar + Prismarine Shard	~200	Diamond, Depth Strider, door trick
Netherite Monstrosity	Nether wastes	Enter central arena in nether brick lair	~300	Full diamond/ netherite, fire resistance
Ender Guardian	End (outer islands)	Eye of Ender on pedestal after Dragon	~400	Endgame gear, Elytra + Slow Falling
Ancient Remnant	Desert	Enter summoning room in large pyramid, insert Nether Star	~350	Blast protection, blocks to pillar
Ignite	Nether (basalt deltas)	5×5 hollow obsidian sphere + Blaze Rod center, wait 10 sec	~250	Fire resistance, water buckets

General strategy:

1. Scout the arena first
2. Bring food buffs (Farmer's Delight + Brewin' And Chewin' stack)
3. Apply runes (Lifesteal or Ice)
4. Set a Waystone nearby
5. AE2 wireless terminal for mid-fight resupply

Leviathan: Phase 1 — dodge charges, attack body, jump shockwaves. Phase 2 (50%) — avoid whirlpools, attack head between spawns. Drops Leviathan Blade (trident-like water weapon).

Netherite Monstrosity: Phase 1 — attack legs to slow. Phase 2 (60%) — enrages, faster attacks + fire breath. Sprint behind pillars. Drops Monstrosity tools (efficient netherite-level mining).

Ender Guardian: Phase 1 — dodge tracking energy bolts, hit wall crystals to stun. Phase 2 (50%) — teleports, summons endermites, break all crystals to lower shield. Phase 3 (20%) — energy nova, pillar up to avoid ground AOE's. Drops high-damage weapons + teleportation gear.

Ancient Remnant: Phase 1 — sandstorm blinds, attack during cooldown. Phase 2 (40%) — sinks into sand, pops up elsewhere, shoot exposed weak point. Drops tools with unique enchantability.

Ignite: Phase 1 — fireball barrage, block with shield or dodge behind pillars. Phase 2 (40%) — absorbs lava from arena, stay on high ground, attack when emerging. Drops fire weapons (permanent fire aspect).

Create Big Cannons — Artillery Engineering

Functional cannons built from Create materials, using rotational power for assembly and gunpowder for ammo.

Getting started:

1. Craft cannon components: barrel (bronze/steel), mount (Create materials), charger
2. Build: place mount, attach barrel, connect charger
3. Arm: insert powder charges + cannonballs into GUI
4. Fire: right-click or redstone signal

Ammo	Effect	Use
Solid cannonball	High damage, block-breaking explosive	Siege, defense
Canister shot	Spread of small projectiles	Anti-personnel
Grape shot	Medium spread	Crowd control
Incendiary	Fire on impact + lingering fire	Area denial
Smoke shell	Smoke screen	Concealment

Integration: Gated behind bronze/steel (Create progression) + gunpowder. Mount on colony walls for defense. Redstone-controlled pulse extender for automatic salvo fire.

Dependency: Ritchie's Projectile Library (auto-installed) — provides custom trajectories and hit detection. No separate config.

Create Big Cannons: Advanced Technologies — Bigger Guns

An addon for Create Big Cannons adding 110 new cannon blocks and new ammunition, using the same casting/assembly workflow as base CBC.

Addition	What it does
Muzzle Brake	Reduces cannon recoil
Silencer / Suppressor	Reduces noise and recoil
Fume Extractor	Hides the muzzle smoke cloud — good for concealed emplacements
Rifled Barrel	Tighter spread than a normal barrel
Twin Autocannon	Double-barreled rapid fire (vertical variant can be hidden in leaves)
Heavy Autocannon	Bigger-caliber autocannon for heavier rounds
Rocket Pod / Rocket Rail	Launches rockets; the medium rocket rail handles four medium rockets
New ammunition	Autocannon & heavy autocannon cartridges/caseless rounds, rockets (AP / HE / HEI / Flak), medium rockets (AP / HE / HEAT / HEF), fluid shells, grapeshot

All tiers (bronze → cast iron → steel → nethersteel) follow the standard CBC casting workflow with cast moulds. No config required.

Dependency: Create Big Cannons (above) + Create. Install the 1.21.1 NeoForge build (0.1.4c, CurseForge/Modrinth).

Torchmaster — Conquering the Land

The **Mega Torch** suppresses hostile mob spawning in a 64-block radius — but each torch is **earned**: crafting requires an Ignitium Ingot from a Cataclysm boss (KubeJS gate, [kubejs/server_scripts/wave5_torchmaster.js](#)):

Item	Gated Recipe	Gate
Mega Torch	Ignitium Ingot (Cataclysm Ignite drop) + Create brass ingots + blaze rods	Phase 2 — first Cataclysm boss + brass age

The conquest loop: scout a hostile region by air/ship/train → clear its structures → defeat the Cataclysm boss for the Ignitium Ingot → craft the Mega Torch → plant it. The area is permanently peaceful. Conquest is renewable — Cataclysm arenas re-summon bosses (Eye item), so there's no world exhaustion.

Config (mod defaults — no override needed; `config/torchmaster-common.toml` is generated on first launch):

Setting	Default	Notes
<code>megaTorchRadius</code>	64	Cube $\pm 64 \rightarrow 129 \times 129 \times 129$ zone: village + farms + perimeter
<code>feralFlareRadius</code>	16	Cube $\pm 16 \rightarrow 32 \times 32 \times 32$ zone, capped at 255 invisible lights
<code>blockOnlyNaturalSpawns</code>	<code>true</code>	Spawner + structure spawns exempt — YUNG's dungeons and Cataclysm content untouched

Why these values, and why not bigger:

- **Mega Torch 64** — the radius is a *cube* (each direction), so 64 covers a village plus its hinterland. Each torch costs one Cataclysm boss kill (KubeJS gate), and the conquest loop is designed to repeat (arenas re-summon bosses). A 96+ cube would let a single kill pacify an entire region and collapse the loop; the original 48 draft underpays the conquest.
- **Feral Flare 16** — it's a *clearing aid*, not a territory tool: it sprinkles invisible light (level 15) in its cube until it hits the 255-light hardcap. At 16 the lights sit ~ 5 blocks apart, keeping block light ≥ 10 ; at 24 they thin to ~ 7.5 -block spacing and dark gaps reappear — worse than 16. Raising the hardcap to compensate trips the mod's own world-corruption warning (badly compressed packet errors).
- **Spawner exemption** — the Mega Torch never blocks `SPAWNER/STRUCTURE` spawns, so YUNG's dungeons and Cataclysm arenas keep their mobs at *any* radius. Torches pacify the surface, never the content.

Session Plan — Wave 5

Session	Focus	What to Do
Session 1 — Combat feel	Better Combat works automatically. Craft/find a Simply Swords weapon. Test daggers (fast stab) vs greatswords (wide sweep). Equip a rune	Learn your weapon
Session 2 — First dungeon	Explorer's Compass $\rightarrow$ YUNG's Better Dungeons:	Dungeon complete

Session		Focus		What to Do	
		clear a dungeon or spider cave			
Session 3 — First Cataclysm boss		Easiest = Leviathan (deep ocean) or Netherite Monstrosity (Nether). Prepare: good weapons, fire resistance, healing food		Epic boss beaten	
Session 4 — Cannons		Craft bronze (zinc + copper in heated mixer). Build Small Cannon. Test fire. Mount near base entrance		Artillery online	
Session 5+ — Boss gauntlet		Remaining Cataclysm bosses. Each drops unique endgame materials. Revisit Twilight Forest bosses with Wave 5 gear		Full endgame gear	
Session 6 — First conquest		Scout a region by plane, clear its structures with guns/artillery, defeat the Cataclysm boss, craft the Mega Torch from its Ignitium Ingot, plant it. The region is yours		First pacified territory	

Tip: Better Combat changes melee fundamentally. Don't skip Session 1 even as a ranged player.

Wave 6 — Building & Decoration

Core Decoration

Mod	Role
Rechiseled	600+ decorative block variants with connected textures
Supplementaries	Vanilla+ decoration — jars, signposts, faucets, weather vanes
Macaw's Bridges	Bridges in all wood types
Macaw's Doors	Vanilla-style doors for every wood
Macaw's Roofs	Roofing blocks
Macaw's Windows	Window variants
Building Wands	Faster large-scale construction
Handcrafted	Furniture — chairs, tables, shelves, curtains
Fetzi's Displays	Item displays — glass cloches, katana stands, wall holders, racks
Laser Bridges & Doors	Laser platforms, bridges, doors, and fences powered by redstone — length scales with signal strength, dye-customizable colors
Diagonal Fences	Connect vanilla fences, fence gates, and walls diagonally — seamless corners

Rechiseled Addons

Mod	Role
Rechiseled: Create	100+ decorative variants of Create blocks + mechanical chisel
Rechiseled: Applied Energistics 2	AE2 autocrafting for Rechiseled blocks via Chiseling Pattern Encoder

Rechiseled — Decorative Block Variants

600+ decorative block variants with connected textures.

How:

1. Craft **chisel** (iron + 2 sticks, check EMI)
2. Right-click with chisel → chiseling screen
3. Place base block in input slot → browse all variants
4. Click variant to preview (cycles through single block, row, 3×3 grid)
5. Take result from output slot
6. **Left-click** placed blocks with chisel to cycle through random variants in-world

Key features:

- **Connected textures** — toggle with bottom-right button
- **Stairs & slabs** — every variant has stair/slab forms
- **No durability** — chisel never breaks

Block types (30+ variants each):

- Stone (bricks, tiles, carved, pillars, panels, runed, mosaic)
- Glass (all colors + connected clear, borders, panels)
- Wood (grain variants, carved panels, ornate borders)
- Wool & Carpet (woven, knitted, cross-stitched)
- Concrete (smooth, paneled, tiles, bordered)
- Nether materials (glowstone, nether bricks, blackstone)

Addons:

- **Rechiseled: Create** — 100+ decorative Create block variants + mechanical chisel
- **Rechiseled: AE2** — autocraft Rechiseled blocks via AE2's Chiseling Pattern Encoder

Supplementaries — Vanilla+ Functional Decoration

Decoration blocks that serve a purpose:

Block	Function	How
Jar	Stores 1 item visually	Right-click with item to put in, empty hand to take out
Signpost	Directional sign pointing to named location	Right-click with tag item to set destination name. Arm points toward Waystone
Faucet	Fluid flow decoration	Place above fluid tank. Right-click to toggle flow
Weather Vane	Shows wind direction/speed	Place outside. Rotates in real-time
Item Shelf	Wall-mounted display (3 items)	Right-click with item to place, sneak-right-click to remove
Rope	Climbable, place in mid-air	Right-click to place knot, add more rope below. Breaks when top knot broken
Flute	Craftable instrument	Right-click to play note. Villagers react

Practical builds:

- Kitchen: Item shelves for ingredients, jars for flour/sugar, faucet over sink
- Entrance: Signposts to “Mine”, “Nether Portal”, “Village”

- Garden: Weather vane, rope swing from tree
- Study: Jars with glowing items as lamps

Macaw's — Complete Building Set

Five modules covering structural elements in every wood type.

Bridges: 6 slabs + 1 stick in bridge pattern. Right-click to extend, shift-right-click to shorten. Straight or rope variants. All 10+ wood types.

Doors: Special doors per wood — glass, slatted, stable (top half opens independently), cottage, shop, paper. Craft with matching wood + relevant material.

Roofs: Roofing blocks — slopes, ridges, gutters. 3 slabs in triangle pattern for slope piece. Auto-connects corners.

Windows: Diamond pane, framed, barred, tropical, stained glass. Craft with glass + wood.

Fences: Picket, stockade, horse, wired, highley gate. Each distinct look.

Building tip: Macaw's roofs + Rechiseled stone + Macaw's windows = cohesive aesthetic.

Diagonal Fences — Seamless Corners

Connects vanilla fences, fence gates, and walls **diagonally** — the classic corner-post look without gaps. Client-side rendering, no config.

Integration: Complements Macaw's building set (Macaw's fences are separate blocks, unaffected) and Forgematica schematics — diagonal corners render correctly in-place. Requires Puzzles Lib (already in pack).

Building Wands — Large-Scale Construction

Place many blocks at once. Closest thing to WorldEdit without a server plugin.

Craft: Stick + 2 blocks of same type in + pattern. Block type determines tier (stone = basic, iron = better, diamond = best).

Modes (right-click wand to open GUI):

Mode	Does	When to use
Build Wall	Wall between 2 clicked points	Enclosing rooms, perimeters
Build Line	1-block-wide line	Railings, borders, paths
Fill Area	3D cuboid between 2 corners	Foundations, clearing rooms
Build Circle	Ring of blocks	Towers, wells
Build Cylinder	Filled cylinder	Silos, water tanks
Build Sphere	Hollow or filled sphere	Domes, decorative globes

How: Left-click to set first corner (green sparkle). Right-click to set second corner (structure builds). Uses blocks from your inventory.

Tips:

- Shift+Right-click to undo last placement
- Use **Undo wand** (stick + slimeball) to reverse recent actions
- Combine with AE2 wireless terminal for inventory access while building

Handcrafted — Furniture

Furniture	Function	Craft
Chairs	Right-click to sit	Slabs + planks
Tables	Right-click to place items on top	Varies by type
Shelves	Wall-mounted item display	Wood + relevant material
Curtains	Right-click to open/close	Wool + stick
Desks	Table + drawer storage	Slabs + planks
Cupboards	Wall-mounted aesthetic storage	Wood + glass
Sofa	Multi-block seating	Wool + wood
Dresser	Tall drawer storage	Wood + relevant material

Tip: Place in your most-used rooms — kitchen, dining, bedroom. Use shelves to display favorite loot (Twilight Forest trophies, Apotheosis mythic gear, rare Simply Swords weapons).

Fetzi's Displays — Item Showcases

Display your hard-earned items in style. Craft a **Workbench** (check EMI) to make all display types.

Display Type	What it does	Best for
Glass Cloches	Glass dome on pedestal — 16 colors	Trophy items, rare drops, gems
Katana Stands	Vertical weapon rack	Simply Swords weapons, boss drops
Wall Holders + Plaques	Wall-mounted display with name tag	Named weapons, “GOD SWORD” vibes
Racks (4 types)	Horizontal/vertical item holders	Tool organization, collection display

Scale, rotation, position: Items in displays can be scaled, rotated, and repositioned. Check the [Wiki](#) for exact controls.

Integration with Apotheosis (Wave 3): Glass cloches and katana stands are ideal for showcasing mythic drops. Mount your perfect gem rolls and affix-rolled

weapons as trophies. Wall holders with plaques let you name your legendary weapons — “Flamebane, Bane of the Ender Dragon.”

Lili’s Pottery — Decorative Earthenware

Part of the [Let’s Do](#) cozy mod series. Adds pottery blocks for interior decoration.

What it adds:

Category	Variants
Clay	6 new clay types found in various biomes
Pots & Vases	7 variants in old and new clay styles
Urn & Amphore	Functional storage — save items from death

How:

1. Find **new clay types** in world (beaches, rivers, various biomes)
2. **Knead** clay at crafting table → raw pot forms
3. **Bake** in furnace → hardened pottery
4. **Paint** patterns → finished decorated pottery

Integration with Handcrafted: Both add decorative blocks but cover different aesthetics. Handcrafted is rustic furniture (chairs, tables, shelves). Lili’s Pottery is earthenware (pots, vases, urns). Together they provide a full interior decoration toolkit.

Laser Bridges & Doors — Energy Barriers

Futuristic laser barriers created by powering Laser Source blocks with redstone.

What it adds:

Block	Function
Laser Source Block	Projects a horizontal laser bridge when powered by redstone
Laser Fence Source Block	Projects a vertical laser barrier

How it works:

1. Craft a **Laser Source Block** — End Crystal + 7 iron ingots + glass (check EMI)
2. Power with **redstone** — laser length scales with signal strength (max 15 blocks by default, configurable)
3. **Dye** with right-click to customize color — any dye works, dispenser-compatible too
4. Convert between laser bridge ↔ laser fence in a crafting grid

Key features:

- **Dynamic length** — changes in real-time as redstone signal strength varies (use comparators, levers at different levels)
- **Blocks cut through** — configurable list of blocks lasers can replace. Default: air only. Tune in [config/laserbridges.json](#)
- **Dispenser dyeing** — automate color changes with redstone and dispensers

Recipe gate: Requires an End Crystal — this is post-End-dragon content. Appropriate for Phase 2+ base building.

Integration with Redstone & Tech:

- **AE2 level emitters** — emit redstone when item quantities hit thresholds → auto-close laser gates when storage is low or auto-open when conditions are met
- **Create redstone links** — wireless redstone across your base to control laser arrays from a central panel
- **Comparators** — use container fullness to control bridge length dynamically

Chill living fit: Purely decorative/functional — no combat interaction, no progression pressure. Dye-customizable to match base aesthetics (warm amber lasers for cozy builds, cool blue for tech areas).

Session Plan — Wave 6

Session	Focus	What to Build
Session 1 — Rechiseled basics	Craft chisel. Convert stone, wood, glass into decorative variants. Try connected textures	Decorative block library
Session 2 — Structural elements	Craft Macaw's roofs, windows, doors in your base wood type. Build a cohesive room	Cohesive room aesthetic
Session 3 — Furniture	Handcrafted chairs, tables, shelves in kitchen and bedroom. Item shelves for loot display	Functional interior
Session 4 — Large-scale	Building Wands — wall mode for perimeter, fill mode for foundation. Build a tower with circle mode	Rapid construction

Session		Focus		What to Build	
Session 5 — Details		Supplementaries jars, signposts, weather vanes. Laser Bridges & Doors for energy barriers and decorative lighting. Rechiseled: Create for factory aesthetics		Decorative details	

Wave 7 — Space Exploration

Core Space Mod

Mod	Role
	Northstar Redux 0.6.4

Northstar Redux (by Astronauts-of-Create) is the maintained continuation of Create: Northstar — space exploration built on the Create mod. Rockets are player-built, superglued Create contraptions; progression runs through a science system (Telescope → Astronomical Readings → Astronomy Table → Space Atlas) that unlocks planets. Oxygen and temperature management, Titanium/Martian Steel/Tungsten materials, and four reachable planets: Moon, Mars, Mercury, Venus.

No worldgen: Northstar generates no Overworld structures — the launch infrastructure is entirely player-built, and planets are separate dimensions. The “Create: Northstar - Structures” addon (planet-surface craters/dungeons) exists but is **not installed** in this pack.

Dependencies: none new — Create 6.0+ and GeckoLib are already in the pack (Wave 1). No Potentials API or Sky Aesthetics needed (those shipped with Stellaris only).

Update policy: Northstar is in active early development (0.6.x). Updates may change item IDs and invalidate Return Tickets. Pin 0.6.4; back up worlds before updating; after an update, re-run the quest validation pipeline ([tools/ftbquests/generate.py](#) + [validate.py](#)) so stale item IDs surface immediately.

Integration with Tech Mods

|| Layer | Northstar Redux ↔ Tech Mod Interaction | ||-----|

-----| || Power |

Mekanism cables can power Northstar machines. Create rotational → Mekanism → Northstar | || Energy | Mekanism universal cable connects to Northstar machines (energy API compat) | || Storage | AE2 ME drives store Northstar materials; autocrafting builds rocket components | || Ore Processing | Mekanism 5x ore multiplication works on Northstar ores (titanium, martian iron, etc.) | || Fuel | TFMG diesel/gasoline compatibility for rocket fuel (native compat) | || Food | Farmer’s Delight meals eatable in space when oxygen is present |

Northstar — Interplanetary Travel

Create-native space exploration. Build a rocket from Create blocks and Northstar parts, glue it together, fuel it, and fly.

Core concepts:

Concept	What it means
Rocket Station	Assembles the rocket and sets the destination (unlocked via science)
Rocket Controls	Interface with the rocket's systems — also accepts redstone for launch automation
Science	Telescope → Astronomical Readings → Astronomy Table (spend as science points) → Space Atlas unlocks planets
Oxygen	No atmosphere off-world. Separator + Filler + Sealer keep you and your base alive
Heat	Heat shielding scales with rocket size; Temperature Regulator + Martian Steel suit for hot planets
Planets	Moon → Mars → Mercury → Venus. Each requires more science and better shielding

Phase 1: Getting to Space

Step 1 — Gather resources:

- Collect iron, copper, redstone, glass, and Create materials (andesite alloy, brass)
- Process **Titanium** (raw titanium ore → ingot) for advanced parts — check EMI for the chain

Step 2 — Build the launch site:

Machine	Function	Priority
Rocket Station	Assembles the rocket and picks the destination	First
Rocket Controls	Rocket system interface (redstone-controllable)	Second
Oxygen Separator	Extracts breathable oxygen	Third
Oxygen Filler	Refills suits and contraptions with oxygen	Fourth
Oxygen Sealer	Pressurizes sealed rooms	Fifth

Step 3 — Build the rocket:

1. Build the hull from Create blocks + **Rocket Thruster** + **Rocket Combustion Chamber**
2. Superglue the whole assembly into one contraption
3. Place **Rocket Controls** on board
4. Use the Rocket Station to assemble and set the destination

Step 4 — Science unlocks:

1. Craft a **Telescope** and take **Astronomical Readings**
2. Spend them at the **Astronomy Table** as science points
3. The **Space Atlas** shows unlocked planets — Moon is first

Step 5 — Launch:

1. Equip the **Iron Space Suit** (full set; fill oxygen at a Filler)
2. Enter the rocket, launch via the Rocket Controls
3. Land on the Moon — watch your oxygen

The Moon — First Destination

Safest first destination. No atmosphere, low gravity, no mobs.

What to do:

- Mine **moon stone** and **lunar sapphire** (shards → polished)
- Find **glowstone ore** and rich veins of standard ores
- Place a **Waystone** immediately; set up a sealed base (Oxygen Sealer + Filler + Temperature Regulator if needed)
- Return with a **Return Ticket** or by launching from the Moon

Other Planets

Planet	Hazards	Key Resource	Notes
Moon	None (oxygen required)	Lunar Sapphire	First destination. Safe practice
Mars	Worms, toads, cobras, moths	Martian Steel, Iron	Best permanent off-world base
Mercury	Extreme heat, raptors, roaches	Tungsten, Titanium	Heat shielding required
Venus	Acid plumes, mimics, scorpions	Volcanic Ash, ores	Hottest planet — endgame target

Off-World Machines

Machine	Function	Why you need it
Oxygen Filler	Refills suits/contraptions	Required on every planet
Oxygen Sealer	Pressurizes sealed rooms	Keeps bases breathable
Oxygen Detector	Reads oxygen levels (redstone)	Automation for life support
Temperature Regulator	Regulates heat/cold	Mercury, Venus, and any cold base

Machine	Function	Why you need it
Atmospheric Concentrator	Concentrates breathable air	Denser life support for large bases
Rocket Waypoint	Precise landing coordinates	Land at your base, not wherever

Instance swap checklist (Stellaris → Northstar Redux)

In the game instance:

1. Remove the `stellaris` jar, `potentials` jar, and `sky-aesthetics` jar (Stellaris-only dependencies)
2. Add the `northstar-redux` 0.6.4 jar (Create 6.0+ and GeckoLib are already installed)
3. Do **not** install “Create: Northstar - Structures”
4. Existing worlds: Stellaris structures remain as inert blocks and its items vanish from inventories — a fresh world is recommended for the swap

Integration With Tech Base

|| System | How it connects |

||-----|-----|

|| **Power** | Mekanism generators at Moon base. Universal cables power Northstar machines | || **Ore processing** | Bring planet ores (titanium, martian iron, tungsten) back to Mekanism line. 5x works on Northstar ores | || **Storage** | AE2 ME drives store Northstar materials. Autocraft rocket components. Quantum link or P2P tunnel to off-world base | || **Transport** | Create trains move rocket materials to the launch site. Conveyors feed the Rocket Station | || **Fuel** | TFMG diesel/gasoline compatibility for Northstar rocket fuel (native compat) | || **Backpacks** | Netherite-upgraded backpacks carry oxygen tanks and supplies | || **Food** | Farmer’s Delight meals eatable in space with oxygen present | || **Gear** | Apotheosis affixes may apply to space suit (if item tags align) | || **MekaSuit** | Pending compat bridge — modules make the MekaSuit a space suit once it tracks Redux 0.6.4+ |

Progression Placement

Space belongs in **Phase 2** (hours 40–200). Prerequisites:

- Create rotational power (brass age — rockets and machines need it)
- Mekanism basic processing (Enrichment Chamber)
- Small AE2 network
- Good weapons/armor (Wave 3/5 gear helps with planet mobs)

Start with Moon missions and a sealed Moon base, then Mars. Mercury and Venus are late Phase 2.

Session Plan — Wave 7

|| Session | Focus | What to Build | ||—————|

|| **Session 1 — Ground prep** | Build Rocket Station + Rocket Controls. Process Titanium. Assemble and superglue the rocket | Launch infrastructure | || **Session 2 — First launch** | Craft Telescope, take Readings. Craft Iron Space Suit, fill oxygen. Launch to the Moon | First off-world base | || **Session 3 — Moon base** | Sealed base: Oxygen Sealer + Filler. Waystone. Mine lunar sapphire. Chunk Loader | Sustainable Moon operation | || **Session 4 — Mars** | Smelt Martian Steel. Land on Mars, mine martian iron and titanium. Temp base + weapons for hostile mobs | New resources | || **Session 5+ — Deep space** | Mercury (tungsten, heat). Venus (endgame). Martian Steel Space Suit. Space Atlas + Return Tickets | Interplanetary logistics |

Tip: Space is the most resource-intensive activity. Don't rush it. Each planet should feel like a meaningful expansion. Note: the MekaSuit is **not** a Northstar space suit by default — you still need the Iron/Martian Steel suits for oxygen. The *Mekanism* × *Create: Northstar* compat mod adds MekaSuit space-suit modules; it's queued for the pack once it tracks Redux 0.6.4+.

Create World

All mods installed and every wave configured — here's how to start a proper world. These settings keep the pack survival-pure, the chat clean, and the worldgen fully modded.

Before Creating

1. **Install the quest book** — copy `config/ftbquests/quests/` into the instance's `config/ftbquests/` folder (merge/overwrite) so the pre-authored book is ready from world one (see *Installing the Quest Book* in Wave 4.5).
2. **Optional data packs** — for the structure-spacing tweak, place `rarer-better-dungeons/` in `saves/<world>/datapacks/` after creating the world (see `datapacks/README.md`).

World Creation Settings (Singleplayer → Create New World)

Setting	Value	Why
Game Mode	Survival	The pack is survival-only — every block placed and item crafted must be earned
Difficulty	Normal	Easy-to-standard curve. The real difficulty ramp is Apotheosis mobs and boss fights, not vanilla mobs
Allow Commands (Cheats)	OFF	Survival purity — no <code>/gamemode</code> escapes, no creative shortcuts; quest rewards stay earned. Cheats can't be enabled later without opening the world to LAN, so set everything you need now
World Type	Default	Terralith + Tectonic overhaul the overworld automatically. Don't pick Superflat, Single Biome, or Amplified — they break modded worldgen
Seed	Leave empty	Random is fine — worldgen mods guarantee interesting terrain either way
Structure Generation	ON	YUNG's collection, Waystones, and every dimension need it

Game Rules → Chat → disable all options (Command Feedback, Command Block Output, Reduced Debug Info, Show Death Messages, Show Recipe Messages, Announce Advancements). Keeps the chat clean — progression popups still appear via Advancement Plaques and the quest book still tracks everything.

After First Launch

1. First world load takes a few extra minutes while Terralith, Tectonic, and YUNG's generate the initial chunks — that's normal.

Mod Count Summary

Wave		Mods	Deps	Total	Notes	
Wave 0 — Foundation		12	10	22	Dependencies (12), Performance & Rendering (12) + FastWorkbench –Sodium Extras –Sodium Options API (Aug 2026 — broken dep chain: sodiumoptionsapi→RSO→Sodium 0.8.12+→incompatible with sodiumoptionsapi), Infrastructure (3) + Simple Recall Potion (travel) — all the nuts & bolts that make the game run (+1 shaderpack, +20 resource packs). Wave 0.5 UI swap: –Immersive UI –Smooth GUI –Tooltip Overhaul –Controlling –First Person Model +Modern UI	
Wave 0.5 — Resource Packs		5	0	5	EMF + ETF + Continuity + Sinytra Connector + Cull Leaves — client-side support mods for Armored Legacy (EMF/ETF), Overlay's (Continuity/Connector), Better Leaves perf (Cull	

Wave		Mods	Deps	Total	Notes
					Leaves); +11 resource packs –Patricx +Faithful –FA Extensions (Aug 2026 resource pack review; removed — Details addon corrupted animal models, user report)
Wave 1 — Tech		20	5	25	Create + 14 addons (incl. Power Grid, Gunsmithing, Protection Pixel, Ornithopter Glider, Train Utilities, Railways Navigator, Interiors, Train Parts, Threaded Trains, Central Kitchen), Mekanism + Generators, AE2, Advanced Finders, Advanced Chimneys + NTGL, GeckoLib, Framework, ForgeEndertech, Kleiders deps
Wave 1.5 — Colony		9	—	9	MineColonies (4 deps counted as mods — they’re library mods; CurseForge-only) + 4 addons: Byzantine Styles, Stylecolonies, Create: Colony Logistics, ColonyLink
Wave 2 — Exploration		35	8	44	YUNG’s (12), Terralith/ Tectonic, Serene Seasons + Serene Seasons Plus + GlitchCore + Gabou’s Libs, Darker Depths, Upgrade Aquatic, dimensions (3), End overhaul (3 + Nullscape dp + 5 deps), Structory, navigation,

Wave		Mods	Deps	Total	Notes
					aircraft, ships, hang glider, MoMP addon, Incendium (Nether biome overhaul), Immersive Machinery (utility machines), Field Guide + 2 required companions (Immersive Overlays, Item Descriptions), Bumblezone (bee dimension) + Resourceful Lib
Wave 3 — Equipment Magic		10	12	22	Skills Mastery Reimagined, Pufferfish's Skills, Simply Swords, Simply More, Too Many Bows, Relics, Enchanting Runes, Immersive Armors, Apotheosis, Apothic Compats + 12 deps (Placebo, Apothic modules, Patchouli, Simply Tooltips, Fzzy Config, Ranged Weapon API, Spell Engine, Bundle API, Pufferfish's Attributes, Curios API)
Wave 4 — Food & Farming		14	—	14	Farmer's Delight + 8 addon mods + Neo Bee Fix + Comfortable Campfires + Smarter Farmers + Create: Café + Spice of Life: Carrot Edition
Wave 4.5 — Quests		6	0	6	FTB Questing ecosystem: FTB Library + FTB Quests + FTB Teams + KubeJS + FTB XMod Compat + MineColonies Questline. Bountiful removed.

Wave		Mods	Deps	Total	Notes
					ExtraQuests removed (Aug 2026 — quest book uses only standard FTB Quests types; its ExtraLib dependency was never in the pack).
Wave 5 — Combat & Mobs		6	5	11	Better Combat, L_Ender's Cataclysm, Enchantment Descriptions, Create Big Cannons + Advanced Technologies addon, Torchmaster
Wave 6 — Building		11	3	14	Rechiseled, Supplementaries, Macaw's (4), Building Wands, Handcrafted, Fetzi's Displays, Lili's Pottery, Laser Bridges & Doors, Diagonal Fences + Rechiseled: Create, Rechiseled: AE2 + Moonlight, Resourceful, Fusion
Wave 7 — Space Exploration		1	0	1	Northstar Redux (Create 6.0+ and GeckoLib already in pack — no new dependencies; no worldgen structures)
Total		181	47	228	All confirmed NeoForge 1.21.1 (–Stellaris –Potentials API –Sky Aesthetics –CC:Tweaked +Northstar Redux, Aug 2026; +Real Camera; Day Counter Plus reverted — Day Counter + Hud Texts restored; +BetterGrassify + Forgified Fabric API + Sky Aesthetics; +Serene Seasons Plus + GlitchCore

Wave		Mods	Deps	Total	Notes	
					+ Gabou's Libs - Chunky - When Dungeons Arise - Structory: Towers, Aug 2026; +Smarter Farmers (Sodium Extras + Sodium Options API removed Aug 2026 — broken dep chain on NeoForge 1.21.1) + Diagonal Fences + Create: Café, Aug 2026 mod review; +FastWorkbench + MoreCulling + BadOptimizations + Searchables + Loot Beams Refork + Nirvana Library + Common Network + Auto HUD + Create Central Kitchen + Spice of Life Carrot + Apothic Compats + Bliss shader, Aug 2026 weapon-QoL review; -Create: Misc and Things, removed by user; +Modern UI -Smooth GUI -Tooltip Overhaul - Controlling -Immersive UI -Smooth Font texture pack -First Person Model, Aug 2026 Modern UI adoption; +EMF +ETF +Continuity +Sinytra Connector +Cull Leaves +9 resource packs -Patricx +Faithful, Aug 2026 resource pack review; -Complementary Unbound -Euphoria Patches +Bliss as main shader, Aug 2026 shader swap; +Rekindled CTM (Fire Rekindled CTM	

Wave		Mods	Deps	Total	Notes	
					addon), Aug 2026; +Faithful 32x AppleSkin Addon, Aug 2026; +Mod Descriptions (Field Guide companion), Aug 2026; +Bumblezone +Resourceful Lib, Aug 2026; -The RCP (1.21.1 pack_format incompatible), Aug 2026)	

Keybindings Reference

Keybind		Suggested		Mod		Wave
View Recipes		R		EMI		0
View Uses		U		EMI		0
Open Fullscreen Map		J		JourneyMap		0
Toggle Minimap		Check default		JourneyMap		0
Open Waystone Menu		Ctrl+Shift+W		Waystones		0
Open Backpack		B		Sophisticated Backpacks		0
Zoom		C		Sodium		0
Open Action Center		Ctrl+K		Modern UI		0.5
Toggle Dynamic Lights		Ctrl+L		LambDynamicLights		0
Open Curios Inventory		Y		Curios API		0
Open Skill Tree		K		Pufferfish's Skills		3
View Weapon Ability		Ctrl (hold)		Simply Swords		3
Open Rune Pouch		Set manually		Runes		3
Open Create Ponder		W (hold over block)		Create		1
Open Town Hall GUI		Right-click		MineColonies		1.5

Pro tip: Open **Options** → **Controls** → **Key Binds** and browse categories at the bottom. Every mod's binds are grouped under its name. Set the ones above first, then explore others as you install new waves.

Modern UI note: it also binds zoom to **C** (OptiFine-style). Disable it in the Action Center (**Ctrl+K**) so Sodium's zoom keeps the key.

Progression Through All Waves

Phase	What you'll be doing
Phase 2 (hours 40–200)	Full Create factory (conveyors, deployers, sequenced crafters, trains). Mekanism 3x–5x ore processing. AE2 ME network with autocrafting. Explore Twilight Forest bosses. Apotheosis Rare/Epic gear. First Cataclysm boss. TFMG oil refining and diesel engines. Build with Rechiseled, Macaw's, Handcrafted. Small Ships for ocean travel. Immersive Aircraft for aerial exploration. Complete quest chapters. Northstar Redux Moon and early planets
Phase 3 (hours 200+)	Mekanism fusion reactor + MekaSuit. AE2 full autocrafting CPU chains. Endgame Apotheosis Mythic gear with perfect gem sockets. Twilight Forest final bosses. Cataclysm endgame bosses. End dimension exploration. Mars and Venus space bases. Monumental building projects. Create mega-factories. Perpetual optimization

General Tips

Finding Help

- **EMI (R/U)** — always your first stop. Hover any item, press R for recipe, U for uses
- **Ponder (W over Create blocks)** — interactive animated tutorials. Better than any wiki
- **Mods screen** — every installed mod's config button (Better Modlist adds badges/search)
- Most mods link to their wiki or Discord from their Mods screen entry

Losing Progress

You won't. GraveStone Mod preserves your items on death. No creeper grief data pack prevents base damage. The pack is designed so you can walk away for a month and come back to exactly the world you left.

Java & Performance

Minecraft 1.21.1 requires **Java 21**. See [Wave -1](#) for JDK installation and JVM arguments.

Performance tips:

- ZGC garbage collector with generational mode (`-XX:+UseZGC -XX:+ZGenerational, -Xms = -Xmx`) — full argument set and RAM table in [Wave -1](#)
- Distant Horizons handles far rendering — keep vanilla render distance at 10
- ModernFix `dynamic_resources = true` for fastest startup
- If you see missing textures, disable `dynamic_resources` temporarily

Cross-Wave Integration Cheat Sheet

If you want to...	Combine these mods
Auto-build a blueprint	Create Schematicannon + Blueprint and Quill
Power Create from Mekanism	Create Crafts & Additions electric motor/alternator
Bridge TFMG oil to AE2	TFMG Energy Converter + Mekanism universal cable
Store colony resources digitally	AE2 Storage Bus on MineColonies warehouse
Feed colony kitchen automatically	Create conveyor → builder's chest + Farmer's Delight meals
Autocraft Mekanism chemicals	Applied Mekanistics + AE2 Pattern Provider
Cook from your ME network	AE2 Export Bus on Farmer's Delight cooking pot
Get combat buffs before bosses	Farmer's Delight meal + Brewin' And Chewin' drink (effects stack)
Find the best loot	Explorer's Compass → YUNG's structure → loot → Apotheosis Rare+ gear
Build a train network	Steam 'n' Rails tracks → Train Utilities stations → Railways Navigator boards
Fly between Aether islands	Immersive Aircraft airship + Aether portal
Defend colony perimeter	Create Big Cannons on walls + Guard Towers

If you want to...	Combine these mods
Explore oceans efficiently	Small Ships cog → Ocean's Delight ingredients → YUNG's Ocean Monuments
Process space ores	Northstar titanium/martian steel → Mekanism 5x line → AE2 storage
Build a megabase	Building Wands (structure) + Rechiseled blocks (aesthetic) + Handcrafted furniture (interior)